

Machine Learning for Realised Volatility Forecasting

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Abstract

We assess the predictive power of machine learning (ML) models for forecasting realised volatility using information from HAR model variables, limit order book (LOB) data, and news sentiment. Training and robustness checks on nearly seven million ML models show that high-dimensional ML models outperform HAR models in 90% of the out-of-sample period, except during extreme volatility. Explainable AI analysis identifies mid prices, mean bids, and mean asks as key predictors. Notably, incorporating ML into ensemble frameworks enhances HAR model performance, though caution is needed when using ML models as direct substitutes, since they may yield unreliable forecasts under certain market conditions.

Keywords: Realised Volatility Forecasting, Machine Learning, Big Data, Long Short-Term Memory, Heterogeneous Autoregressive Models, Explainable AI.

JEL: C22, C45, C51, C53, C55, C58

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1 Introduction

Volatility forecasting plays a critical role in financial modelling and decision-making. This study investigates the effectiveness of machine learning (ML) models combined with a rich feature set in forecasting volatility for 23 NASDAQ tickers over a sample period from 27 July 2007 to 27 January 2022. It is vital to consider ML for several reasons. First, classical econometric models, without undergoing additional modifications, cannot handle a large number of input variables in big datasets, such as limit order book (LOB) data and news stories, within a single model. Second, ML models are usually more efficient in capturing nonlinear relationships in high dimensions, a task that standard econometric models generally struggle to accomplish. In this study, we aim to explore the impact of a wide array of variable groups originating from the well-known HAR-family of models, high-frequency LOB data, and sentiment variables extracted from news stories on the forecasting accuracy of realised volatility (RV) under different market conditions, achieved through the utilisation of ML models, notably long short-term memory (LSTM) models.

ML models have been used in various areas of finance for several years. Gu et al. (2020) showed the superior performance of ML models for empirical asset pricing, Gu et al. (2021) introduced an autoencoder asset pricing model and produced smaller out-of-sample pricing errors compared to classical leading factor models, Jiang et al. (2023) produced more accurate stock return predictions based on ML image analyses, and Chen et al. (2024) employed ML in the estimation of the stochastic discount factor. A subgroup of ML models has been developed for sequential data such as video, music, text, and, in our case, financial time series. The recurrent neural network (RNN) and its extension, LSTM (Hochreiter and Schmidhuber, 1997), are among the most widely used pioneering ML models in both academia and industry for sequential data, paving the way for more advanced ML models (see Vinyals et al. (2019) and Andrychowicz et al. (2020) for recent recurrent use cases). Through their recurrent structure and, in the case of LSTM, gated memory mechanisms, these models can perform sequential processing and learn long-term dependencies in time series.

The availability of high-frequency financial data makes RV a popular model-free proxy for actual volatility (Andersen et al., 2001; Barndorff-Nielsen and Shephard, 2001). With the popularity of RV, the heterogeneous autoregressive (HAR) model (Corsi, 2009) and its variations are commonly used in forecasting RV. Well-known variants include HAR-J (HAR with jumps) and CHAR

(continuous HAR) (Andersen et al., 2007), Patton and Sheppard (2015) SHAR (semivariance-HAR), which separates the impact of negative and positive returns on subsequent RV, and HARQ (Bollerslev et al., 2016), which adjusts forecasts for measurement error based on realised quarticity (RQ).

Within the realm of literature centred on ML models for RV forecasting, Hillebrand and Medeiros (2010), Fernandes et al. (2014), Audrino and Knaus (2016), Branco et al. (2024), and Audrino and Chassot (2025) demonstrated that ML models exhibit similar or lower forecasting performance when compared to the HAR-family of models. On the other hand, Bucci (2020), Christensen et al. (2023), Zhu et al. (2023), and Li and Tang (2025) reported improvements in RV forecasting using ML models. Motivated by these contradictory findings, this study aims to establish a more systematic comparison between the two classes of models, with the objective of better identifying their respective strengths and weaknesses under different market conditions. Also, apart from fitting ML models with a large, comprehensive dataset, it is very important to fully explore the hyperparameters and tuning of ML models. In contrast to these previous ML studies that utilised a relatively small number of variables, our study focuses on leveraging a vast array of data features and testing near seven million ML models. This approach enables us to explore thoroughly whether ML (with a wide and varied range of variables) outperforms the HAR-family of models in RV forecasting.

The three variable sets, i.e., LOB, news sentiments and HAR, are commonly used in the literature for volatility forecasting, though rarely all three at the same time. The high dimensionality of ML models allows us to test all these variables together in a single model. Using combinations of 147 predictors (6 HAR, 9 news sentiments, and 132 LOB variables), this study provides strong statistical evidence that ML models outperform the HAR-family of models in RV forecasting according to MSE, quasi-likelihood (QLIKE), and mean directional accuracy (MDA) loss functions, as well as the reality check (RC). However, and more importantly, this considerable and statistically significant enhancement applies to 90% of out-of-sample daily forecasts, except when the actual RV reaches high levels. On high volatility days, the HAR-family of models generally outperform ML models. This finding is potentially concerning and highlights the need for cautious evaluation before considering ML models as direct substitutes for the HAR-family of models. Moreover, LOB data, in general, provides stronger volatility forecasting performance when compared to news sentiment variables. A key discovery of our study is the importance of

incorporating a wide range of input variables in a single model to improve RV forecasting performance. We noted that even a simple ML model with HAR variables can outperform HAR models in forecasting RV. As emphasised by Christensen et al. (2023), this could be due to the ability of ML models to capture nonlinear relationships in RV forecasting. However, the performance improvement becomes particularly pronounced when a rich set of predictors is included in the ML models.

Recently, there has been increased attention in the fields of accounting and finance towards deploying explainable AI (XAI) techniques to evaluate ML models; see, for example, Erel et al. (2021), Bali et al. (2023), and Chronopoulos et al. (2024). By applying XAI, we find that, among the 147 input variables, mid prices at all LOB levels, mean bid, and mean ask are the most informative variables for forecasting RV. We also find that the relative importance of different groups of information sets changes over time, with 2018 emerging as a structural breakpoint. Prior to 2018, news-based variables constitute the most informative group. However, in the post-2018 period, characterised by heightened market volatility, LOB variables become more prominent. Additionally, HAR variables demonstrate greater importance during periods of high volatility. This alternation in the importance of different predictors over time once again underscores the critical relevance of the time-varying choice of optimal predictors for RV forecasting. The results also demonstrate that HAR variables maintain their importance, particularly for forecasting RV during high volatility days.

Extensive robustness checks also confirmed that, for normal volatility days in the out-of-sample period, the superiority of ML over the HAR-family of models remained unchanged. We also trained and tested LSTM models for individual tickers using different ML hyperparameters, i.e., the number of units and the number of epochs. For 90% of the out-of-sample forecasting period, when the actual RV level is less extreme, the optimal number of units and epochs varies, but the overall ML structure is simpler. For the remaining 10% of the out-of-sample period, when the actual RV level is extreme, a more complex ML structure with a larger number of units and epochs is needed. The full out-of-sample is also influenced by high volatility days and leans towards more complex models. We further reinforced our robustness checks by changing the model type, altering the input data, employing alternative loss functions, and implementing additional techniques to mitigate overfitting. Across all specifications, the results consistently support our main conclusions. Our findings also suggest that a simple ensemble method, which

combines the outputs of the HAR-family of models and ML models, offers a practical and effective way to leverage the strengths of both model classes simultaneously. This method achieved robust RV forecasting performance across varying market conditions. Importantly, the results indicate that exclusive reliance on ML models may be suboptimal in certain scenarios, particularly during high volatility days, a limitation that can be mitigated through this simple method.

The literature offers conflicting evidence on whether ML models systematically outperform the HAR-family of models in forecasting realised volatility. We show that these discrepancies largely reflect fundamental differences in sample composition, market conditions, and modelling design rather than the existence of a universally superior approach. Relative forecasting performance is inherently sample- and regime-dependent: variations in the underlying asset universe and sample timeframe directly affect the frequency, severity, and composition of high volatility days, which can influence forecast evaluation and model rankings. ML models are flexible function approximators with large hyperparameter spaces, and differences in model architecture, feature construction, tuning strategies, and validation schemes can also generate variation in out-of-sample performance across studies. By contrast, the HAR-type specifications are deliberately parsimonious in structure and estimation, which tends to yield more stable and comparable performance across samples. Finally, a relatively small number of extreme realised volatility observations, often associated with stock-specific jumps followed by periods of elevated market-wide volatility, can disproportionately affect forecast comparisons, with ML models being more sensitive to such tail events, while HAR-type specifications are relatively more robust.

The remainder of this study is organised as follows: Section 2 gives a brief review of RV and the HAR-family of models. Section 3 describes the data and provides variable definitions. Section 4 provides the empirical framework: RNN, LSTM, and regularisation techniques in Subsection 4.1, and the structure of our proposed ML models in Subsection 4.2. Section 5 presents the results of the primary experiments, Section 6 evaluates the predictive power of input variables, and Section 7 performs a series of robustness checks. Section 8 also introduces a simple ensemble method for forecasting RV. Finally, Section 9 concludes with a discussion of the findings from this study.

2 Realised Volatility and HAR-Family of Models

Suppose that P_t is the ticker price process with the following dynamics:

$$d \log(P_t) = \mu_t dt + \sigma_t dW_t + J_t dQ_t, \quad (1)$$

where μ_t is the drift (a continuous function), σ_t is the volatility process (a càdlàg function), J_t is the jump size, W_t is the standard Brownian motion, and Q_t is a Poisson process. For time $t - 1$ to t , the integrated variance is defined as follows:

$$IV_t = \int_{t-1}^t \sigma_s^2 ds. \quad (2)$$

This integrated variance is not observable; therefore, the realised variance is defined by the summation of squared high-frequency returns as follows:

$$RV_t \equiv \sum_{i=1}^M r_{t,i}^2, \quad (3)$$

where $M = 1/\delta$, and the δ -period intraday return is defined as $r_{t,i} \equiv \log(P_{t-1+i\delta}) - \log(P_{t-1+(i-1)\delta})$. In the absence of jumps, it provides a consistent estimator as the number of intraday observations increases, or equivalently, as $\delta \rightarrow 0$ (Barndorff-Nielsen and Shephard, 2002). Building on existing research, this study seeks to forecast realised variance.

2.1 HAR-Family of Models

Introduced by Corsi (2009), the HAR model, defined below, encompasses the most popular RV forecasting model:

$$RV_{t+1} = \beta_0 + \beta_1 RV_t + \beta_2 \overline{RV}_t^w + \beta_3 \overline{RV}_t^m + \epsilon_{t+1}, \quad (4)$$

where RV_t is the daily RV at time t , $\overline{RV}_t^w$ is the daily average over the last week (last 5 days), and $\overline{RV}_t^m$ is the daily average over the last month (last 21 days). Corsi (2009) showed that this easy-to-estimate linear model with a simple set of historical RVs produced remarkable forecasting performance. Since the work of Corsi (2009), research on improving RV forecasting performance has gained momentum. Researchers expanded the basic HAR model with various information sets and high-frequency data to enhance RV forecasting performance.

Andersen et al. (2007) and Corsi and Reno (2009) analysed the impact of adding a jump component to the basic HAR model. The jump component at time t is defined as $J_t = \max[RV_t - BPV_t, 0]$, with bipower variations:

$$BPV_t = \frac{\pi}{2} \sum_{i=1}^{\overline{M}-1} |r_{t,i}| |r_{t,i+1}|, \quad (5)$$

where $\overline{M}$ denotes the number of intraday returns per day and $r_{t,i}$ is the i -th intraday return on day t , for $i = 1, \dots, \overline{M}$. The HAR-J model is specified as follows:

$$RV_{t+1} = \beta_0 + \beta_1 RV_t + \beta_2 \overline{RV}_t^w + \beta_3 \overline{RV}_t^m + \beta_j J_t + \epsilon_{t+1}. \quad (6)$$

As an alternative specification, the CHAR replaces the predictive variables with BPV_t as follows:

$$RV_{t+1} = \beta_0 + \beta_1 BPV_t + \beta_2 \overline{BPV}_t^w + \beta_3 \overline{BPV}_t^m + \epsilon_{t+1}, \quad (7)$$

where BPV_t , $\overline{BPV}_t^w$, and $\overline{BPV}_t^m$ are, respectively, the daily BPV, the daily average over the past week, and the daily average over the past month at time t . Without the jump component, BPV in Equation (7) is better at capturing volatility persistence and long memory than RV in Equation (4).

Patton and Sheppard (2015) proposed the SHAR model, separating the impact of negative and positive intraday returns on subsequent RV. In the SHAR model, the first lag of RV in the HAR model (Equation (4)) is replaced by a positive return RV_t^+ and a negative return RV_t^- , where $RV_t^+ \equiv \sum_{i=1}^M r_{t,i}^2 \mathbb{I}_{(r_{t,i} > 0)}$ and $RV_t^- \equiv \sum_{i=1}^M r_{t,i}^2 \mathbb{I}_{(r_{t,i} < 0)}$. Here, $\mathbb{I}$ denotes an indicator function. The authors found that RV_{t+1} is more strongly related to RV_t^- than to RV_t^+ for the S&P 500 index and 105 individual tickers.

Bollerslev et al. (2016) studied the impact of measurement error on volatility forecasting. First, they defined the integrated quarticity, $IQ_t = \int_{t-1}^t \sigma_s^4 ds$, and its discrete-time equivalent, realised quarticity $RQ_t \equiv \left(\frac{M}{3}\right) \sum_{i=1}^M r_{t,i}^4$. Next, they introduced the ARQ, HARQ, and HARQ-F models. The HARQ model is defined as follows:

$$RV_{t+1} = \beta_0 + \beta_1 RV_t + \beta_{1Q} RQ_t^{1/2} RV_t + \beta_2 \overline{RV}_t^w + \beta_3 \overline{RV}_t^m + \epsilon_{t+1}, \quad (8)$$

where $RQ_t^{1/2}$ is demeaned for easier interpretation. For $\beta_{1Q} < 0$, RV_t has a lower impact when the measurement error is larger, and a higher impact when the measurement error is smaller. When the coefficients on the weekly and monthly components are set to zero (i.e., $\beta_2 = \beta_3 = 0$), Equation (8) reduces to the ARQ model. Also, HARQ-F is defined by adding the daily average of RQ over the past week and past month to the HARQ model specification. Bollerslev et al. (2016) found the HARQ model to have better forecasting performance, producing more volatility persistence in normal times and quicker volatility mean reversion in erratic times for the S&P 500 index and 27 Dow Jones constituent tickers. Finally, Rahimikia and Poon (2020) compared a range of alternative variables aimed at enhancing the forecasting performance of HAR-family of models. They showed that augmenting standard HAR specifications with only the previous day’s average LOB depth and news count yields statistically and economically significant improvements in RV forecasting performance, without requiring any modification to the underlying model structure.

3 Variables

In this study, 23 NASDAQ tickers continuously traded between 27 July 2007 and 27 January 2022 were chosen for analysis.¹ RV is calculated according to Equation (3) using 5-minute ticker returns. Table 1 provides the RV descriptive statistics for these 23 tickers. Subsection 3.1, Subsection 3.2, and Subsection 3.3 describe the three variable sets from, respectively, HAR-family, news stories, and LOB². The LOB variables for these 23 tickers are compiled using information extracted from LOBSTER³ (Huang and Polak, 2011). These are the three main variable sets commonly used in the literature for volatility forecasting. The fourth variable set is simply the amalgamation of all three variable sets above.

3.1 HAR-Family Variables

Table 2 lists the variables used in the HAR-family of models. The first column (‘Description’) contains the names of the variables. ‘RV’, ‘BPV’, ‘BPV jump’, ‘negative RV’, ‘positive RV’,

¹Nearly seven million ML models were trained and evaluated in this study. To ensure the analysis remained computationally manageable, a limited number of tickers was selected, specifically those with the highest market capitalisations over the sample period and complete data coverage across all trading days. Model training and evaluation were carried out on high-performance computing servers. Despite the limited number of tickers, the model development process was highly computationally intensive, requiring several months of continuous training and evaluation.

²Throughout this study, we use the terms LOB and order book (OB) interchangeably.

³LOBSTER stands for ‘limit order book system – the efficient reconstructor.’

Table 1: Descriptive Statistics of Realised Volatility

Ticker	Min	Max	1 st quantile	Median	3 rd quantile	Mean	STD	Kurtosis	Skewness
AAPL	0.102	229.420	0.899	1.733	3.680	4.623	12.596	111.012	9.124
MSFT	0.067	216.181	0.829	1.449	2.814	3.237	8.125	194.004	11.275
INTC	0.030	318.697	1.103	1.873	3.577	4.299	11.628	294.963	13.982
CMCSA	0.004	237.387	0.910	1.632	3.320	3.821	9.697	192.169	11.462
QCOM	0.122	373.543	1.024	1.975	4.129	5.073	15.380	200.609	12.100
CSCO	0.047	343.946	0.886	1.561	3.028	4.115	13.160	212.453	12.258
EBAY	0.205	252.608	1.319	2.271	4.356	5.082	12.592	142.684	10.009
GILD	0.064	259.489	1.167	1.892	3.379	4.304	12.930	182.820	12.063
TXN	0.177	287.897	1.047	1.905	3.748	4.014	9.820	311.666	14.242
AMZN	0.065	547.030	1.305	2.336	4.808	6.200	19.359	242.205	12.735
SBUX	0.052	265.094	0.864	1.594	3.423	4.201	11.237	161.435	10.626
NVDA	0.159	1104.351	2.282	4.358	9.084	9.756	30.117	586.612	20.058
MU	0.292	484.388	3.570	6.246	11.912	12.818	25.734	89.141	7.960
AMAT	0.292	531.579	1.783	3.028	5.712	6.005	14.632	532.194	18.338
NTAP	0.119	462.821	1.503	2.587	5.154	6.289	18.008	201.510	11.934
ADBE	0.119	569.720	1.099	2.020	3.908	4.947	15.003	588.095	18.867
XLNX	0.229	265.374	1.296	2.363	4.787	5.005	11.941	194.718	11.764
AMGN	0.032	214.156	0.969	1.593	2.872	3.398	9.612	183.759	11.898
VOD	0.055	219.033	0.687	1.342	3.137	3.933	10.869	122.252	9.601
CTSH	0.189	485.894	0.984	1.764	4.161	5.288	15.757	325.214	14.287
KLAC	0.154	499.808	1.456	2.710	5.416	5.919	16.878	354.626	16.033
PCAR	0.039	389.930	1.157	2.162	4.633	5.125	12.108	313.338	13.010
ADSK	0.268	693.772	1.644	2.765	5.167	6.644	22.377	388.131	16.554

Table 2: HAR-Family Variables

Description	#	Characteristic
RV	1	$RV_t \equiv \sum_{i=1}^M r_{t,i}^2$
BPV	1	$BPV_t = \frac{\pi}{2} \sum_{i=1}^{\bar{M}-1} r_{t,i} r_{t,i+1} $
BPV jump	1	$J_t = \max(RV_t - BPV_t, 0)$
Positive; negative RV	2	$RV_t^+ \equiv \sum_{i=1}^M r_{t,i}^2 \mathbb{I}_{(r_{t,i} > 0)}; RV_t^- \equiv \sum_{i=1}^M r_{t,i}^2 \mathbb{I}_{(r_{t,i} < 0)}$
Realised quarticity	1	$RQ_t \equiv (\frac{M}{3}) \sum_{i=1}^M r_{t,i}^4$

Notes: The first column ('Description') contains the names of the variables. Section 2 describes 'RV', 'BPV' and 'BPV jump' (as defined in Barndorff-Nielsen and Shephard (2004)), 'negative RV' and 'positive RV' (as defined in Patton and Sheppard (2015)), and 'realised quarticity' (as defined in Bollerslev et al. (2016)). The second column ('#') lists the number of variables included. The formula used to compile each defined variable is shown in the last column ('Characteristic'). This group has six defined variables in total. The most commonly used 5-minute sampling frequency is used for calculating these variables. $\mathbb{I}$ denotes an indicator function.

and ‘realised quarticity’ are described in Section 2. The second column (‘#’) lists the number of variables included. The formula used to compile each defined variable is shown in the last column (‘Characteristic’). This group has six defined variables in total. The most commonly used 5-minute sampling frequency is used for calculating these variables.

3.2 News Variables

The Dow Jones Newswires covers the Wall Street Journal, MarketWatch, Barron’s news, etc. Every news story is tagged with ‘significant’, ‘about’, or ‘mention’. ‘Significant’ denotes a news story that is important to a specific ticker; ‘about’ denotes a news story about a ticker but of no particular significance, while ‘mention’ denotes cases where the ticker is referenced but is not the main subject of the news story. As ‘significant’ is introduced much later and is not available for most of our sample period, the tag ‘about’ is used for extracting company-related news for constructing the nine daily news variables listed in Table 3. Apart from ‘News count’, the other sentiment variables, ‘Negative’, ‘Positive’, ‘Uncertainty’, ‘Litigious’, ‘Weak modal’, ‘Moderate modal’, ‘Strong modal’, and ‘Constraining’, are compiled according to the LM dictionary (Loughran and McDonald, 2011).⁴

The steps for preprocessing the news data and calculating the sentiment variables follow those in Loughran and McDonald (2011). The sentiment measure for a particular company on day t is equal to the weighted word counts summed over all relevant words and all related news stories published about the company on day t . The averaging is done across sentiment measures of all news stories for a company on a specific day. The proposed term-frequency-inverse-document-frequency (tf-idf) weighting scheme in Loughran and McDonald (2011) and Loughran and McDonald (2016) is applied when calculating the news sentiment variables to take into account that some words appear more often than others.⁵

⁴The LM dictionary is downloaded from Software Repository for Accounting and Finance, University of Notre Dame.

⁵A note of caution is due here, since the LM dictionary was explicitly developed in the context of 10-K filings. Therefore, its direct usage in other types of financial textual datasets may not yield the desired results (see Loughran and McDonald (2020) for more details). Bearing in mind this limitation, the LM dictionary is still one of the most widely used for sentiment analysis in finance at the time of writing.

Table 3: News Variables

Description	#	Characteristic
News count	1	<i>Number of news stories</i>
Positive sentiment	1	<i>Average of positive sentiments</i>
Negative sentiment	1	<i>Average of negative sentiments</i>
Uncertainty sentiment	1	<i>Average of uncertainty sentiments</i>
Litigious sentiment	1	<i>Average of litigious sentiments</i>
Weak modal sentiment	1	<i>Average of weak modal sentiments</i>
Moderate modal sentiment	1	<i>Average of moderate modal sentiments</i>
Strong modal sentiment	1	<i>Average of strong modal sentiments</i>
Constraining sentiment	1	<i>Average of constraining sentiments</i>

Notes: The first column ('Description') contains the variable names. The second column ('#') lists the number of variables. The formula used to compile each defined variable is shown in the third column ('Characteristic'). All sentiment measures in this table are calculated based on the LM dictionary (Loughran and McDonald, 2011). 'News count' is the number of stories on that day. The average value is the daily average of the specific sentiment measure in the first column for all news stories for a company on that day.

3.3 LOB Variables

The LOBSTER dataset contains LOB and message data for NASDAQ-listed tickers, the latter including the execution, submission, cancellation, and deletion of orders. The data cleaning steps are described in Appendix A, and the CRSP dataset is used to correct stock price and volume for stock splits, stock dividends, spin-offs, stock distributions, and rights issues.^{6 7} Table 4 lists the LOB variables inspired by Kercheval and Zhang (2015) for up to ten LOB levels, giving rise to 132 variables in total and representing a comprehensive set of LOB features. The flexibility of ML makes it possible to include such a large and comprehensive set of LOB variables in a single model.

These variables are designed to describe the state and dynamics of the LOB at high-frequency. Specifically, we include basic variables such as the best bid and ask prices and volumes, which reflect the immediate available liquidity on both sides of the LOB. In addition, we incorporate higher-order variables that measure depth beyond the best level, spreads between quotes, mid

⁶We performed supplementary analysis on the LOB data by omitting the cleaning procedures to evaluate their effect on the calculation of RV. Detailed descriptive statistics and the correlation between RVs, computed with and without the cleaning steps, are provided in Table A2. The results show that the uncleaned data yield descriptive statistics that closely mirror those reported in Table 1. In all instances, we observed an exceptionally high correlation, thereby confirming the robustness of the RV calculations.

⁷Table C1 further examines whether differences in high-frequency data sources may contribute to discrepancies in empirical findings. This table reports the correlation between daily RV computed from LOBSTER data and RV constructed from NYSE TAQ data over the common training period. For the TAQ-based RV, only cleaning steps in Appendix A that do not rely on LOB information are applied, and the same sampling frequency and aggregation methodology are used as in the main analysis. The table documents consistently high correlations across tickers, indicating a close alignment between RV measures derived from the two data sources despite differences in market microstructure content and timestamp resolution. Overall, this evidence suggests that the empirical results are unlikely to be driven primarily by the choice of data source.

Table 4: Limit Order Book Variables

Description	#	Characteristic ^a	Parameter
Bid-ask spreads	10	$[(P_l^{ask} - P_l^{bid})]_{l=1}^N$	-
Mid prices	10	$[(P_l^{ask} + P_l^{bid})/2]_{l=1}^N$	-
Price differences	18	$[P_l^{ask} - P_1^{ask}]_{l=2}^N, [P_1^{bid} - P_l^{bid}]_{l=2}^N$	-
Absolute price differences	18	$[P_{l+1}^{ask} - P_l^{ask}]_{l=1}^{N-1}, [P_l^{bid} - P_{l+1}^{bid}]_{l=1}^{N-1}$	-
Mean prices	2	$[\frac{1}{N} \sum_{l=1}^N P_l^{ask}, \frac{1}{N} \sum_{l=1}^N P_l^{bid}]$	-
Mean volumes	2	$[\frac{1}{N} \sum_{l=1}^N V_l^{ask}, \frac{1}{N} \sum_{l=1}^N V_l^{bid}]$	-
Price/volume accumulated differences	2	$[\sum_{l=1}^N (P_l^{ask} - P_l^{bid}), \sum_{l=1}^N (V_l^{ask} - V_l^{bid})]$	-
Price/volume changes	40	$[\Delta P_l^{ask}, \Delta P_l^{bid}, \Delta V_l^{ask}, \Delta V_l^{bid}]_{l=1}^N$	$\Delta t = 1 \text{ day}$
Event-type proportions ^b	10	$[\pi^{(E^V)bid(ask)}, \pi^{(E^H)bid(ask)}, \pi^{(S)bid(ask)}, \pi^{(C)bid(ask)}, \pi^{(D)bid(ask)}]$	$\Delta_t = 1 \text{ day}$
Relative proportions ^c	10	$[1_{\{\pi_{\Delta t}^{(E^V)bid(ask)} > \pi_{\Delta T}^{(E^V)bid(ask)}\}}, 1_{\{\pi_{\Delta t}^{(E^H)bid(ask)} > \pi_{\Delta T}^{(E^H)bid(ask)}\}}, 1_{\{\pi_{\Delta t}^{(S)bid(ask)} > \pi_{\Delta T}^{(S)bid(ask)}\}}, 1_{\{\pi_{\Delta t}^{(C)bid(ask)} > \pi_{\Delta T}^{(C)bid(ask)}\}}, 1_{\{\pi_{\Delta t}^{(D)bid(ask)} > \pi_{\Delta T}^{(D)bid(ask)}\}}]$	$\Delta_T = 1 \text{ day}, \Delta_t = 15 \text{ minutes}$
Accelerations ^d	10	$[\Delta \pi^{(E^V)bid(ask)}, \Delta \pi^{(E^H)bid(ask)}, \Delta \pi^{(S)bid(ask)}, \Delta \pi^{(C)bid(ask)}, \Delta \pi^{(D)bid(ask)}]$	$\Delta t = 1 \text{ day}$

Notes: This table contains LOB variables inspired by Kercheval and Zhang (2015). The first column ('Description') contains the variable names. The second column ('#') lists the number of variables included from 10 LOB levels. The formula used to compile each defined variable is shown in the third column ('Characteristic'). The fourth column ('Parameter') specifies the time parameter. When there is no time parameter, the last snapshot of the LOB for a particular day is used. N denotes the number of LOB levels used per side (bid and ask); throughout we set $N = 10$. Depending on the construction, level-based variables contribute $2N$ components when defined separately for bid and ask, and N components when bid and ask are combined into a single statistic per level. Variables defined as deviations from the best quote (level 1) are indexed by $l = 2, \dots, N$ and thus contribute $2(N - 1)$ components. Also, aggregates over levels yield a fixed number of components (e.g., two when computed once per side). ^a (E^V) and (E^H) denote visible and hidden executions, while (S), (C), and (D) stand for submission, cancellation, and deletion of orders, respectively. 'P' and 'V' stand for price and volume.

^b Proportion of messages of each event type and side within the window, defined as the count of events of that type divided by the total number of messages in the window. ^c This value is a one-or-zero binary number. ^d First difference of the event-type proportions from 'Event-type proportions'.

prices, and volume imbalances, providing a richer description of the cross-sectional configuration of liquidity. By summarising both the best levels and deeper layers of the LOB, these features allow us to capture variations in trading pressure and market tightness. This selection encompasses variables capturing not only the instantaneous structure of the LOB, but also its recent dynamics, such as changes in bid and ask prices and volumes over short time intervals. We also include event-based proportions and their short-horizon changes to capture recent trading activity.

4 Empirical Framework

RNN is a type of artificial neural network designed to recognise patterns in data sequences that have a temporal dimension, such as financial time series. Subsection 4.1 provides a brief review of RNN, their extension, LSTM, and regularisation techniques. Subsection 4.2 presents the structure of the proposed ML models used in this study.

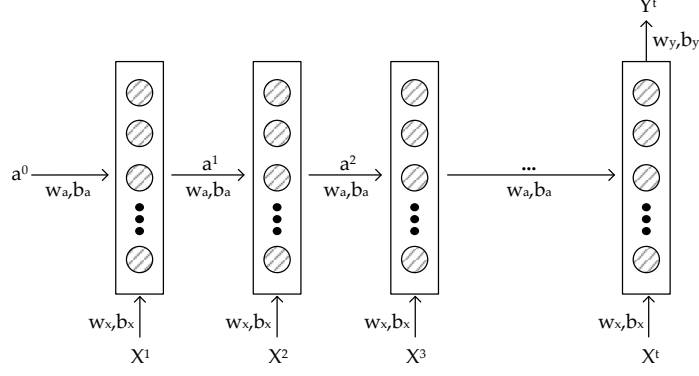


Figure 1: Recurrent Neural Network Abstract Representation

Notes: This representation of the RNN has t input vectors through time ($X^1, X^2, X^3, \dots, X^t$) and one output (Y^t). W_x and b_x are the shared weights and biases between inputs and the neural network layers for time steps 1 to t , W_a and b_a are the shared weights and biases between different layers, and W_y and b_y are the shared weights and biases between the last layer of the neural network and the single output (Y^t). For the ease of exposition, this illustration portrays an RNN for a single input variable. Also, in this representation, every layer has an arbitrary number of units, which are represented by the hatched circles. a^1 to a^t are the transferred information from one layer to the subsequent layer, and a^0 is the input vector for the first layer.

4.1 RNN and LSTM

Figure 1 is a representation of the RNN with t input vectors ($X^1, X^2, X^3, \dots, X^t$) and one output (Y^t). W_x and b_x are the shared weights and biases between inputs and the neural network layers for time steps 1 to t , W_a and b_a are the shared weights and biases between layers, and W_y and b_y are the shared weights and biases between the last layer of the neural network and the single output (Y^t). For the ease of exposition, Figure 1 portrays an RNN for a single input variable. Also, in this representation, every layer has an arbitrary number of units⁸ which are represented by the hatched circles, a^1 to a^t are the transferred information from one layer to the subsequent layer, and a^0 is the input vector for the first layer. a^t is defined as follows:

$$a^t = \tanh(w_{ax}[a^{t-1}, X^t] + b_{ax}), \quad (9)$$

where w_{ax} is the stacked matrix of w_a and w_x , b_{ax} is the stacked matrix of b_a and b_x , and $\tanh$ is the hyperbolic tangent activation function. Also, the output (Y^t) is defined as follows:

$$Y^t = \tanh(w_y a^t + b_y), \quad (10)$$

⁸A unit (neuron) is a basic building block that takes in a set of inputs, performs mathematical operations on them, and produces a single output. Generally, an ML model with more units can be considered more complex.

where w_y and b_y are the output weights and biases, and $\tanh$ is the hyperbolic tangent activation function. The weights and biases in Equation (9) and Equation (10) are estimated by minimising MSE using gradient descent with backpropagation. RNN suffers from the vanishing-gradient problem: the influence of early inputs on later states can decay toward zero as sequences grow longer, limiting the ability to capture long-range dependencies.

Hochreiter and Schmidhuber (1997) use gated cells with their own sets of weights to store, read, or erase historical information. To tackle the vanishing gradient problem, a more constant error rate is maintained in LSTM to allow the model to continue learning over many steps. First, define the candidate memory cell ($\tilde{c}^t$) as:

$$\tilde{c}^t = \tanh(w_c[a^{t-1}, X^t] + b_c), \quad (11)$$

where w_c and b_c are the weights and biases of the candidate memory cell, and $\tanh$ is the hyperbolic tangent activation function. Then, the input, forget, and output gates are defined as:

$$G_i = \sigma(w_i[a^{t-1}, X^t] + b_i), \quad (12)$$

$$G_f = \sigma(w_f[a^{t-1}, X^t] + b_f), \quad (13)$$

$$G_o = \sigma(w_o[a^{t-1}, X^t] + b_o), \quad (14)$$

where w_i and b_i are the weight and bias of the input gate, w_f and b_f are the weight and bias of the forget gate, w_o and b_o are the weight and bias of the output gate, and σ is the sigmoid activation function. Taken together, the updated memory cell (c^t) is calculated as:

$$c^t = G_i \odot \tilde{c}^t + G_f \odot c^{t-1}, \quad (15)$$

where c^{t-1} is the memory cell at time $t - 1$, and $\tilde{c}^t$ is the candidate memory cell at time t from Equation (11). The input gate controls the flow of the input activation into the memory cell, while the forget gate scales the internal state of the cell before adding it to the memory cell. Finally, a^t , the control for information flowing into the next layer, is calculated as:

$$a^t = G_o \odot \tanh(c^t), \quad (16)$$

where G_o is the output gate in Equation (14), c^t is the memory cell in Equation (15), and $\tanh$ is the hyperbolic tangent activation function. The role of the output gate is to control the output flow of the cell activation into the rest of the neural network. All the weights and biases are to be learned during training. Collectively, LSTM avoids the vanishing gradient by suitably memorising and forgetting some past states, so theoretically, it can capture long dependencies in sequential data. Appendix B demonstrates that, under fixed gate configurations, the LSTM architecture collapses to standard models, illustrating its role as a unifying framework.

One typical issue with ML is overfitting, i.e., when the error rate in the training set is artificially low, but the error rate in the test data is high. In this study, we implemented L^2 regularisation and dropout, two of the most widely used regularisation techniques. The L^2 regularisation works by adding a regularisation term, $\lambda||w||^2$, to the loss function, $\mathcal{L}$, as follows:

$$\mathcal{L}'(w; X, Y) = \mathcal{L}(w; X, Y) + \lambda||w||^2, \quad (17)$$

where $\mathcal{L}$ and $\mathcal{L}'$ are the initial and modified loss functions, and w , X , and Y are, respectively, the weights, inputs, and output of the ML model; $||\cdot||^2$ is the squared L^2 norm; and λ is the regularisation factor. The regularisation term penalises a model with larger weights. The larger the regularisation factor, λ , the more severe the penalty. Together, $\lambda||w||^2$ helps mitigate overfitting.

Moving to another regularisation technique, dropout works by randomly removing some units along with their incoming and outgoing connections during training. As a rule of thumb, the optimal dropout rate typically ranges between 20% and 50% of the input and hidden units (Srivastava et al., 2014). The purpose of dropout is to introduce noise into the optimisation process, making training more challenging and thereby preventing the ML model from overfitting.⁹

4.2 ML Model Structure

In this study, we follow the classical framework to evaluate forecasting performance by producing out-of-sample forecasts using only *ex-ante* information. This means repeatedly re-estimating the ML model using only historical information available at the time and rolling forward one step at a time over the out-of-sample period. Such a process is extremely laborious, even without the

⁹In Subsection 7.6, as part of the robustness checks, the regularisation techniques are extended to include early stopping, and its impact on the performance of ML models in forecasting RV is examined.

complexity of testing different ML models using many combinations of hyperparameters and 147 input variables for each of the 23 tickers. Instead of testing several ML models, we have decided here to focus on LSTM and concentrate on gaining a deeper understanding of its strengths and weaknesses in volatility forecasting.¹⁰ Recent studies have found neural networks to be superior among some popular ML models, and among neural networks, LSTM has a structure resembling a time series model. We believe that a detailed investigation of LSTM is crucial at this juncture for its importance in time series analysis. Furthermore, an even more essential objective revolves around assessing the importance of the model’s ability to forecast RV using a rich and diverse set of predictors in a single setting.

As explained in Section 1 and Subsection 4.1, the LSTM has many unique features making it ideal for handling sequential data, including financial time series. This model is one of the top ML choices for modelling time series data in academia and industry. Hence, our goal here is to focus on the LSTM, testing a wide range of hyperparameters, in order to better understand the tuning strategy for achieving the best possible performance. We also believe that testing many models with limited variations in tuning parameters could lead to misleading conclusions. Furthermore, we have chosen a single-layer LSTM instead of a more complex LSTM. A simple vanilla LSTM is chosen as it is more tractable for follow-up analyses. A simpler LSTM model is easier to train; substantially more time, computing capacity, and data are needed for training a more complex LSTM model. Finally, complex ML models have many more combinations of hyperparameters to test and choose from, making the analyses more complicated and less intuitive.¹¹

Figure 2 presents the structure of the LSTM model used in this study. For every explanatory or predictive variable X , the sequence $(X^t, X^{t-1}, \dots, X^{t-n})$ is included as input to the LSTM model. The number of outputs from the LSTM (D_O) is equal to the number of units in this model. A FCNN is used for converting D_O outputs into a single forecast (RV^{t+1}) .¹² Two regularisation techniques, L^2 regularisation and dropout, as described in Subsection 4.1, are implemented to

¹⁰As a robustness check, Subsection 7.5 examines the performance of a FCNN model in forecasting RV, serving as a replacement for the LSTM model.

¹¹Although the theoretical benefit of a deeper sequential architecture is not clear, deeper models could lead to better performance in some specific tasks (Goldberg, 2016). With our objectives in mind and all the constraints at hand, a multi-layer (stacked) model is left for future research.

¹²As highlighted by Andersen et al. (2007), considering logarithmic RV could serve as a viable approach for controlling RV’s extreme distributions. However, as emphasised in Patton and Sheppard (2015), this alternative approach centres on predicting log-RV rather than volatility in levels, despite the latter typically being more relevant for economic applications. In alignment with the approach taken by Bollerslev et al. (2016) and consistent with the prevailing literature on the HAR-family of models, this study uses the raw RV formulation instead of log-RV.

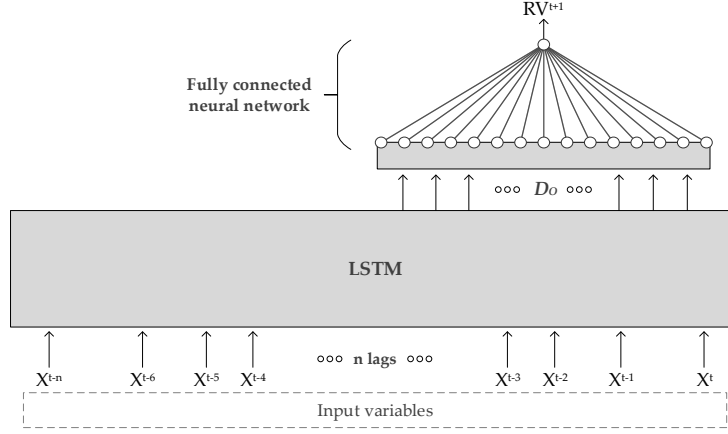


Figure 2: ML Model Structure

Notes: For every input variable (X), n lags of that variable ($X^t, X^{t-1}, \dots, X^{t-n}$) are included as input variables to the LSTM model. The number of outputs of the LSTM (D_O) is equal to the number of units in the LSTM model. An FCNN converts the D_O outputs of the LSTM to a single RV^{t+1} .

avoid overfitting.

The ML technical specifications implemented here are as follows: the optimisation algorithm is ADAM (Kingma and Ba, 2014). ADAM is an adaptive learning rate optimisation algorithm based on stochastic gradient descent. The initial learning rate is set to 0.001, and a separate learning-rate schedule with decay coefficient 10^{-5} is applied. For both the LSTM and FCNN, kernel and bias L^2 regularisation are applied with regularisation coefficient 10^{-4} . The second regularisation technique, dropout, is applied between the LSTM and FCNN with a rate of 0.5. For the LSTM, the sigmoid activation function is used for the input, forget, and output gates, while the hyperbolic tangent activation function is used for the candidate cell update and for the cell-state output when computing the hidden state. Also, for FCNN, the rectifier activation function¹³ is chosen.¹⁴ For the primary experiments in Section 5, MSE is used as the loss function, and the number of epochs is set to 50. The LSTM is tested with 5, 10, 15, 20, and 25 units. The FCNN consists of a single layer.

In the primary experiments presented in Section 5, 21 lags (corresponding to $n = 20$ in Figure 2) are employed for all input variables, aligning with the structure of the HAR model. However, the

¹³A unit with the rectifier activation function is called a rectified linear unit (ReLU).

¹⁴The choice of hyperparameters adopted here is consistent with the default values used in the literature and common practices. Certain hyperparameters, such as the learning-rate schedule coefficient and the L^2 regularisation coefficient, are chosen through trial and error. While the robustness checks in Section 7 provide valuable insights into the sensitivity of the choice of loss function, and the number of units and epochs, there are other hyperparameters that could impact model performance. Because of the extensive number of models trained in this study and the computational intensity involved, investigating the simultaneous impact exerted by all hyperparameters is beyond the scope of this research. Therefore, a potential avenue for future research would be to extend the robustness analysis to include a more comprehensive examination of model performance sensitivity to the choice of hyperparameters.

Table 5: Number of Independent Variables and Parameters of Models

Group	HAR-Family of Models							
Model	AR(1)	HAR	HAR-J	CHAR	SHAR	ARQ	HARQ	HARQ-F
Number of Variables	1	3	4	3	4	2	4	6
Number of Parameters	2	4	5	4	5	3	5	7

Group	ML Models			
Model	HAR-ML	News-ML	OB-ML	News/OB-ML
Number of Variables	$6 \times \text{NoL}^a$	$15 \times \text{NoL}$	$138 \times \text{NoL}$	$147 \times \text{NoL}$
Number of Parameters (5 Units) ^b	246 ^c	426	2,886	3,066
Number of Parameters (10 Units)	691	1,051	5,971	6,331
Number of Parameters (15 Units)	1,336	1,876	9,256	9,796
Number of Parameters (20 Units)	2,181	2,901	12,741	13,461
Number of Parameters (25 Units)	3,226	4,126	16,426	17,326

Notes: HAR-ML contains only the HAR-family variables described in Subsection 3.1. News-ML augments the HAR-family variables with all news variables described in Subsection 3.2. OB-ML augments the HAR-family variables with all LOB variables described in Subsection 3.3. Finally, News/OB-ML includes the HAR-family, news, and LOB variables simultaneously. Hence, all ML models are built on the HAR-family baseline, and the additional blocks capture the incremental contribution of news and LOB information. ^a Number of lags (i.e., 21 for the primary experiments). ^b Number of units in the LSTM model. ^c In a single-layer LSTM, let p represent the number of input features and h the number of hidden units. The total number of trainable parameters is given by $4h(p + h + 1) + (h + 1)$, because each of the four gates (input, forget, cell, and output) contributes $h(p + h + 1)$ parameters: comprising an input-weight matrix of size $p \times h$, a recurrent-weight matrix of size $h \times h$, and a bias vector of length h . Additionally, the final dense (output) layer adds h weights and one bias. Consequently, for the HAR-ML case where $p = 6$ inputs and $h = 5$ units, the total number of trainable parameters is $4 \times 5(6 + 5 + 1) + 5 + 1 = 246$.

number of variables included in the ML model is far greater than that included in HAR, even with the restriction to 21 lags. All predictive variables are standardised by removing the mean and scaling to unit variance. The standardisation is executed every day when the window is rolled forward, using only the data in the relevant window.¹⁵

Table 5 shows the number of variables and parameters for the HAR-family of models (top panel) and the ML model variants (bottom panel). The ML models clearly use more input variables than the HAR models. Combining all datasets, the News/OB-ML model uses 147 features per time step and a sequence length of 21 lags, corresponding to 147×21 input elements per observation, whereas the AR(1) model relies on a single input variable. The ability to deal with nonlinear relationships in a high-dimensional environment is a vital feature of ML models. Note also that adding more lags to the ML models does not change the number of parameters in the LSTM model; only the number of variables and the number of units determines the number of parameters. For each of the 23 tickers, only the data for that ticker are used for training. The same seed in the random number generator (RNG) is also used for all models to ensure reproducible results.

¹⁵In accordance with the HAR model structure, the input sequence length for the primary experiment is set to 21. For the robustness checks in Subsection 7.1, we experimented with sequence lengths set to 5 and 1.

5 Out-of-Sample Forecasting Results

The entire sample period of 27 July 2007 to 27 January 2022 was separated into an in-sample training period from 27 July 2007 to 11 September 2015 (2,046 days) and an out-of-sample forecasting period from 14 September 2015 to 27 January 2022 (1,604 days).¹⁶ Following Poon and Granger (2003), the model’s forecasting power is judged only according to the out-of-sample forecasting performance. To maintain consistency with the estimation and implementation procedures of the HAR-family of models, all ML models in this study are retrained daily using a rolling window approach. To better understand the forecasting performance on normal versus high volatility days, we classify high volatility days whenever the RV is greater than $Q3 + 1.5 \times IQR$, where $Q1$ and $Q3$ are the first and third quartiles, respectively, and $IQR = Q3 - Q1$. Normal volatility days are defined as all trading days excluding those classified as high volatility. This classification is applied separately for each ticker, using only data from the out-of-sample period. Based on this definition and averaging across all tickers, about 10% (160 days) of the out-of-sample period (1,604 days) are classified as high volatility days. We have purposefully avoided parametric jump estimation as it is sensitive to the assumption of the stock price dynamics and the bandwidth adopted for data frequency.¹⁷ We consistently noticed that the forecasting of normal and high volatility requires different ML specifications and different information sets. Thus, as long as the out-of-sample forecasts are separated into a main set of normal observations and a smaller set of extreme observations (i.e., high volatility days), regardless of the method used to separate them, our findings and conclusions should continue to hold.

To better understand the characteristics of high volatility days, Figure 3 illustrates the monthly distribution of high volatility days over the out-of-sample period. High volatility days are identified at the individual-ticker level based on the RV distribution and aggregated to monthly counts for each asset. The horizontal axis denotes calendar months, while the vertical axis reports the number of high volatility days per month for a given ticker. Black markers correspond to individual assets, the solid line reports the cross-sectional monthly average, and the shaded band represents one standard deviation around this average. Consistent with the empirical classifica-

¹⁶To ensure comparability, both the in-sample estimation and the out-of-sample forecasting should reflect similar data characteristics. Hence, the selection of in-sample and out-of-sample periods is also driven by the desire to include days with extreme volatility in both periods. In particular, the in-sample period encompasses the 2008 financial crisis, while the out-of-sample period includes the COVID-19 disruptions.

¹⁷The distinction between normal and high RVs is not known *a priori*; it is solely made *ex post* for the purpose of comparing out-of-sample performance.

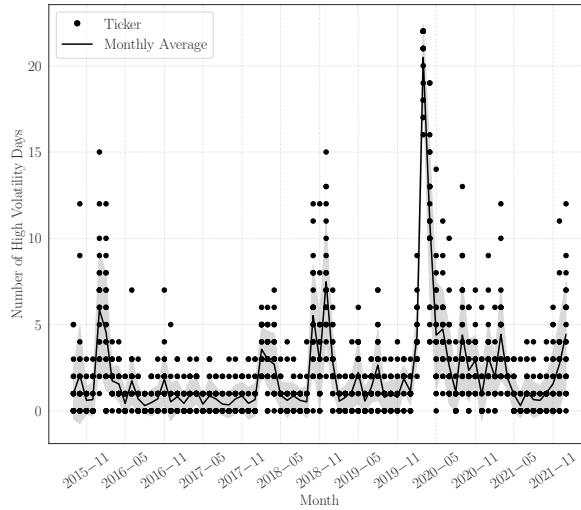


Figure 3: Number of High Volatility Days per Month over the Out-of-Sample Period

Notes: The figure shows the monthly frequency of high volatility days for individual tickers over the out-of-sample period. The horizontal axis corresponds to calendar months, and the vertical axis reports the number of days within each month classified as high volatility for a given asset. Black-filled markers represent individual tickers, illustrating cross-sectional dispersion in volatility dynamics. The solid line denotes the cross-sectional average number of high volatility days per month, while the shaded band represents one standard deviation across tickers.

tion, high volatility days account for a relatively small fraction of the out-of-sample observations. Nevertheless, the figure shows that these days are observed throughout the sample and across assets, with pronounced variation during time and substantial cross-sectional heterogeneity. While their frequency increases during periods of heightened market-wide RV like COVID-19 around 2020, high volatility days are not confined to a small number of isolated episodes. This pattern suggests that elevated RV reflects both market-wide risk factors, such as shifts in aggregate uncertainty or risk premia, while the majority of observations are attributable to asset-specific episodes of heightened volatility driven by idiosyncratic events, for example firm-level information arrivals, liquidity shocks, and flash crashes.

We use the reality check (RC) to test the forecasting performance of the ML model against every model in the HAR-family. In line with White (2000) and Bollerslev et al. (2016), the stationary bootstrap of Politis and Romano (1994) with 999 re-samplings and an average block length of 5 is used for this test.¹⁸ The hypotheses of the test are defined as follows:

$$\begin{aligned}
 H_0 : \min_{k=1, \dots, n} \mathbb{E}[L^k(RV, X) - L^0(RV, X)] &\leq 0, \\
 H_1 : \min_{k=1, \dots, n} \mathbb{E}[L^k(RV, X) - L^0(RV, X)] &> 0,
 \end{aligned} \tag{18}$$

where L^k is the loss measure of the k^{th} out of n benchmark HAR models, which include AR(1),

¹⁸Our analysis shows that the results are not sensitive to the choice of block length.

HAR, HAR-J, CHAR, SHAR, ARQ, HARQ, and HARQ-F, and L^0 is the loss measure of the target ML model. Rejection of the null hypothesis indicates that $L^k > L^0$ for all k , meaning the ML model outperforms the HAR-family of models jointly.

Following Patton (2011), the MSE and QLIKE loss functions are chosen for measuring the RV forecasting performance, in addition to the mean directional accuracy (MDA) metric. According to Patton (2011), the MSE and QLIKE loss functions are among the family of robust and homogeneous loss functions for volatility forecasting comparison.¹⁹ Rankings of volatility forecasts based on these loss functions are robust to noise in the proxy, and the rankings are invariant to the choice of units of measurement. On the other hand, MDA is a useful measure complementing these two cardinal measures, especially for monitoring the overfitting of the models. The three loss functions are defined as follows:

$$MSE(RV_t, \widehat{RV}_t) \equiv \frac{1}{N} \sum_{t=1}^N (RV_t - \widehat{RV}_t)^2, \quad (19)$$

$$QLIKE(RV_t, \widehat{RV}_t) \equiv \frac{1}{N} \sum_{t=1}^N \left(\frac{RV_t}{\widehat{RV}_t} - \log\left(\frac{RV_t}{\widehat{RV}_t}\right) - 1 \right), \quad (20)$$

$$MDA(RV_t, RV_{t-1}, \widehat{RV}_t) \equiv \frac{1}{N} \sum_{t=1}^N \mathbf{1}\{\text{sign}(RV_t - RV_{t-1}) = \text{sign}(\widehat{RV}_t - RV_{t-1})\}, \quad (21)$$

where RV_t is the true RV at time t , $\widehat{RV}_t$ is the forecast RV at time t , N is the number of days in the out-of-sample period, and $\text{sign}(\cdot)$ and $\mathbf{1}\{\cdot\}$ denote the sign and indicator functions, respectively. Following Bollerslev et al. (2016), if the forecast RV is larger (smaller) than the maximum (minimum) of the RV in the estimation series, it is replaced by the average of the RV in the estimation series.

To select the best-performing model among the HAR-family of models, we assess both in-sample and out-of-sample forecasting performance across all HAR-type specifications. Table C2 reports parameter estimates and in-sample evaluation metrics aggregated across 23 stocks, including adjusted R^2 , MSE, QLIKE, and MDA. The results indicate that the CHAR model provides the strongest and most stable in-sample performance, achieving the highest average and median adjusted R^2 , as well as the lowest average and median MSE and the lowest average QLIKE among competing HAR-family of models. While differences in MDA across models are generally

¹⁹See Poon and Granger (2003) for a review of the volatility performance metrics.

modest, CHAR remains competitive in terms of directional accuracy. The estimated coefficients across models are broadly consistent with theoretical expectations and prior empirical findings. Table C3 reports forecasting performance over the full out-of-sample period, as well as separately for normal and high volatility days. Consistent with the in-sample evidence, CHAR delivers the lowest average and median MSE and QLIKE over the full out-of-sample period and dominates alternative HAR specifications according to the RC across both volatility regimes, while also exhibiting competitive MDA across subsamples. Although some models, such as HAR-J or HARQ-F, perform well under specific metrics or market conditions, their performance gains are not systematic. This evidence identifies CHAR as the best-performing model within the HAR-family, and we therefore adopt it as the benchmark. We also observe a clear pattern in Table C3: the out-of-sample performance of all HAR-family of models deteriorate substantially during high volatility days. This is evidenced by elevated MSE and QLIKE values, indicating that these models systematically fail to forecast high volatility days prior to their realisation.

We compare the out-of-sample forecasting performance of four groups of ML models, differing by their information sets, against the benchmark CHAR model as the best-performing HAR model. If we define, for ticker i , and for each ML model j ,

$$\rho_{MSE,i,j} = \frac{MSE_i^{ML} (Model_j)}{MSE_i^{OLS} (CHAR)}, \quad (22)$$

$$\rho_{QLIKE,i,j} = \frac{QLIKE_i^{ML} (Model_j)}{QLIKE_i^{OLS} (CHAR)}, \quad (23)$$

$$\rho_{MDA,i,j} = \frac{MDA_i^{OLS} (CHAR)}{MDA_i^{ML} (Model_j)}, \quad (24)$$

where $j = 1, 2, 3, 4$ is one of the four groups of ML models in Table 5, and $i = 1, \dots, 23$ identifies each of the 23 tickers. For the ML models, $j = 1$ corresponds to HAR-ML, and the HAR variables are described in Subsection 3.1; $j = 2$ is News-ML, and the news variables are described in Subsection 3.2; $j = 3$ is OB-ML, and the LOB variables are described in Subsection 3.3; and finally, $j = 4$ is News/OB-ML, i.e., the news and LOB variables combined. It is worth noting that News-ML, OB-ML, and News/OB-ML also contain the HAR variables. In this context, a value of one indicates no difference in performance relative to the benchmark. A value below one reflects better performance compared to the benchmark, whereas a value above one represents worse performance relative to the benchmark.²⁰

²⁰In Equation (24), the numerator and denominator for the MDA metric are inverted to ensure that lower

Next, the ML model j 's average $\rho_{\text{MSE/QLIKE/MDA},j}$ for the 23 tickers are:

$$\text{Average } \rho_{\text{MSE},j} = \frac{1}{23} \sum_{i=1}^{23} \rho_{\text{MSE},i,j}, \quad (25)$$

$$\text{Average } \rho_{\text{QLIKE},j} = \frac{1}{23} \sum_{i=1}^{23} \rho_{\text{QLIKE},i,j}, \quad (26)$$

$$\text{Average } \rho_{\text{MDA},j} = \frac{1}{23} \sum_{i=1}^{23} \rho_{\text{MDA},i,j}, \quad (27)$$

while Median $\rho_{\text{MSE},j}$, Median $\rho_{\text{QLIKE},j}$, and Median $\rho_{\text{MDA},j}$ are the median equivalents of Equation (25), Equation (26), and Equation (27), respectively.

5.1 Full Out-of-Sample Period

Table 6 reports the average and median $\rho_{\text{MSE/QLIKE/MDA}}$ for the four groups of ML models, with 21 lags of every group of variables included and different numbers of units (5, 10, 15, 20, and 25). The corresponding RC value is the percentage of tickers with better ML performance, in terms of MSE, QLIKE, or MDA, at the 5% and 10% significance levels against every model in the HAR-family of models (AR(1), HAR, HAR-J, CHAR, SHAR, ARQ, HARQ, and HARQ-F). For cardinal forecasts evaluated using MSE and QLIKE, and considering the RC results, improvement in forecasting performance is clear, especially for the HAR-ML and OB-ML groups. However, the results in Table 6 are mixed. In all cases and for all groups of ML models, the average and median values show degradation in performance compared to the CHAR model as a benchmark. For directional forecasts evaluated using MDA, ML outperformed CHAR in most specifications, delivering the largest improvement: about 4% overall and 100% for RC, primarily within the OB-ML and News/OB-ML groups. Taken together, except for the MDA loss function, consistent improvements are noticeable only from the RC results.²¹ However, by separating

values correspond to better forecasting performance.

²¹Table 6 indicates that, over the full out-of-sample period, ML underperformed in terms of average and median MSE and QLIKE relative to the strongest benchmark, but outperformed in terms of RC when evaluated against the full set of benchmark models. The ρ -metrics quantify the magnitude of the forecast-error gap between each ML specification and the single strongest benchmark (CHAR). In contrast, the RC statistic assesses, for each ticker, whether the ML model significantly outperforms all members of the entire HAR-family of models (AR, HAR, HAR-J, CHAR, SHAR, ARQ, HARQ, HARQ-F), and reports the percentage of tickers for which the null hypothesis is rejected. Hence, an ML specification may yield ρ -values slightly above one, indicating a marginal loss to CHAR, yet still record a high RC if it dominates several other HAR variants across most tickers. The two statistics, therefore, capture distinct but complementary dimensions of forecast performance. When both metrics align, the ranking is unambiguous; when they diverge, ρ quantifies the shortfall against the top-performing benchmark, while RC reflects the extent of statistically significant gains across the broader benchmark set. Following standard practice in forecast evaluation, we retain both metrics, as this dual perspective mitigates model-selection bias and

Table 6: Out-of-Sample Volatility Forecasting Performance
14 September 2015 to 27 January 2022 (1,604 days)

		HAR-ML					News-ML					OB-ML					News/OB-ML				
Units		5	10	15	20	25	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25
Full Out-of-Sample Period																					
MSE ^a	Avg	1.225	1.199	1.182	1.161	1.146	1.229	1.204	1.182	1.166	1.150	1.209	1.179	1.156	1.144	1.174	1.216	1.178	1.157	1.151	1.138
	Med	1.198	1.180	1.160	1.139	1.119	1.198	1.179	1.160	1.139	1.123	1.178	1.174	1.136	1.125	1.151	1.186	1.154	1.145	1.147	1.123
RC ^d	5%	13.04	34.78	39.13	56.52	73.91	17.39	26.09	39.13	52.17	73.91	26.09	39.13	56.52	73.91	78.26	17.39	39.13	47.83	69.57	78.26
	10%	47.83	60.87	82.61	86.96	100	52.17	60.87	82.61	91.30	95.65	60.87	73.91	91.30	100	100	65.22	69.57	86.96	91.30	95.65
QLIKE ^b	Avg	8.081	3.881	2.517	1.742	1.391	7.787	4.258	2.600	2.004	1.526	4.816	2.335	1.656	1.525	2.082	6.174	2.210	1.678	1.636	1.455
	Med	6.879	3.321	2.185	1.569	1.305	7.596	4.154	2.474	1.743	1.352	3.178	1.996	1.504	1.442	1.922	5.142	1.878	1.528	1.473	1.386
RC	5%	0.00	0.00	0.00	0.00	13.04	0.00	0.00	0.00	4.35	8.70	0.00	4.35	0.00	8.70	13.04	0.00	0.00	0.00	0.00	8.70
	10%	0.00	4.35	17.39	34.78	60.87	0.00	0.00	4.35	8.70	43.48	4.35	8.70	13.04	17.39	21.74	0.00	8.70	13.04	21.74	17.39
MDA ^c	Avg	1.015	0.993	0.973	0.969	0.969	1.016	0.999	0.974	0.964	0.961	0.967	0.946	0.938	0.942	0.963	0.992	0.943	0.938	0.944	0.952
	Med	1.001	0.988	0.969	0.968	0.972	0.999	0.990	0.970	0.964	0.963	0.952	0.945	0.941	0.947	0.964	0.969	0.937	0.928	0.945	0.953
RC	5%	65.22	91.30	95.65	100	100	65.22	82.61	95.65	95.65	100	78.26	91.30	100	100	100	69.57	91.30	100	100	100
	10%	69.57	95.65	95.65	100	100	65.22	86.96	95.65	100	100	86.96	95.65	100	100	100	73.91	91.30	100	100	100

Notes: The best value in each row is marked in bold. ^a The mean (or median) MSE ratio of ML to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^b The mean (or median) QLIKE ratio of ML to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^c The mean (or median) MDA ratio of CHAR to ML for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^d Percentage of tickers with the outstanding performance of ML against the HAR-family of models at the 5% and 10% significance levels.

the out-of-sample actual RVs into normal and high volatility days, a clearer picture emerged in Subsection 5.2 below.²²

5.2 Normal vs High Volatility Days

Table 7 reports the results for normal and high volatility days separately. First, consider the top panel of Table 7, which corresponds to normal volatility days and, on average, accounts for 90% of the out-of-sample forecast period. The four groups of ML models outperform the CHAR model, as the average and median MSE and QLIKE values are mostly below one. When using the MSE measure, RC values are 100% in all but one case. When using QLIKE, RC is near 100% in many cases. This provides strong evidence supporting the superior forecasting power of ML models compared to all HAR-family of models on normal volatility days. The substantial improvement is also evident in the MDA loss function, with directional forecasting performance increasing by around 11% in the best-performing model. Among the four ML model groups, OB-ML and News/OB-ML perform the best. In general, a higher number of units corresponds to better forecasting performance on normal volatility days.²³

The results for high volatility days in the bottom panel of Table 7 present a completely different picture. All four groups of ML models underperformed the CHAR model, as the average/median MSE/QLIKE values are all above one. When the MSE measure is used, the RC values are low, most of which are below 50%, and even worse when QLIKE is used. Likewise, the MDA loss function results show deterioration in forecasting performance. These sharply contrasting results

strengthens the robustness of our inference.

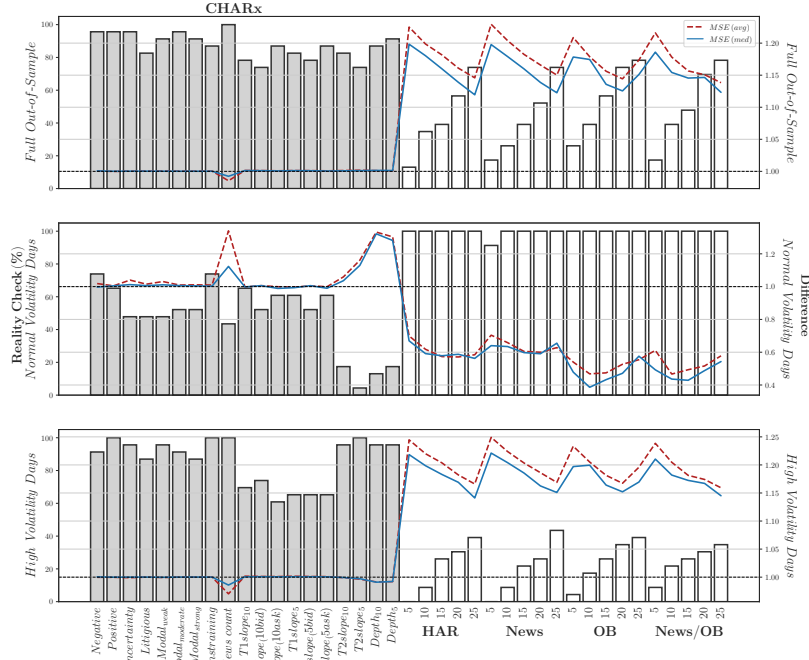
²²To better understand the impact of the rule governing the replacement of RV forecasts that fall outside the estimation-sample range on the results in Section 5, Table C4 reports summary statistics for the HAR-family of models across 23 tickers. The table shows that such forecast replacements occur very infrequently for all HAR specifications, with zero or near-zero counts for most models. Table C5 reports the corresponding statistics for the ML models across alternative model classes and sizes. While forecast replacements occur more frequently for ML models, particularly for smaller models and richer information sets, their frequency declines markedly as model complexity increases. Overall, these results indicate that extreme forecast replacements are more common for ML models than for HAR-family of models, but remain limited relative to the total number of out-of-sample observations (1,604 days) and vary systematically with model specification rather than occurring uniformly across ML architectures.

²³The finding that adding LOB variables to the ML models substantially improves forecasting performance, particularly on normal volatility days, contradicts Rahimikia and Poon (2020), which found that news variables had stronger predictive power than LOB variables. The contradictory finding could be due to the fact that the ML model here has a large number of predictors, whereas in Rahimikia and Poon (2020), 10 LOB variables and 9 news sentiment measures were added, only one at a time, to the CHAR model. In contrast, News/OB-ML, the largest ML model tested here, has 147 predictors, i.e., 132 LOB variables, 9 news variables, and 6 HAR variables, all included and tested in a single ML model. Hence, apart from the ability to capture nonlinear relationships as highlighted in Christensen et al. (2023), the ability of the ML model to fit a large number of predictors jointly significantly amplifies its forecasting accuracy, particularly on normal volatility days.

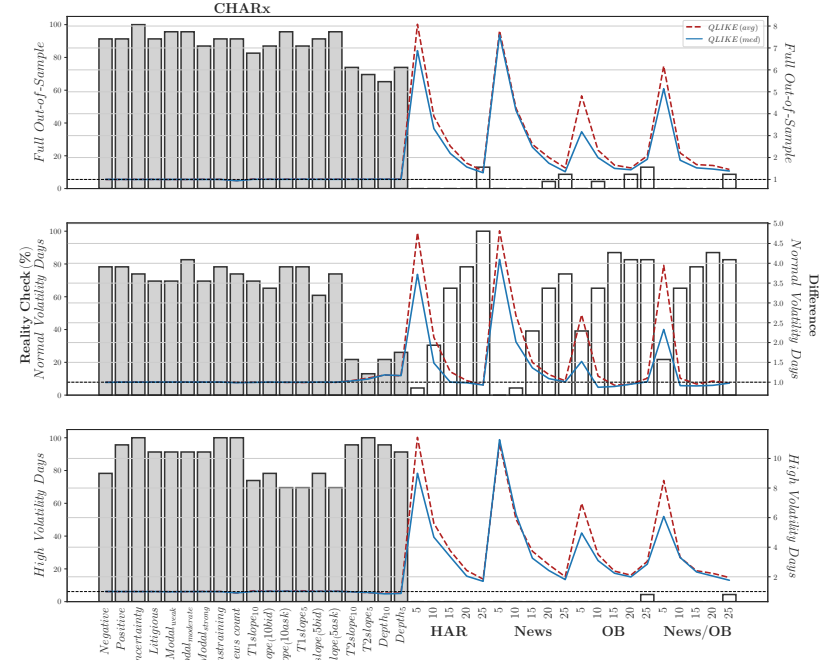
Table 7: Out-of-Sample Volatility Forecasting Performance: Normal vs High Volatility Days
14 September 2015 to 27 January 2022 (1,604 days)

		Units	HAR-ML					News-ML					OB-ML					News/OB-ML				
			5	10	15	20	25	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25
Normal Volatility Days																						
MSE ^a	RC ^d	Avg	0.699	0.618	0.573	0.571	0.584	0.704	0.658	0.605	0.600	0.628	0.540	0.468	0.474	0.527	0.554	0.611	0.467	0.493	0.516	0.578
		Med	0.669	0.591	0.578	0.587	0.562	0.640	0.635	0.597	0.591	0.655	0.478	0.386	0.433	0.471	0.576	0.492	0.436	0.429	0.488	0.543
		5%	100	100	100	100	100	91.30	100	100	100	100	100	100	100	100	100	100	100	100	100	100
	10%	100	100	100	100	100	91.30	100	100	100	100	100	100	100	100	100	100	100	100	100	100	
QLIKE ^b	RC	Avg	4.770	2.132	1.263	1.041	0.926	4.819	2.682	1.499	1.192	1.036	2.701	1.153	0.924	0.951	1.099	3.948	1.100	0.953	1.029	0.976
		Med	3.721	1.484	1.001	0.985	0.928	4.098	2.018	1.362	1.090	1.010	1.524	0.871	0.892	0.950	1.010	2.331	0.913	0.906	0.921	0.977
		5%	4.35	30.44	65.22	78.26	100	0.00	4.35	39.13	65.22	73.91	39.13	65.22	86.96	82.61	82.61	21.74	65.22	78.26	86.96	82.61
	10%	4.35	34.78	69.57	82.61	100	0.00	4.35	39.13	78.26	86.96	39.13	65.22	86.96	82.61	82.61	21.74	73.91	82.61	91.30	86.96	
MDA ^c	RC	Avg	0.954	0.941	0.925	0.928	0.936	0.956	0.944	0.925	0.920	0.923	0.904	0.888	0.890	0.900	0.918	0.928	0.885	0.890	0.898	0.916
		Med	0.951	0.943	0.927	0.933	0.936	0.938	0.935	0.925	0.917	0.923	0.899	0.892	0.892	0.896	0.921	0.905	0.882	0.887	0.907	0.915
		5%	86.96	95.65	100	100	100	82.61	91.30	100	100	100	100	100	100	100	100	100	91.30	100	100	100
	10%	86.96	95.65	100	100	100	86.96	91.30	100	100	100	100	100	100	100	100	95.65	100	100	100	100	
High Volatility Days																						
MSE	RC	Avg	1.245	1.220	1.204	1.182	1.166	1.249	1.224	1.203	1.186	1.169	1.233	1.205	1.181	1.167	1.196	1.238	1.205	1.181	1.174	1.159
		Med	1.218	1.199	1.183	1.169	1.141	1.221	1.204	1.186	1.163	1.151	1.197	1.200	1.164	1.152	1.170	1.210	1.182	1.172	1.167	1.145
		5%	0.00	8.70	26.09	30.44	39.13	0.00	8.70	21.74	26.09	43.48	4.35	17.39	26.09	34.78	39.13	8.70	21.74	26.09	30.44	34.78
	10%	30.44	39.13	47.83	56.52	69.57	26.09	39.13	39.13	47.83	73.91	30.44	43.48	52.17	56.52	69.57	26.09	43.48	52.17	60.87	65.22	
QLIKE	RC	Avg	11.425	5.613	3.759	2.428	1.844	11.027	5.907	3.728	2.819	2.007	6.963	3.512	2.383	2.108	3.040	8.513	3.307	2.406	2.228	1.948
		Med	8.998	4.679	3.348	2.044	1.695	11.270	6.232	3.261	2.426	1.810	4.962	3.096	2.241	1.992	2.857	6.077	3.304	2.326	2.045	1.765
		5%	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.35	0.00	0.00	0.00	0.00	4.35
	10%	0.00	4.35	4.35	0.00	13.04	0.00	0.00	0.00	4.35	17.39	4.35	4.35	0.00	4.35	4.35	4.35	0.00	0.00	4.35	4.35	
MDA	RC	Avg	1.070	1.066	1.063	1.059	1.062	1.070	1.064	1.062	1.059	1.057	1.069	1.065	1.063	1.059	1.059	1.069	1.064	1.060	1.056	1.050
		Med	1.058	1.058	1.058	1.049	1.050	1.058	1.058	1.058	1.054	1.054	1.058	1.058	1.054	1.054	1.050	1.058	1.058	1.054	1.056	1.050
		5%	0.00	0.00	0.00	0.00	4.35	8.70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.35
	10%	0.00	0.00	0.00	4.35	13.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.35	

Notes: The best value in each row is marked in bold. ^a The mean (or median) MSE ratio of ML to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^b The mean (or median) QLIKE ratio of ML to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^c The mean (or median) MDA ratio of CHAR to ML for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^d Percentage of tickers with the outstanding performance of ML against the HAR-family of models at the 5% and 10% significance levels.



(a) Forecast Evaluation under MSE



(b) Forecast Evaluation under QLIKE

Notes: The bar chart shows the percentage of tickers with outstanding performance considering the MSE loss function in Figure 4a and the QLIKE loss function in Figure 4b at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for each specified CHARx model (grey bars) and the ML model (white bars). The values for the bar chart can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MSEs and QLIKES of the CHAR relative to the CHARx models (left group) and ML models (right group) across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

Figure 4: Comparison of ML and CHARx Models

to normal volatility days suggest that the ML models fitted to the full in-sample period, dominated by normal volatility days, are not appropriate for forecasting the infrequent high volatility days. One possible explanation for these contrasting results could be due to RV’s persistence differing on high versus normal volatility days. It is widely documented that volatility is highly persistent, but not when volatility is extreme. Volatility half-life is much shorter for extreme volatility compared to the normal level of volatility.²⁴ This is the first and most important clue that separate modelling considerations are needed for normal versus high volatility regimes. Here, as we did not separate the normal and high volatility days, the optimisation was driven by the 90% of normal volatility days’ forecast evaluations. Nevertheless, although the high volatility days constitute only 10% of the out-of-sample period, the poor performance of the ML models during high volatility days strongly indicates that these models, particularly in their basic form without additional adjustments to the RV definition and/or the model itself, might be economically less useful. This is because precise forecasts of high volatility days hold greater importance in a risk management scenario compared to days with low volatility. Our findings show that ML models trained using the full in-sample period did not perform well on the 10% of (non-consecutive day) daily volatility forecasts when the actual (non-consecutive day) daily RVs are very high. This is different from the finding in Bucci (2020), which studied monthly S&P 500 volatility and found good forecasting performance from ML models during the Great Recession high volatility period from September 2007 to June 2009, which may include many days when the index volatility was at a ‘normal’ level. Our sample of individual tickers will have more idiosyncratic volatility that does not persist and is more challenging to forecast.²⁵

We further extend our analysis by comparing ML models with the CHARx model proposed by Rahimikia and Poon (2020), which augments the original CHAR model in Subsection 2.1 by incorporating variables derived from news and LOB data from the previous day to forecast daily RV. In Figure 4, the white (grey) bar reports the RC values for a specific ML model (CHARx)

²⁴In the individual tickers analyses and the robustness checks later, we also find very different ML modelling considerations for normal versus high volatility days.

²⁵As discussed in Section 5, the results are based on a rolling window approach with a fixed in-sample training period of 2,046 days. To further assess the robustness of our findings, we re-estimated all models using an increasing window approach; the corresponding results are presented in Table C7. In a second experiment, we shortened the in-sample training period from 2,046 days to 1,604 days, holding all other parameters constant; the results of this robustness check are reported in Table C8. In both cases, the CHAR benchmark model was re-estimated to match the respective training setup, using an increasing window in the first case and the shortened in-sample period in the second. Both sets of results are consistent with our main conclusions. Forecasting performance improves on normal volatility days, but deteriorates on high volatility days, which in turn negatively impacts the full out-of-sample performance. Consistent with earlier findings, the OB-ML and News/OB-ML models continue to exhibit the strongest performance among the four ML model groups.

against all HAR-family of models. The dashed (solid) line represents the out-of-sample average (median) ρ_{MSE} in Figure 4a, and the average (median) ρ_{QLIKE} in Figure 4b. The news variables in the CHARx model are defined in Subsection 3.2. The LOB variables include the ‘type 1 modified slope’ (Næs and Skjeltorp, 2006), the ‘type 2 slope’ (Kalay et al., 2004), and the ‘LOB depth’. A variable name, ‘T1slope(5bid)’, means ‘type 1 slope’ aggregated from the first five levels on the bid side of the LOB. If a LOB variable name does not contain ‘ask’ or ‘bid’, it means that it is calculated using both the bid and ask sides of the LOB. The results are reported for the full out-of-sample period (top panel), for days with normal volatility (middle panel), and for days with high volatility (bottom panel).

The most striking result in Figure 4 is the superior forecasting performance of the ML models compared to CHARx for normal volatility days. In contrast, all four groups of ML models performed poorly on high volatility days. News count and LOB depth are the only variables that, when added to the CHAR model, help to forecast RV on high volatility days. In the full out-of-sample period, the CHARx models demonstrate superior performance, particularly in terms of RC metrics, with notable improvements observed when the news count variable is incorporated into the CHAR model. The results for MDA presented in Figure C1 broadly support these findings, viz., a significant improvement in RV forecasting performance on normal volatility days when switching from CHARx to ML, but significant degradation in performance on high volatility days. For the full out-of-sample evaluation, interestingly, the ML models substantially improve performance in terms of both magnitude and RC metrics compared to the CHARx model. These findings from the MDA provide evidence that the ML models are effectively capturing the dynamics of RV, suggesting that the risk of overfitting is relatively low.

Combining all results, the findings suggest that no single model consistently outperforms others in forecasting RV. While ML models demonstrate strong performance on days characterised by normal volatility, their effectiveness diminishes during periods of high volatility, which becomes evident over the full out-of-sample evaluation period. Moreover, the choice of performance metric significantly influences model assessment. In particular, when using MDA, ML models emerge as clear outperformers not only during normal volatility days but also across the entire out-of-sample period. However, this advantage comes at the expense of reduced forecasting accuracy during high volatility days. These results indicate that, in their current architectures, ML models may not serve as direct substitutes for HAR-family of models. Instead, they can be viewed as

complementary models that enhance forecasting performance when used alongside HAR-family of models. Moreover, following the characteristics of high volatility days documented in Section 5, we find that excluding these rare high volatility episodes that are predominantly driven by asset-specific shocks, while the remaining episodes are largely attributable to systematic factors, leads to a significant improvement in RV forecasting performance when employing ML models. Although numerous ML models have already been tested, further analysis is necessary to ensure the generalisability of the results. This includes exploring variations in input variables, model architectures, training parameters, and model types, among others, to confirm that the findings are not specific to this particular evaluation framework.²⁶

5.3 ML with Limited Feature Set

This subsection examines three distinct limited information sets. Beginning with the first two, the first set comprises the past 21 lags of RV, while the second includes the previous day’s RV, the average RV over the past week, and the average RV over the past month. The results are presented in Figure 5, where $\Phi_t = \{RV_t, \dots, RV_{t-20}\}$ for the LSTM model in Figure 5a, and $\Phi_t = \{RV_t, \overline{RV}_{t-1}^w, \overline{RV}_{t-1}^m\}$ for the simpler FCNN model in Figure 5b.²⁷ The FCNN with three input variables is equivalent to the HAR model structure but with a potentially nonlinear relationship. The MSE is presented on the left and the QLIKE loss on the right. The full out-of-sample period is shown at the top, followed by normal volatility days in the middle, and high volatility days at the bottom.

In both Figure 5a and Figure 5b under MSE for normal volatility days, the ML model, generally with a higher number of units, continues to outperform CHAR and the HAR-family of models. The average and median ρ_{MSE} are below one in all cases. The RC values are also close to 100%

²⁶Table C6 repeats the main out-of-sample comparison in Table 6 and Table 7, but replaces CHAR with the standard HAR model as the benchmark. Relative to the CHAR-benchmarked ratios, HAR-benchmarked ratios are uniformly closer to one, reflecting that HAR is a weaker baseline than CHAR and therefore easier to outperform. Importantly, this change in benchmark does not alter the substantive conclusions drawn from the CHAR-based analysis. First, the same regime dependence is preserved: ML models continue to deliver sizable improvements on normal volatility days, while underperforming during high volatility days, with the latter capable of dominating performance over the full out-of-sample period. Second, the relative ordering across model variants and sizes is largely preserved.

²⁷The FCNN has a simple structure, unlike the LSTM. It consists of only three layers, viz. input, hidden, and output layers. The size of the input layer is the same as the number of input variables. The hidden layer size varies from 5, 10, 15, 20, to 25. The size of the output layer is equal to the number of output(s), which is one in this study. The dropout rate between the hidden layer and the output layer is set equal to 0.5. The activation functions are sigmoid and rectifier, respectively, for the hidden and output layers. The other ML model specifications remained unchanged.

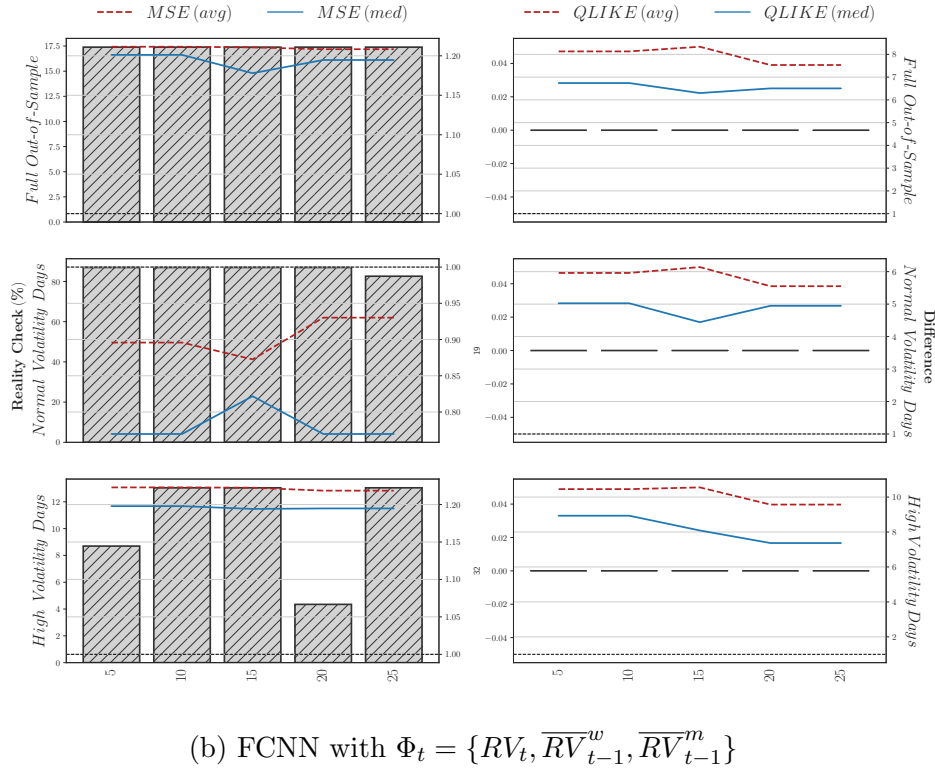
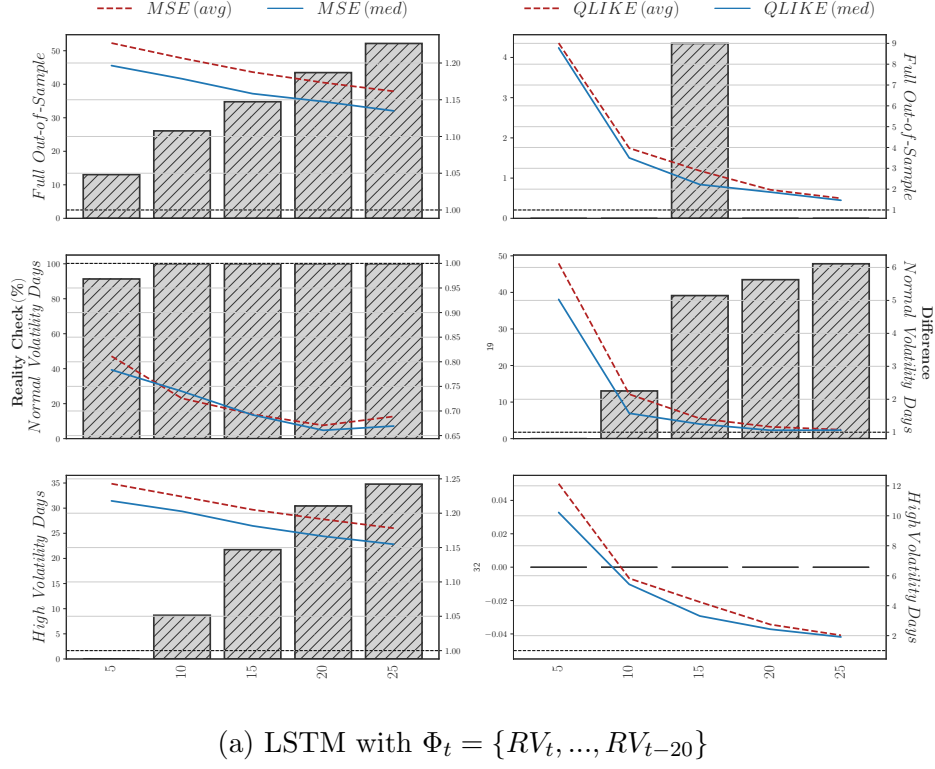


Figure 5: ML Models with Limited Information Sets

Notes: The bar chart shows the percentage of tickers with outstanding performance considering the MSE (left figures) and QLIKE (right figures) loss functions at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for each specified number of units of the ML model. The values for the bar chart can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MSEs and QLIKEs of the ML model relative to the CHAR model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

in many cases. As before, the ML models perform poorly on high volatility days. However, results under QLIKE are less marked compared to those under MSE, especially for FCNN (as compared to LSTM). Furthermore, a comparison between Figure 5 and Figure 4 clearly indicates that limiting the predictors to include only the history of RV and HAR variables significantly degrades forecasting performance for normal volatility days. This deterioration is evident not only in contrast to models with a greater number of predictors such as OB-ML and News/OB-ML, but also when compared to the smaller scale HAR-ML model. Hence, consistent with the outcomes presented in Subsection 5.2, aside from nonlinearity, a diverse range of data features plays a crucial role in the ML model’s RV forecasting performance. When extending the analysis to the full out-of-sample period, the results remain broadly consistent with previous findings. However, since substantially less data are used as input, the performance deterioration is more severe, both in terms of magnitude and the RC metric. The MDA results in Figure C2 for LSTM and in Figure C3 for FCNN produce findings consistent with the discussion here.²⁸ These findings again suggest that while ML models are effective under normal market conditions, their performance weakens during periods of high volatility, ultimately limiting their performance over the full out-of-sample period.

Turning to the third restricted information set, a key question is whether the superior performance of the OB-ML group, as documented in Subsection 5.1 and Subsection 5.2, can be attributed to the greater number of variables incorporated in the model. To examine this, we constrain the set of LOB variables employed in the OB-ML model to correspond with the CHARx specification, as detailed in Figure 4. Specifically, keeping all model specifications and parameters consistent with those outlined in Subsection 4.2, we include only a subset of LOB variables: the LOB depth on both bid and ask sides, type 1 slope from the bid side, ask side, and jointly, as well as type 2 slope. These variables are computed using both 5 and 10 levels of the LOB, and are also used in Figure 4 and Figure C1, where each variable is added individually to the base CHAR model in separate specifications. As mentioned, we use all input variables collectively in the LSTM model. Consequently, to construct the limited version of OB-ML, as before, we also

²⁸ML provides a substantial MDA improvement over the HAR-family of models on normal volatility days. In some cases, the improvement exceeds 6%, and the RC values approach 100%. Nevertheless, ML performance is poor on high volatility days, and this is more marked when using only three variables in the FCNN compared to the LSTM model with the past history of RV. Extending the analysis to the full out-of-sample period, the improvement is clearly noticeable in Figure C2, but not in Figure C3, which uses only three input variables. We also see that, compared to the MDA results for the primary experiment in Figure C1, this obtained improvement in performance is weaker. These findings again highlight the critical role of using a diverse range of data features in enabling ML models to achieve gains.

include the HAR-family variables, which decreases the total number of input variables from 138 in Table 5 to 16.

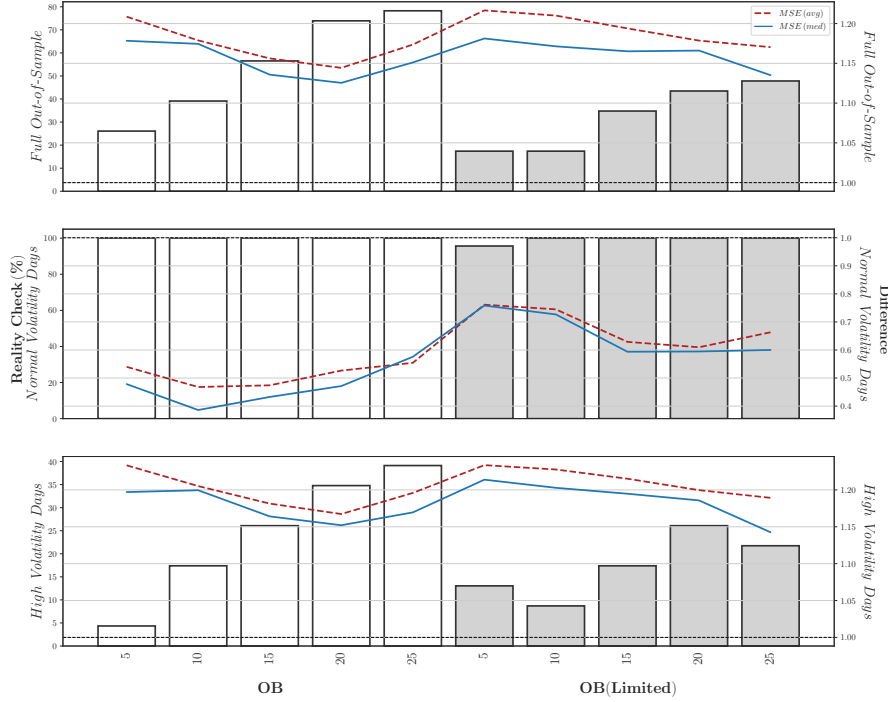
Figure 6 presents the results for the OB-ML, as defined in Subsection 3.3, alongside those for a restricted version of the OB-ML that incorporates a reduced set of variables, as defined above. The first five bars represent the results obtained using the LOB information set, while the second five bars are those derived from the limited LOB information set. It is evident that across the full out-of-sample period, as well as when normal and high volatility days are considered separately, the OB-ML group of models with access to the full information set consistently demonstrates superior forecasting performance. This finding is strongly supported by the RC results in nearly all cases. Although the magnitude of improvement, as reflected in both average and median values, exhibits some variation, the overall enhancement in forecasting performance remains evident when transitioning from the limited to full LOB information set. Furthermore, the MDA results presented in Figure C4 further support this conclusion. Overall, even when considering a limited number of LOB variables, an improvement in forecasting performance is observed; however, the gains are not as substantial as those achieved when a larger set of LOB variables is utilised.

5.4 Evaluation of Individual Tickers

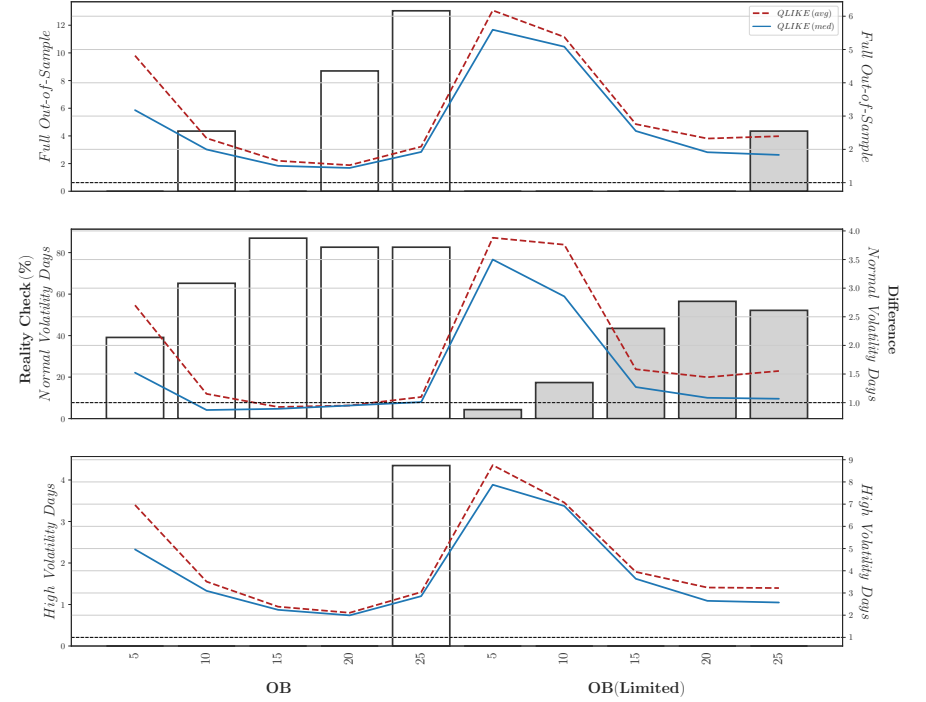
To understand the changes in forecasting performance at the ticker and model levels, this subsection analyses the $\rho_{MSE,i}$ and $\rho_{QLIKE,i}$ between the ML model (OB-ML with 15 units as one of the best-performing ML specifications in Table 7) and each of the HAR-family of models²⁹ using radar plots in Figure 7. The MSE results are shown on the left and the QLIKE results on the right. The top figure corresponds to full out-of-sample days, the middle figure to days with normal volatility, and the bottom figure to high volatility days. A ρ below one indicates that the ML model outperformed, while a ρ above one indicates that it underperformed. The bold inner circle represents $\rho = 1$, i.e., no difference in performance.

For normal volatility days in Figure 7b, it is apparent that OB-ML outperformed every model in the HAR-family of models, as ρ_{MSE} and ρ_{QLIKE} are mostly lower than one and lie inside the bold circle for most of the tickers. For high volatility days, presented in Figure 7c at the bottom, most ρ_{MSE} and ρ_{QLIKE} are above one and lie outside the bold circle, i.e., ML underperformed

²⁹That is, AR(1), HAR, HAR-J, CHAR, SHAR, ARQ, HARQ, and HARQ-F.



(a) Forecast Evaluation under MSE



(b) Forecast Evaluation under QLIKE

Notes: The bar chart is the percentage of tickers with the outstanding performance considering the MSE loss function in Figure 6a and the QLIKE loss function in Figure 6b at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for every specified model. The first five bars represent the results obtained using the LOB information set in Subsection 3.3, while the second five bars are those derived from the limited LOB information set. The dashed (solid) line shows the average (median) out-of-sample MSEs and QLIKES of the ML model relative to the CHAR model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

Figure 6: Comparison of OB-ML and OB-ML (Limited) Models

all HAR-family of models. The ρ_{MSE} and ρ_{QLIKE} are also large for six tickers, viz. ‘NVDA’, ‘AMAT’, ‘ADSK’, ‘MU’, ‘NTAP’ and ‘KLAC’. From the descriptive statistics of RV in Table 1, the RV of ‘MU’ has a very high standard deviation, while the RVs of ‘NVDA’, ‘AMAT’, ‘ADSK’, ‘NTAP’ and ‘KLAC’ exhibit very high skewness. When moving to the full out-of-sample period in Figure 7a, a clear degradation in performance is observed for most tickers, as well as across the various benchmark models. The MDA results presented in Figure C5 align with those from MSE and QLIKE above. However, in line with the previous findings, for the full out-of-sample period, the improvement is clear for many tickers.³⁰

These results again confirm the previous findings that the ML model dominates the HAR-family of models on normal volatility days but underperforms on high volatility days, especially for tickers exhibiting extreme volatility distributions. Moreover, in line with the results in Section 5, the red dotted line, representing CHAR, lies closest to the bold circle, suggesting that CHAR has the best forecasting performance among the HAR-family of models, albeit not as good as ML. Additionally, there are some substantial differences in the forecasting performance of HAR-family of models on both normal and high volatility days, with some exhibiting much worse performance than CHAR. Once again, the results reveal a consistent pattern: ML models perform well under certain volatility conditions, while the HAR-family of models performs better under others, indicating that each approach excels under distinct volatility dynamics. Also, the MDA again clearly shows an improvement in performance for the full out-of-sample period, which is not the case for the MSE and QLIKE loss functions.

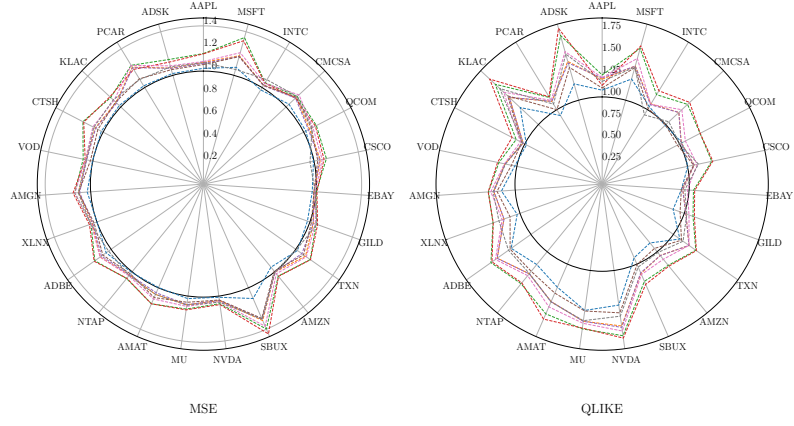
Figure 8 uses a box plot to present the distribution of true RV and the forecast RVs from the OB-ML model with 15 and 25 units, as well as from CHAR. For clarity, the RV values on the y-axis are truncated at 100%. The points above each box plot indicate high volatility days. Several important observations can be made here. CHAR produced some forecasts that are classified as high volatility days, whereas the ML models rarely generated forecasts that fall into this category. CHAR’s RV forecasts were more conservative compared to the actual RV, and the ML models produced even more conservative forecasts, capped at around 20%. Consequently, both performed

³⁰Consistent with the previous results, Figure C5 shows that for normal volatility days (middle figure), all the ρ_{MDA} values lie inside the bold circle, indicating that ML substantially outperformed every model in the HAR-family of models. In some cases, this improvement approaches 20%. In contrast, on high volatility days (right figure), ML underperformed the HAR-family of models for many tickers. Over the full out-of-sample period (left figure), all points fall within the circle, again demonstrating improved performance for the majority of tickers and benchmark models.

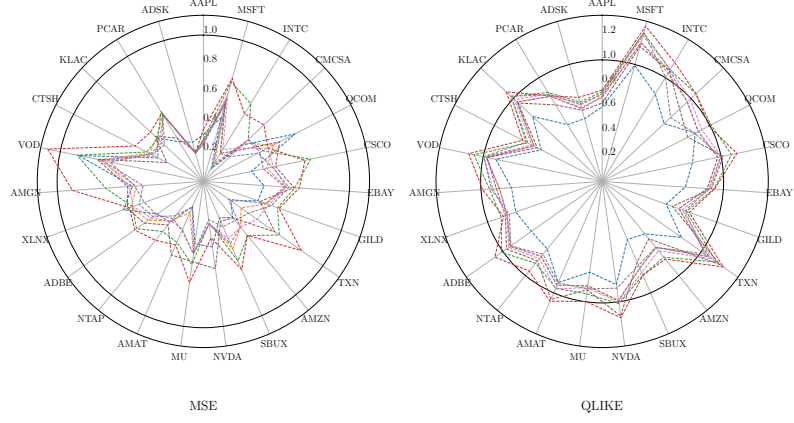
poorly on high volatility days. Closer inspection of other HAR-family of models revealed a similar pattern. In general, the HAR-family of models produced forecasts with higher means and standard deviations during the out-of-sample period than those of the ML models.³¹ This is one of the reasons for the poor forecasting performance of the ML models on high volatility days. Increasing the number of units in the ML model raises both the mean and standard deviation of the RV forecasts, resulting in better performance on such days.

This behaviour can be better understood through an analogy with large language models (LLMs). For instance, image-generation models often produce clocks displaying 10:10, an advertising convention common in training data due to its visual symmetry. Analogously, ML models are trained on data dominated by normal volatility days, leading them to focus on forecasting these typical conditions. As a result, they perform well during normal volatility days but struggle on rarer high volatility days. This imbalance in the training data biases model learning and contributes to the performance asymmetry observed. This issue is further exacerbated by the high capacity of ML models to capture complex nonlinear patterns, which makes them more sensitive to such imbalances. In contrast, linear models, while less flexible, are more robust on high volatility days, at the cost of significantly reduced forecasting performance during normal volatility days.

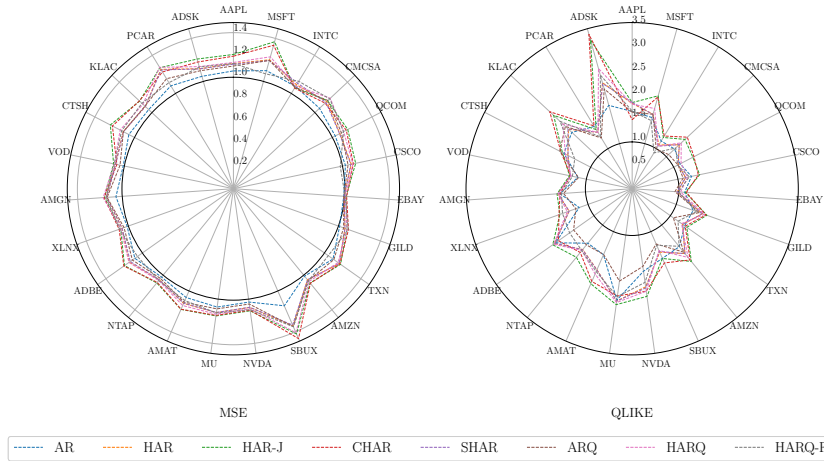
³¹The forecasts from AR(1), HAR, HAR-J, CHAR, SHAR, ARQ, HARQ, and HARQ-F exceed the true RV in, respectively, 77.6%, 78.7%, 79.8%, 80.2%, 78.6%, 78.0%, 78.8%, and 77.6% of cases (average across 23 tickers), while the OB-ML forecasts with 15 and 25 units exceed the true RV in 59.5% and 65.0% of cases, respectively. The standard deviations of forecasts from the eight models under the HAR-family exceed that of OB-ML with 15 units (25 units) in, respectively, 91.3%, 100%, 100%, 100%, 100%, 100%, 100%, and 100% (82.6%, 100%, 100%, 100%, 100%, 100%, 100%, and 100%) of the 23 tickers. This further underscores the fundamentally different behaviour of these two types of models in RV forecasting.



(a) Full Out-of-Sample Period



(b) Normal Volatility Days



(c) High Volatility Days

Figure 7: ρ_{MSE} and ρ_{QLIKE} between OB-ML (15 units) and HAR-Family of Models

Notes: Each radar chart illustrates the performance of the OB-ML model with 15 units relative to the AR(1), HAR, HAR-J, CHAR, SHAR, ARQ, HARQ, and HARQ-F models across the specified tickers. The evaluations are conducted over the full out-of-sample period (Figure 7a), during normal volatility days (Figure 7b), and during high volatility days (Figure 7c). The left and right radar charts present results based on the MSE and QLIKE loss functions, respectively. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The bold circle represents the baseline (no improvement, i.e., a value of one).

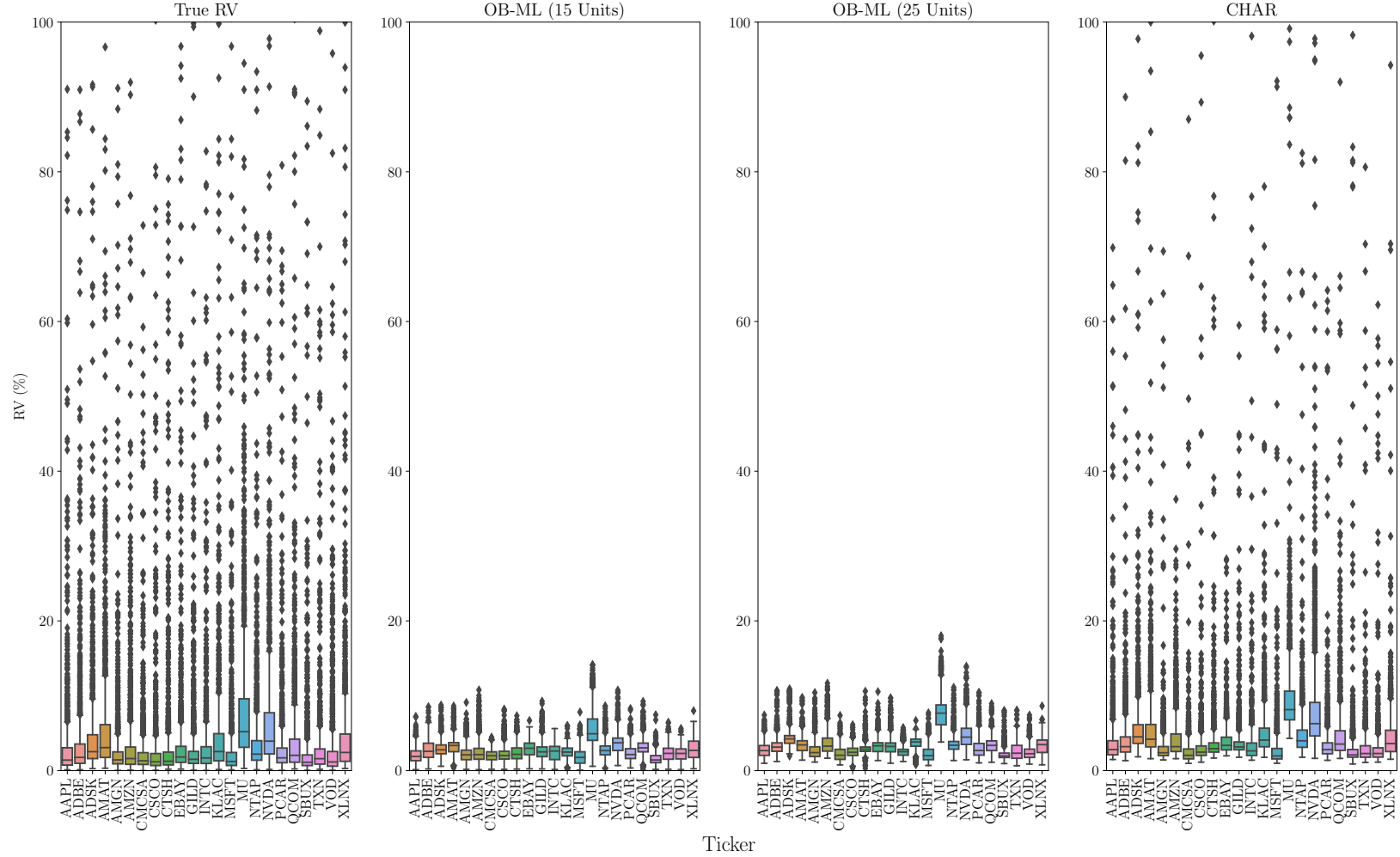


Figure 8: True and Forecast RVs from OB-ML and CHAR

Notes: From left to right, the box plots show the distribution of the true RVs, the forecast RVs from the OB-ML models with 15 and 25 units, and CHAR, respectively. For clarity, the RV on the y-axis is truncated at 100%. The points above each box plot represent high volatility days.

6 XAI

ML models are often described as ‘black boxes’. In this section, an XAI technique is employed to enhance our understanding of the relative importance of various features used in RV forecasting. We calculate the SHAP (SHapley Additive exPlanations) values for the set of 147 (i.e., six HAR, nine news, and 132 LOB) variables included in the News/OB-ML model with 15 units, representing the middle number of units we tested. Instead of using the best-performing ML model, we use News/OB-ML because this model group includes all variables, and the goal here is to examine the explanatory power of each variable in the set. Lundberg and Lee (2017) introduced SHAP as a type of sensitivity analysis within the realm of XAI techniques. The Shapley value ϕ_i , defined below, measures the importance of variable i :

$$\phi_i = \frac{1}{|N|!} \sum_{S \subseteq N \setminus \{i\}} |S|! (|N| - |S| - 1)! [f(S \cup \{i\}) - f(S)], \quad (28)$$

where $N = 147$ is the total number of variables included in the News/OB-ML model, S represents all possible subsets from $N \setminus \{i\}$ (i.e., the set of 146 variables after excluding the i^{th} variable), $f(S \cup \{i\}) - f(S)$ is the change in RV forecast by adding variable i to the set S , and $|N|$ is the number of variables in N . $|S|!$ and $(|N| - |S| - 1)!$ show the number of ways the chosen and the remaining set of variables may be presented. The resulting value is the average contribution to RV forecasts of the input variable i^{th} . In this study, we utilise a high-speed approximation algorithm called Deep SHAP, based on DeepLIFT (Deep Learning Important FeaTures; Shrikumar et al. (2017)).³² For each of the 23 tickers, we calculate SHAP for every day in the out-of-sample period. A variable is among the most important variables for each ticker when it is among the top 10% (i.e., 15 out of 147 variables), and also appears in the list at least 50% of the time (i.e., 802 out of 1,604 days) in the out-of-sample period.

The consolidated results across 23 tickers are reported in Table 8, displaying the number of times each variable is chosen as an important variable. The findings clearly indicate that primary financial variables, including ‘Mid prices’ (at various LOB levels from 1 to 10), ‘Mean bid’, and

³²DeepLIFT attributes neural network outputs to input features by comparing activations to a reference state and propagating resulting differences back through the network. Deep SHAP extends this approach by integrating Shapley values from cooperative game theory, thereby combining the computation efficiency of DeepLIFT with the theoretical rigour of SHAP. By leveraging DeepLIFT’s backpropagation mechanism and averaging over multiple reference baselines, Deep SHAP provides an efficient approximation of SHAP values while preserving their desirable properties.

Table 8: Variable Aggregate Importance

Mid price (L1)	13 ^a	Relative proportion (S/bid)	3	Relative proportion (D/ask)	1
Mid price (L7)	11	News count	2	Relative proportion (C/ask)	1
Mean price (bid)	10	Relative proportion _V (E/ask)	2	Relative proportion _V (E/bid)	1
Mean price (ask)	9	Mean volume (bid)	2	Bid-ask spreads (L2)	1
Mid price (L4)	9	Mean volume (ask)	2	Event proportion _V (E/ask)	1
Mid price (L6)	8	Bipower variation (BPV)	2	Price difference (L7/bid)	1
Mid price (L10)	8	Event proportion (S/ask)	2	Accelerations _V (E/bid)	1
Mid price (L3)	8	Event proportion (C/ask)	2	Price difference (L9/bid)	1
Mid price (L9)	7	Price difference (L4/ask)	2	Volume change (L4/bid)	1
Mid price (L8)	7	Price difference (L2/bid)	2	Volume change (L6/bid)	1
Price difference (L6/ask)	6	Price difference (L10/bid)	2	Price difference (L2/ask)	1
Mid price (L2)	6	Volume change (L10/bid)	2	Price difference (L6/bid)	1
Mid price (L5)	5	Price difference (L6/bid)	2	Event proportion (C/bid)	1
Event proportion _H (E/bid)	4	Price difference (L3/ask)	2	Event proportion (D/bid)	1
Price difference (L2/ask)	4	Price difference (L5/ask)	1	Accelerations _H (E/ask)	1
Price difference (L3/bid)	3	Volume change (L3/ask)	1	Accelerations (S/bid)	1
Relative proportion (S/ask)	3	Price difference (L5/bid)	1	Event proportion (S/bid)	1
Relative proportion (C/bid)	3	Price difference (L1/bid)	1	Accelerations _H (E/bid)	1
Relative proportion (D/bid)	3	Relative proportion _H (E/ask)	1	Uncertainty sentiment	1

Notes: ‘L’ stands for the level of the LOB. ‘E’, ‘S’, ‘C’, and ‘D’ stand for execution, submission, cancellation, and deletion of orders, respectively. For execution, ‘V’ and ‘H’ denote visible and hidden execution, respectively.

^a The number of times the indicated variable is chosen as an important variable among the 23 tickers.

‘Mean ask’, are among the most important variables. It is also dominated by LOB variables. Only three non-LOB variables are included in Table 8, viz., BPV from the list of HAR variables, and ‘News count’ and ‘Uncertainty’ from the set of news variables.³³ Finally, many complex LOB-derived variables are chosen as important variables but only for a few tickers, suggesting idiosyncratic behaviour among individual tickers.³⁴

Next, we made an information set comparison in terms of the groups’ forecasting performance in the out-of-sample period. The top chart in Figure 9 shows the daily SHAP average importance (across 23 tickers) of HAR, LOB, and news variable groups. The y-axis displays the normalised importance of the three variable groups at the daily level, with values summing to one for each day. The bottom chart shows the daily RVs of the NASDAQ-100 (with a 5-minute sampling interval) downloaded from the Oxford-Man Institute of Quantitative Finance Realised Library.³⁵ What stands out from Figure 9 is that the HAR variables exhibited the lowest overall importance, except during sporadic episodes of heightened volatility. HAR’s dominating importance during the COVID-19 disruption and the extremely high volatility period in early 2020 is very prominent. The out-of-sample period can also be separated into a pre-2018 low-volatility period and a post-

³³Similarly, Rahimikia and Poon (2020) found that, among all the news variables, ‘News count’ is the most powerful for forecasting RV.

³⁴Zhu et al. (2023) found that the key variables for RV forecasting within ML models are primarily sourced from high-frequency returns. However, the findings of this study suggest that broadening the range of features may alter this perspective. Specifically, it becomes evident that primary variables from various levels of the LOB are informative for accurate RV forecasting.

³⁵A note of caution is due here, since there are some missing daily RVs in this library.

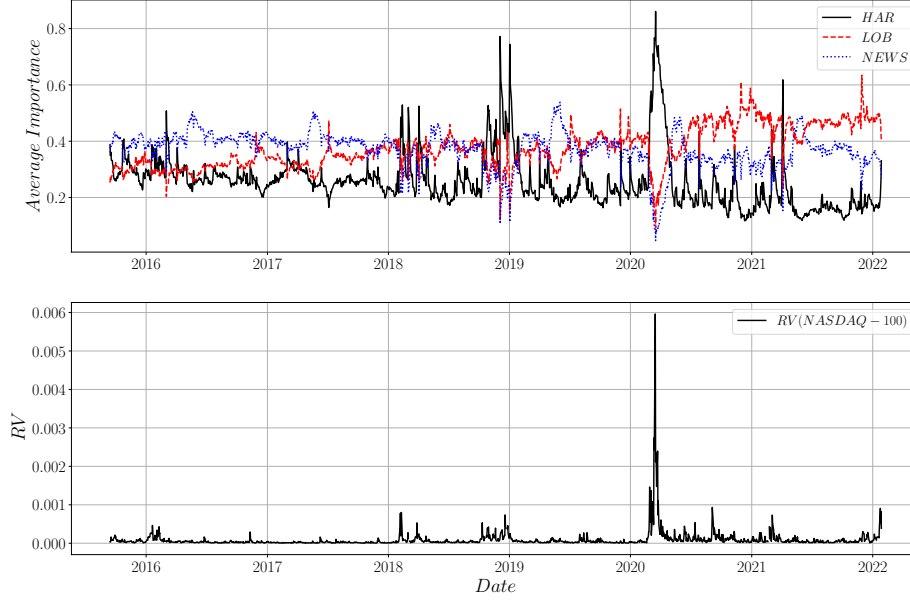


Figure 9: Average Variable Importance vs Market RV (Out-of-Sample)

Notes: The top chart shows the daily SHAP average importance (across 23 tickers) of the HAR variable group in Subsection 3.1, the news variable group in Subsection 3.2, and the LOB variable group in Subsection 3.3. This analysis uses News/OB-ML (15 units) to cover all variables together. The y-axis displays the normalised importance of the three variable groups at the daily level, with values summing to one for each day. The bottom chart shows the daily RVs of the NASDAQ-100 (5-min) downloaded from the Oxford-Man Institute of Quantitative Finance Realised Library.

2018 period populated by many episodes of fluctuating volatility. Before 2018, the variable importance across all three information sets remained largely stable. The variable importance of news sentiment dominated LOB and HAR variables before 2018, when the market was calm. Post-2018, the variable importance of news sentiment steadily declined and was dominated by LOB variables. There were a few short time intervals when the importance of news sentiment variables dominated, but these did not coincide with the timing of the most volatile periods. In summary, it is clear that (i) the importance of the three information sets varied through time, (ii) some predictors are more informative during extremely volatile periods, while other predictors are more informative during normal volatility periods, and (iii) the elevated importance of HAR variables is consistent with the findings reported in Subsection 5.4. Nevertheless, although the ML models assign greater importance to HAR variables during periods of high volatility, its performance still falls short of that achieved by the HAR-family of models under these conditions. The flexible ML structure (where there is no fixed time chain, and past information is flexibly controlled by the LSTM gates at every time step) can accommodate the complex patterns in (i), (ii), and (iii). Moreover, the inherent flexibility of the ML model in integrating a wide array of predictive variables plays a crucial role in driving these results.

7 Robustness Checks

Up to now, all results in Section 5 show that the ML model with LOB variables outperformed the HAR-family of models for forecasting RV on normal volatility days, but not on high volatility days. This is also reflected in the forecasting performance over the full out-of-sample period, where the HAR-family of models generally outperform ML models. In this section, we perform a series of robustness checks for this conclusion by using time-restricted input information in Subsection 7.1, using QLIKE (instead of MSE) as the loss function in Subsection 7.2, testing a large combination of hyperparameters in Subsection 7.3 and Subsection 7.4, changing the model architecture from LSTM to FCNN in Subsection 7.5, and adding additional safeguards against overfitting in Subsection 7.6. Apart from the experimented parameters in each subsection, all other model specifications are the same as those adopted in Subsection 4.2.

7.1 Restricted Number of Lags

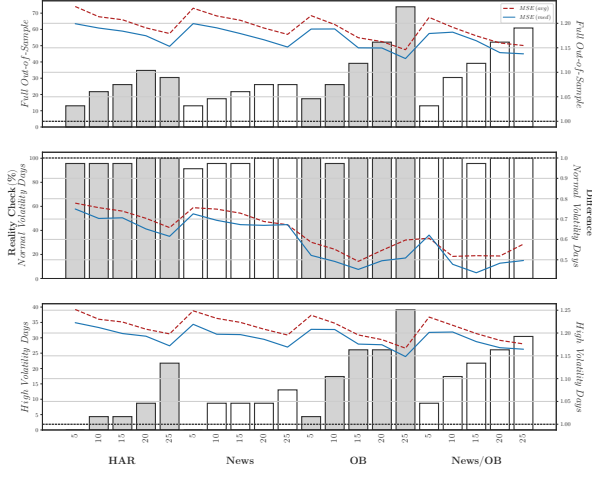
One could argue that the forecasting performance of ML is solely attributable to the extensive historical information it utilises. Here, we restrict all input variables to have five lags (equivalent to one week) and one lag (i.e., the previous day). An LSTM with one lag (one input and one output) reduces to a standard neural network. Figure 10 presents the average (dashed line) and median (solid line) of ρ_{MSE} and ρ_{QLIKE} , as well as the RC values (bar chart).

The results show that restricting the information set from 21 days in Figure 4 to 5 days and 1 day did not change the superior performance of the ML models for normal volatility days when evaluated using MSE. Additionally, the QLIKE results are comparatively weaker for both the 5-day and 1-day cases relative to our primary experiment, particularly when a smaller amount of information is utilised. The 1-day results are also weaker than the 5-day results. Consistent with previous findings, the ML models continue to face challenges in forecasting RV on high volatility days, resulting in weaker performance over the full out-of-sample period. Similar to the results on normal and high volatility days, the achieved performance over the full out-of-sample period is also weaker than that observed in the primary experiment. The MDA results presented in Figure C6 are consistent with previous findings, further reinforcing earlier patterns and highlighting the superior performance of the models in directional forecasting.³⁶ Due to the

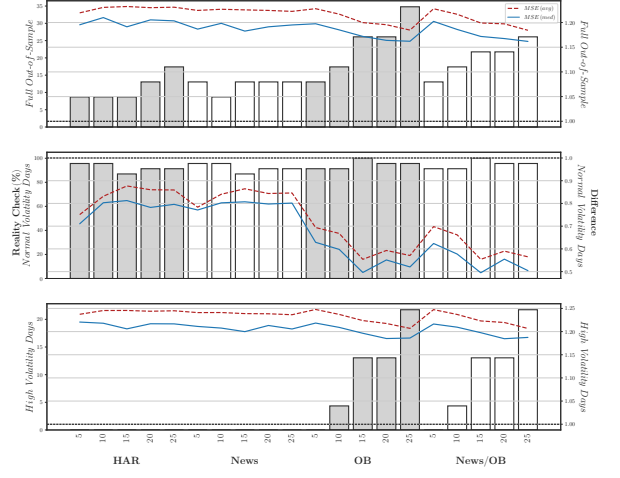
³⁶For normal volatility days, the RC values of ML models are high, and many reach 100%. The improvement in MDA by switching from CHAR to ML varies between 2% to 10% for ML with different numbers of units and

(a) Forecast Evaluation under MSE

(i) Five Lags

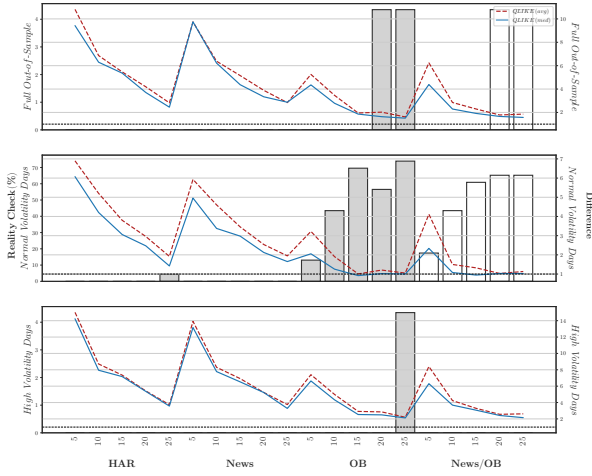


(ii) One Lag



(b) Forecast Evaluation under QLIKE

(i) Five Lags



(ii) One Lag

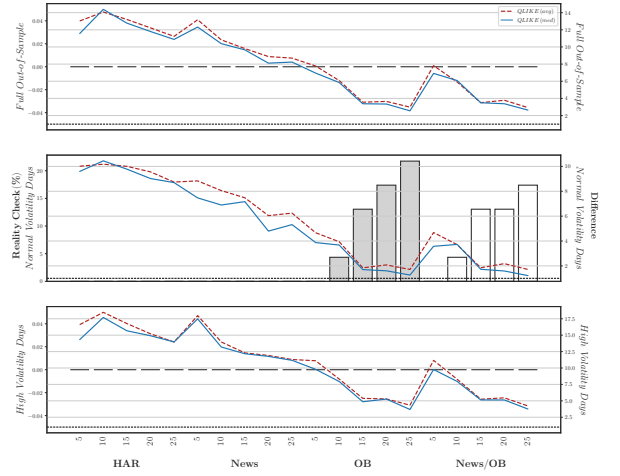


Figure 10: ML Models with Restricted Number of Lags

Notes: The left and right figures show the results considering five lags (last week) and one lag (last day), respectively. The bar chart shows the percentage of tickers with outstanding performance, considering the MSE/QLIKE loss function at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for each HAR-ML, News-ML, OB-ML, and News/OB-ML group. The values for the bar chart can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MSEs and QLIKEs of the ML model relative to the CHAR model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

good performance of ML models even in the case of a restricted number of lags, we also conclude that our findings of superior ML forecasting performance on normal volatility days are due not only to the larger historical information set it has access to but also to its model structure.

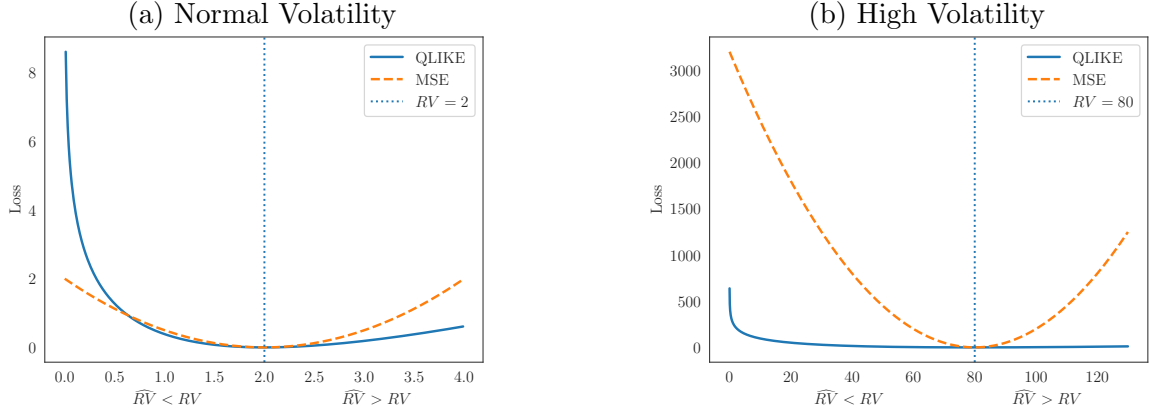


Figure 11: A Representation of MSE and QLIKE Loss Functions

Notes: This figure presents the shape of the MSE and QLIKE loss functions. The vertical dashed line denotes the true RV, which equals 2 in panel (a) and 80 in panel (b). Forecast values located to the left of the true RV correspond to underprediction, $\widehat{RV} < RV$, whereas values to the right correspond to overprediction, $\widehat{RV} > RV$.

7.2 MSE vs QLIKE as Loss Function

All the ML model results produced so far are based on minimising MSE as the loss function in the training period while using MSE and QLIKE in forecast evaluation. Here, we test whether changing the loss function to minimising QLIKE will change the results and conclusions. According to Patton (2011), MSE and QLIKE are members of a family of robust and homogeneous loss functions, $L(\cdot)$, shown below:

$$L(RV, \widehat{RV}; b) = \begin{cases} \frac{1}{(b+1)(b+2)}(RV^{b+2} - \widehat{RV}^{b+2}) - \frac{1}{b+1}\widehat{RV}^{b+1}(RV - \widehat{RV}), & \text{for } b \notin \{-1, -2\} \\ \widehat{RV} - RV + RV(\log(\frac{RV}{\widehat{RV}})), & \text{for } b = -1 \\ \frac{RV}{\widehat{RV}} - \log(\frac{RV}{\widehat{RV}}) - 1, & \text{for } b = -2, \end{cases} \quad (29)$$

where L is the loss function, RV is the true RV, $\widehat{RV}$ is the fitted (forecasted) RV, and b is the scalar parameter. For $b = 0$, L becomes MSE, and L becomes QLIKE if $b = -2$. Figure 11 presents the shape of these two loss functions, setting the true RV (vertical dashed line) equal to 2 (in

different numbers of lagged variables. It is also evident that the results for the 1-day are weaker than those for the 5-day, and that incorporating additional information as input improves the performance of the ML models. In summary, these results show that ML outperforms the HAR-family of models in RV forecasting for normal volatility days, even with only 1 and 5 lags instead of 21 lags. This is further reflected in the full out-of-sample results, which exhibit RC values approaching 100% and up to a 4% improvement in MDA for both the 5-day and 1-day cases.

Figure 11a) and 80 (in Figure 11b). Forecast values located to the left of the true RV correspond to underprediction, $\widehat{RV} < RV$, whereas values to the right correspond to overprediction, $\widehat{RV} > RV$. Compared to MSE, QLIKE is asymmetric and penalises large under-forecasts more than large over-forecasts. Patton and Sheppard (2015) found, in volatility forecasting, that the power of DMW tests (Diebold and Mariano (1995) and West (1996)) is higher when the loss function is QLIKE instead of MSE, suggesting that QLIKE might be a better loss function for ranking competing volatility forecasting models.

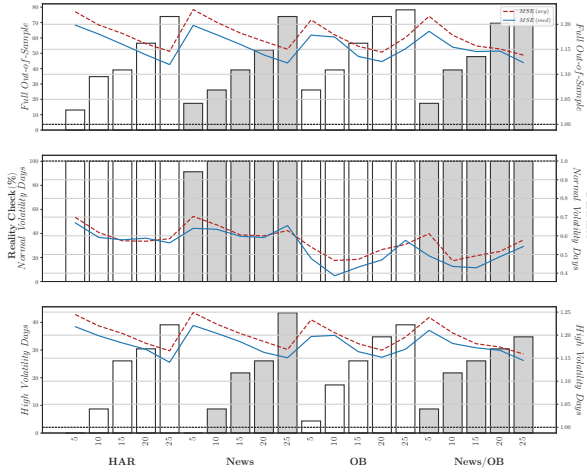
As a robustness check, we compare MSE versus QLIKE as the loss function for training the ML models, while all other model specifications remain unchanged. QLIKE (as in Equation (20) and Equation (29)) is undefined when $\widehat{RV} \leq 0$. To address this limitation and prevent the loss function from becoming undefined during training, we append a Lambda layer after the FCNN. This layer enforces a lower bound on the predicted RV, ensuring that $\widehat{RV} \geq 0.01$ and thus remains strictly positive.³⁷ The results are presented in Figure 12. Figure 12a (Figure 12b) evaluates the forecasts under MSE (QLIKE), while the left (right) figure within each subfigure shows the results for the ML models trained using minimising MSE (QLIKE) as the loss function.

As before, there are substantial gains from ML models, improving forecasts on normal volatility days but with a performance degradation on high volatility days. Figure 12a shows that changing the MSE to the QLIKE loss function causes a substantial degradation in forecasting performance for high volatility days. Considering both RC and the average and median ρ_{MSE} values for normal volatility days, the degradation is evident but not as severe as on high volatility days. Further analysis of forecasts evaluated under the QLIKE loss function in Figure 12b shows a similar but more severe degradation pattern in performance when switching to the QLIKE loss function. It can be seen that the degradation for normal volatility days is high; many RC values are near zero, and all the average and median ρ_{QLIKE} values are above one, indicating substantial degradation in forecasting performance. Moving to high volatility days, incorporating the QLIKE loss function in the training process changes all RC values to zero with substantial degradation in ρ_{QLIKE} values. The results for the full out-of-sample period are also consistent with previous findings and indicate that using QLIKE instead of MSE leads to degradation in performance under both MSE and QLIKE metrics, with the effect being more pronounced for

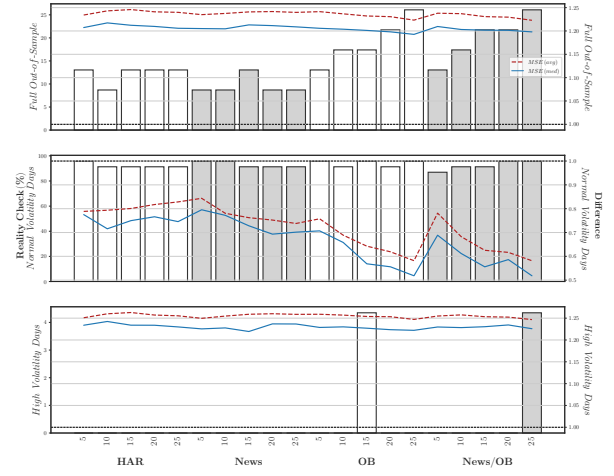
³⁷Another way of ensuring $\widehat{RV} > 0$ is to change the scalar parameter, b , to very close to -2 , such as -1.99 or -2.01 in Equation (29). In order to maintain consistency with the other experiments, this method is not utilised in this study.

(a) Forecast Evaluation under MSE

(i) MSE Loss Function

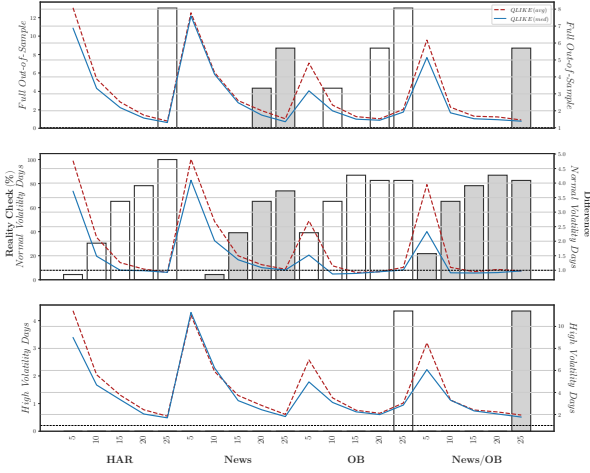


(ii) QLIKE Loss Function



(b) Forecast Evaluation under QLIKE

(i) MSE Loss Function



(ii) QLIKE Loss Function

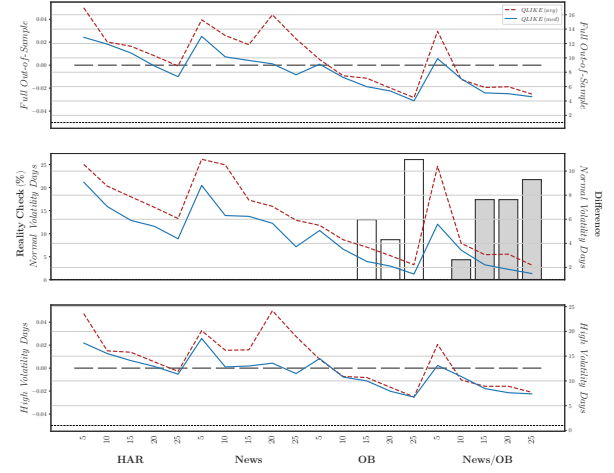


Figure 12: Minimising MSE vs QLIKE as the Loss Function in Training

Notes: For each of the two subfigures, the left (right) figure presents the results for the four groups of ML models (HAR-ML, News-ML, OB-ML, and News/OB-ML) trained using minimising MSE (QLIKE) as the loss function. The other model specifications are the same as in Subsection 4.2. The bar chart is the percentage of tickers with the outstanding performance at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for every specified ML model. The values for the bar chart can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MSEs and QLIKEs of the ML model relative to the CHAR model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

QLIKE. Furthermore, for the full out-of-sample results, a general degradation in performance, particularly in terms of magnitude, is evident. However, RC values still demonstrate moderate, albeit reduced, improvements in performance compared to the primary experiment. The results derived from the MDA loss function align with previous empirical findings.³⁸

However, what remains unclear is why, when minimising QLIKE (instead of MSE) as the loss function, the ML models severely underperform when QLIKE is designed to avoid large under-forecasts. The clue lies in the QLIKE weighting function when true RV is very high. Returning to Figure 11, where the MSE and QLIKE loss functions are presented for true $RV = 2$ on the left and true $RV = 80$ on the right, it is clear that when the true RV is very high, the weights of QLIKE become very flat on both sides of the true RV; only when $\widehat{RV} \ll RV$ does the weight begin to rise. This means that when true RV is very high, QLIKE becomes insensitive to the size of the forecast errors (except for extremely large under-forecasts). Hence, the optimisation algorithm loses the ability to learn to forecast accurately. This evidence suggests that QLIKE, without further modification, is not appropriate as a loss function for training ML models, at least as specified here. Therefore, only the minimisation of MSE is used in this study.³⁹

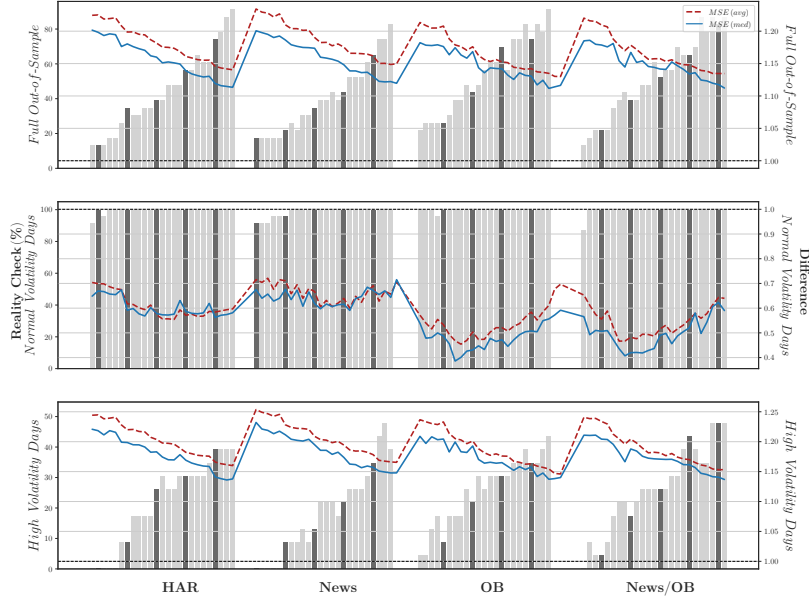
7.3 No. of Units vs No. of Epochs

This subsection tests the sensitivity of the number of units (#units) and the number of epochs (#epochs) in affecting the forecasting performance of ML models. We test the number of units ranging from 5, 10, 15, 20, to 25, and for the number of epochs, from 25, 50, 75, 100, to 125. The other ML model specifications are the same as those in Subsection 4.2. For 23 tickers, four ML groups, and 1,604 days in the out-of-sample period, 3,689,200 ($25 \times 23 \times 4 \times 1604$) ML models are trained and tested, which is substantially higher than the 737,840 ($5 \times 23 \times 4 \times 1604$) ML models trained in Section 5.

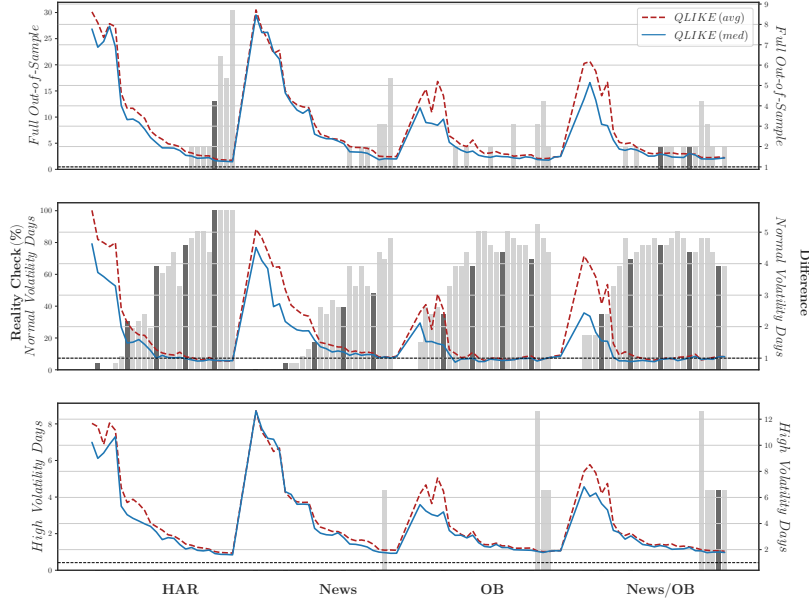
The results are presented in Figure 13, Figure 13a for the MSE loss function, and Figure 13b for the QLIKE loss function, across the four groups of ML models (from left to right: HAR-ML, News-ML, OB-ML, and News/OB-ML). The results are shown in the following order from left to

³⁸From Figure C7, changing the MSE loss function in Figure C7a to the QLIKE loss function in Figure C7b results in a degradation in performance on both normal and high volatility days. This pattern is also evident across the full out-of-sample period.

³⁹The results here also weaken the case for using QLIKE as a measure for forecast evaluation when QLIKE is used as the optimisation algorithm loss, especially on high volatility days.



(a) Forecast Evaluation under MSE



(b) Forecast Evaluation under QLIKE

Figure 13: No. of Units vs. No. of Epochs

Notes: From left to right, each figure consists of the HAR-ML, News-ML, OB-ML, and News/OB-ML groups. For each group, the results are shown in the following order from left to right (#units-#epochs): 5-25, 5-50, 5-75, 5-100, 5-125, 10-25, 10-50, 10-75, 10-100, 10-125, 15-25, 15-50, 15-75, 15-100, 15-125, 20-25, 20-50, 20-75, 20-100, 20-125, 25-25, 25-50, 25-75, 25-100, and 25-125. For the sake of clarity, these values are not shown in the figures. The bar chart shows the percentage of tickers with outstanding performance, considering the MSE loss function in the top part (QLIKE in the bottom part) at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for each specified ML model. The dark grey bars represent the RC values of the primary experiments in Section 5. The RC values can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MSEs and QLIKEs of the ML model relative to the CHAR model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

right (#units-#epochs): 5-25, 5-50, 5-75, 5-100, 5-125, 10-25, 10-50, 10-75, 10-100, 10-125, 15-25, 15-50, 15-75, 15-100, 15-125, 20-25, 20-50, 20-75, 20-100, 20-125, 25-25, 25-50, 25-75, 25-100, and 25-125. For the sake of clarity, these values are not shown in Figure 13. The light grey bars represent the percentage of tickers with outstanding performance at the 5% significance level based on the RC. The dark grey bars represent the RC values from the primary experiments described in Section 5.

First, consider the case of MSE (QLIKE) as the loss function in Figure 13a (in Figure 13b). As before, the results confirm that, for normal volatility days, generally, a higher number of units produced the best performance in terms of average (median) $\rho_{MSE/QLIKE}$ and the RC values. Second, the number of units influences forecasting performance more than the number of epochs. Third, ML outperformed CHAR and the HAR-family of models only on normal volatility days; when switching to the high volatility days, ML models underperformed, with above one average (median) $\rho_{MSE/QLIKE}$ and RC values lower than 50%. Also, increasing the number of units and epochs reduces the amount of underperformance on high volatility days. Fourth, the results for QLIKE as the loss function show the same patterns. Fifth, of the four groups of ML models tested here, OB-ML and News/OB-ML performances are the most outstanding. Finally, consistent with previous findings, the poor performance observed on high volatility days has a significant impact on the full out-of-sample results, leading to a degradation in performance. However, the incorporation of more complex models and an expanded information set helps mitigate this effect. As in earlier results, the degradation in performance is primarily in magnitude relative to the CHAR benchmark. Notably, the RC results continue to show improvements, particularly when more complex models are employed. The results for the MDA in Figure C8 corroborate the findings here.⁴⁰

These results further confirm that the ML models defined in this study exhibit strong volatility forecasting performance during periods of normal market conditions but perform less effectively on high volatility days. This suggests that greater attention is needed when forecasting during episodes of heightened market turbulence. To improve forecasting performance under such conditions, more complex ML models with an increased number of units and training epochs may

⁴⁰Most RC values for normal volatility days are near 100% and ρ_{MDA} reaching around 12% improvement in the OB-ML and News/OB groups. For high volatility days, poor performance is clear for all combinations of the number of units and epochs. However, consistent with prior outcomes, improvements in forecasting performance over the full out-of-sample period are evident, both in terms of mean and average values, as well as in the results produced by the RC.

need to be developed and trained.⁴¹

7.4 Units vs Epochs: Individual Ticker Analysis

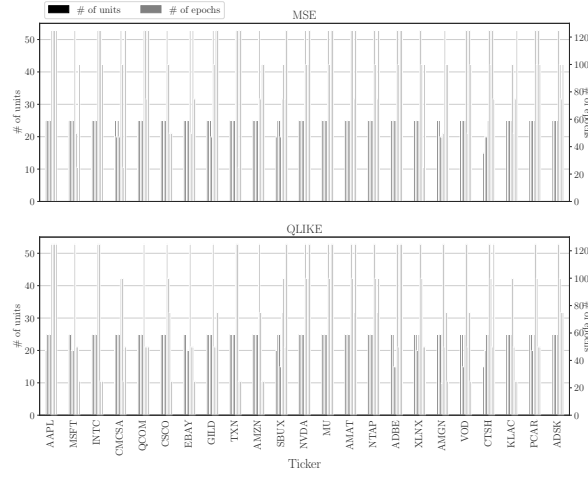
This subsection concerns the best combination of the number of units and epochs for each ticker’s best volatility forecasting performance. The results are presented in Figure 14, Figure 14a, Figure 14b, and Figure 14c, corresponding to the full out-of-sample period, as well as performance on normal and high volatility days, respectively. For each ticker, there are two sets of four bars corresponding to the four groups of ML models, viz., from left to right, HAR-ML, News-ML, OB-ML, and News/OB-ML. The first four black bars represent the optimal number of units for the four groups of ML models, while the next four grey bars represent the optimal number of epochs for the same groups. Within each subfigure, the top (bottom) graph reports the results based on the MSE (QLIKE) loss function. The number of units (# of units) can be read from the left axis, and the number of epochs (# of epochs) can be read from the right axis.

The optimal numbers of units and epochs are generally larger for high volatility days than for normal volatility days. There are also some variations among the tickers, especially for normal volatility days in Figure 14b for both MSE and QLIKE.⁴² For high volatility days in Figure 14c, there are fewer variations across tickers, and the results are in favour of a more complex ML model with about 25 units and 70 to 120 or more epochs. The results for the full out-of-sample period in Figure 14a match this. The corresponding results for the MDA loss function are presented in Figure C9.⁴³ The individual ticker analysis provides important insight into the importance of hyperparameter choice conditioned on the level of actual RV. A more complex ML model specification (with a high number of units and epochs) works well for almost all tickers for forecasting

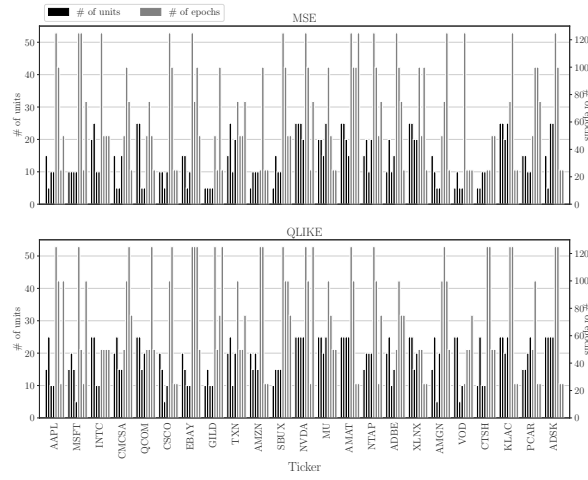
⁴¹We deliberately omit the early stopping mechanism in the training process because of three primary considerations. Firstly, allocating a segment of the time series for early stopping would lead to the exclusion of the most recent data from the training process. This segment could potentially hold crucial insights for more accurate RV forecasts. Secondly, as detailed earlier, we have tested 3,689,200 distinct models, each employing different combinations of the number of units and epochs. The findings unanimously point towards poor performance on high volatility days by all models tested. That is, our conclusion, in general, is not sensitive to the choice of hyperparameters. Thirdly, by omitting early stopping, we can compare performance due entirely to the choice of hyperparameters. Although early stopping is omitted here, in Subsection 7.6, we test its impact with a smaller-scale experiment.

⁴²Looking at Figure 14b, it is apparent that for some of the tickers like ‘NVDA’, ‘MU’, ‘AMAT’, and ‘ADSK’, a more complex model with a higher number of units is required for better forecasting performance, consistent with the findings in Subsection 5.4.

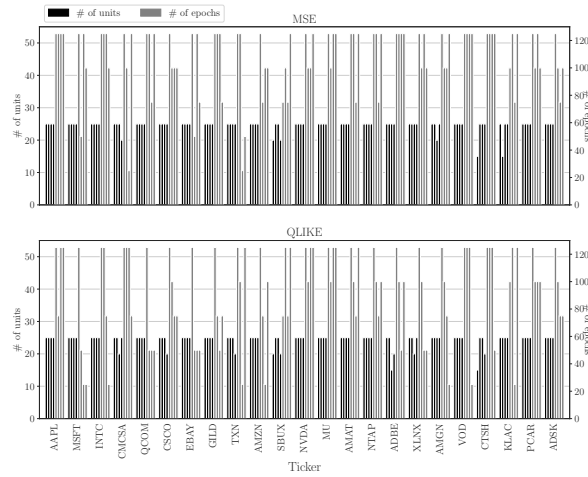
⁴³Interestingly, although some differences are observed across individual tickers, a less complex model with fewer units and fewer training epochs tends to be sufficient. Increasing model complexity or extending the number of training epochs does not result in substantial performance gains when evaluated using the MDA loss function. This further supports the premise that these ML models, many of which are fundamentally designed for directional forecasting, perform effectively even in their most minimal configurations.



(a) Full Out-of-Sample



(b) Normal Volatility Days



(c) High Volatility Days

Figure 14: The Best-Performing Hyperparameters (#Units & #Epochs)

Notes: There are two sets of four bars for each ticker that correspond to the four groups of ML models, viz. HAR-ML, News-ML, OB-ML, and News/OB-ML. The first four black bars correspond to their optimal number of units, while the next four grey bars correspond to their optimal number of epochs. The top, middle, and bottom figures correspond to the full out-of-sample period, normal volatility days, and high volatility days, respectively. Within each subfigure, the top (bottom) graph reports the results considering the MSE (QLIKE) loss function. The number of units (# of units) can be read from the left axis, and the number of epochs (# of epochs) can be read from the right axis.

RV on high volatility days. In contrast, for forecasting RV on normal volatility days, one would expect more idiosyncratic variations, and careful tuning of hyperparameters for each ML group and for each ticker can provide additional improvement to forecasting performance. Additionally, the pronounced impact of high volatility days on full out-of-sample performance supports the use of more complex ML models when evaluating the entire out-of-sample period. Altogether, these variations raise concerns about using ML models as direct substitutes for the HAR-family of models, due to the inherent complexities associated with hyperparameter optimisation.

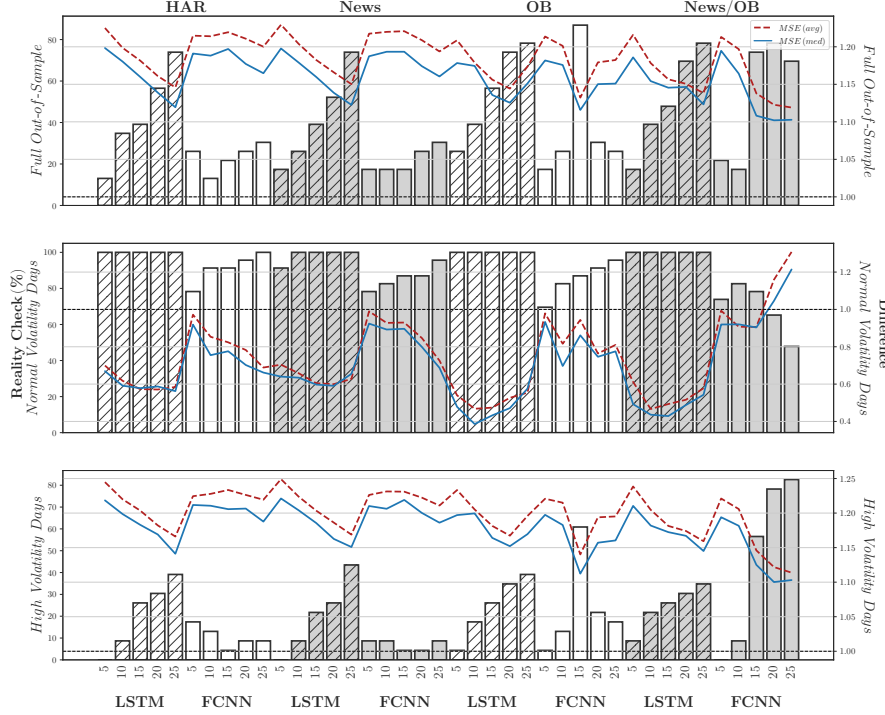
7.5 Fully Connected Neural Network

FCNN is among the most commonly employed alternatives to conventional models in the literature. Accordingly, we investigate the impact of replacing the LSTM model with an FCNN on out-of-sample forecasting performance. To ensure a controlled comparison, we retain the same hyperparameter settings as those used for the LSTM model, as detailed in Subsection 4.2. Furthermore, we ensure that the input variables are consistent with those used in the LSTM experiments. Specifically, in the LSTM model, each variable includes 21 lags, which are naturally accommodated by the LSTM architecture due to its inherent capacity to model sequential dependencies. In contrast, the FCNN architecture does not inherently accommodate sequential data. To preserve the input structure, 21 lags of each variable must be explicitly specified as separate input features for the FCNN. As a result, the number of input variables rises to 126, 315, 2,898, and 3,087 for the HAR-ML, News-ML, OB-ML, and News/OB-ML groups, respectively.

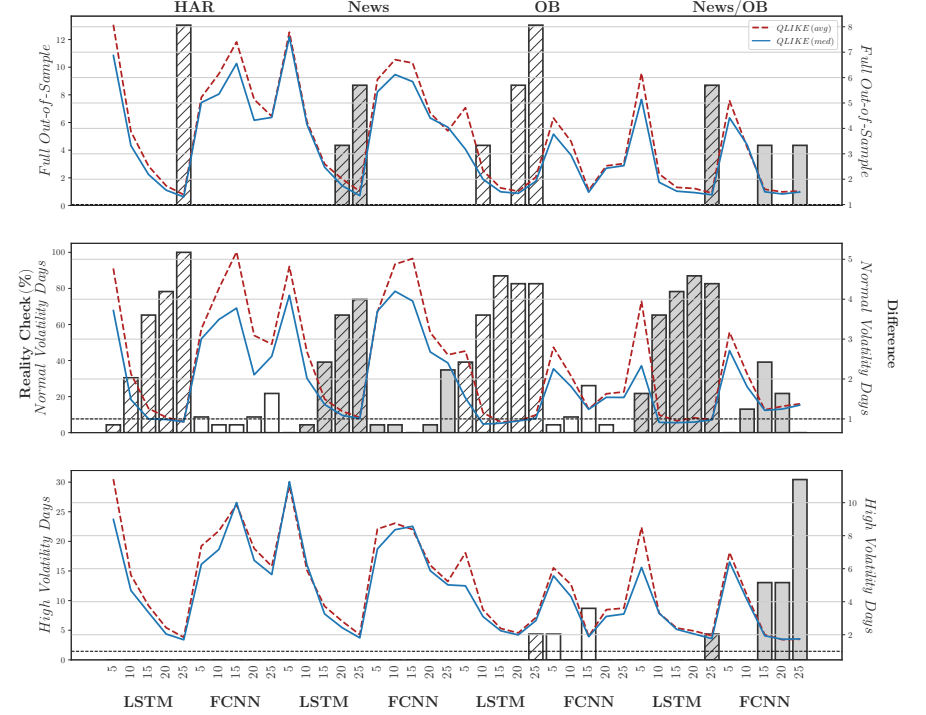
We present the results for HAR-ML, News-ML, OB-ML, and News/OB-ML. Specifically, for 23 tickers, these four ML groups, and 1,604 days in the out-of-sample period, a total of 737,840 ($5 \times 23 \times 4 \times 1604$) ML models, specifically FCNN, are trained and tested. Figure 15 presents a comparison of the LSTM model from Subsection 5.1 and Subsection 5.2 with the results obtained from FCNN. The first, second, third, and fourth sets of ten bars correspond to the HAR, News, OB, and News/OB groups, respectively. Within each set, the first five hatched bars represent results from the LSTM model, while the subsequent five bars represent results from the FCNN model. Each bar within these subsets reflects performance associated with a specific number of units. The values for the bar chart correspond to the left-hand axis. As before, the dashed (solid) line represents the average (median) out-of-sample MSEs and QLIKEs of the ML model relative to the CHAR model across 23 tickers, with values read from the right-hand side. Across

all information sets, the superior performance of the LSTM model relative to the FCNN model is evident. As previously observed, model performance is generally superior on normal volatility days, with the LSTM consistently achieving 100% RC, while the FCNN reaches this level of RC primarily in its more complex forms. A similar pattern emerges in terms of magnitude of performance, as even the simplest LSTM model matches or outperforms the most complex FCNN model. The results under the QLIKE loss function further reinforce these findings. This can be attributed to the sequential structure of LSTM, which provides greater predictive power for RV forecasting compared to FCNN models. Additionally, the comparatively better performance of FCNN on normal volatility days, rather than during high volatility days, further supports our previous conclusions. A similar pattern is also observed over the full out-of-sample period. Consistent with the LSTM, the FCNN exhibits a marked decline in performance, particularly regarding the magnitude of its forecasting performance. The results for MDA in Figure C10 align with the findings presented here, both demonstrating the superior performance of LSTM over FCNN.

Table 9 reports the out-of-sample forecasting performance of the FCNN model and enables a direct comparison with the LSTM results presented in Table 6 and Table 7, using identical information sets and evaluation designs. Overall, the LSTM models outperform their FCNN counterparts. With respect to MSE, in the full out-of-sample period both architectures produce loss ratios exceeding unity across all specifications, although in some cases the FCNN achieves values marginally closer to one for the best-performing configurations. In contrast, for QLIKE, the LSTM models generally deliver more favourable outcomes in the corresponding optimal specifications. The comparison is most pronounced for MDA, where the LSTM consistently attains lower loss ratios than the FCNN in the leading configurations. The RC statistics broadly corroborate these findings, indicating stronger and more persistent dominance of the LSTM specifications. Results for normal volatility days reinforce this pattern, with LSTM exhibiting substantially superior performance. During high volatility days, the FCNN shows a modest advantage in a subset of configurations; however, both architectures perform poorly in this regime, consistent with our earlier evidence that ML models generally perform poorly during periods of elevated volatility. Moreover, these results highlight that the choice of ML architecture, and potentially the associated hyperparameter configurations used in model design, has an impact on RV forecasting performance, particularly for high volatility days, which, as discussed in Section 5, remain diffi-



(a) Forecast Evaluation under MSE



(b) Forecast Evaluation under QLIKE

Notes: The bar chart is the percentage of tickers with the outstanding performance considering the MSE loss function in Figure 15a and the QLIKE loss function in Figure 15b at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for every specified model. The first, second, third, and fourth sets of ten bars correspond to the HAR, News, OB, and News/OB groups, respectively. Within each set, the first five hatched bars represent results from the LSTM model, while the subsequent five bars represent results from the FCNN model. Each bar within these subsets reflects performance associated with a specific number of units. The dashed (solid) line shows the average (median) out-of-sample MSEs and QLIKES of the ML model relative to the CHAR model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

Figure 15: Comparison of LSTM and FCNN Models

Table 9: Out-of-Sample Volatility Forecasting Performance of FCNN Models
14 September 2015 to 27 January 2022 (1,604 days)

		Units	HAR-ML					News-ML					OB-ML					News/OB-ML				
			5	10	15	20	25	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25
Full Out-of-Sample																						
MSE ^a		Avg	1.215	1.214	1.219	1.211	1.200	1.217	1.220	1.221	1.209	1.194	1.214	1.201	1.132	1.179	1.182	1.213	1.197	1.138	1.123	1.119
		Med	1.191	1.188	1.197	1.177	1.165	1.187	1.193	1.193	1.174	1.160	1.182	1.176	1.116	1.150	1.151	1.195	1.164	1.108	1.102	1.103
RC ^d	5%		26.09	13.04	21.74	26.09	30.44	17.39	17.39	17.39	26.09	30.44	17.39	26.09	86.96	30.44	26.09	21.74	17.39	73.91	78.26	69.57
	10%		60.87	60.87	56.52	60.87	65.22	60.87	60.87	52.17	60.87	73.91	56.52	69.57	95.65	86.96	82.61	56.52	78.26	95.65	95.65	100
QLIKE ^b		Avg	5.221	6.152	7.402	5.141	4.474	5.924	6.706	6.573	4.609	3.902	4.406	3.487	1.560	2.527	2.615	5.091	3.254	1.602	1.501	1.535
		Med	5.017	5.352	6.554	4.320	4.433	5.438	6.119	5.841	4.407	4.049	3.773	2.946	1.487	2.431	2.534	4.414	3.358	1.503	1.419	1.495
RC	5%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.35	0.00	4.35
	10%		4.35	0.00	4.35	8.70	0.00	4.35	0.00	0.00	0.00	0.00	4.35	0.00	0.00	0.00	0.00	0.00	0.00	4.35	4.35	4.35
MDA ^c		Avg	1.016	1.009	1.017	0.999	0.992	1.022	1.021	1.035	1.003	0.988	1.014	0.985	0.970	0.974	0.974	1.018	0.987	0.966	0.976	0.979
		Med	1.022	1.013	1.014	0.991	0.972	1.023	1.029	1.033	0.997	0.983	1.003	0.979	0.970	0.971	0.980	1.011	0.982	0.966	0.983	0.983
RC	5%		52.17	69.57	60.87	82.61	82.61	52.17	47.83	47.83	73.91	73.91	69.57	73.91	95.65	95.65	91.30	56.52	82.61	95.65	91.30	100
	10%		65.22	73.91	65.22	86.96	82.61	60.87	56.52	52.17	73.91	78.26	78.26	73.91	95.65	95.65	91.30	56.52	82.61	100	95.65	100
Normal Volatility Days																						
MSE		Avg	0.972	0.853	0.824	0.783	0.689	0.990	0.928	0.930	0.847	0.726	0.979	0.816	0.944	0.764	0.810	0.993	0.907	0.906	1.159	1.308
		Med	0.919	0.755	0.776	0.702	0.662	0.924	0.892	0.896	0.799	0.687	0.933	0.697	0.861	0.746	0.775	0.919	0.920	0.903	1.046	1.214
RC	5%		78.26	91.30	91.30	95.65	100	78.26	82.61	86.96	86.96	95.65	69.57	82.61	86.96	91.30	95.65	73.91	82.61	78.26	65.22	47.83
	10%		78.26	91.30	95.65	95.65	100	82.61	86.96	86.96	91.30	95.65	69.57	82.61	86.96	95.65	95.65	73.91	82.61	78.26	65.22	52.17
QLIKE		Avg	3.255	4.271	5.184	3.091	2.873	3.677	4.876	5.018	3.165	2.608	2.796	2.076	1.248	1.630	1.678	3.175	2.130	1.228	1.316	1.378
		Med	3.008	3.492	3.776	2.106	2.571	3.700	4.200	3.954	2.678	2.403	2.255	1.813	1.241	1.541	1.538	2.711	1.805	1.215	1.247	1.347
RC	5%		8.70	4.35	4.35	8.70	21.74	4.35	4.35	0.00	4.35	34.78	4.35	8.70	26.09	4.35	0.00	0.00	13.04	39.13	21.74	0.00
	10%		8.70	4.35	4.35	8.70	21.74	4.35	4.35	0.00	4.35	34.78	4.35	8.70	30.44	4.35	4.35	0.00	13.04	43.48	21.74	0.00
MDA		Avg	0.965	0.955	0.962	0.944	0.938	0.970	0.967	0.981	0.948	0.934	0.959	0.930	0.940	0.925	0.926	0.964	0.936	0.935	0.952	0.957
		Med	0.977	0.964	0.955	0.953	0.923	0.977	0.980	0.979	0.954	0.929	0.954	0.931	0.936	0.916	0.932	0.964	0.943	0.930	0.951	0.957
RC	5%		82.61	86.96	86.96	86.96	95.65	73.91	86.96	82.61	91.30	91.30	86.96	95.65	100	100	100	78.26	95.65	100	95.65	91.30
	10%		86.96	86.96	91.30	91.30	95.65	82.61	86.96	86.96	95.65	100	86.96	95.65	100	100	100	78.26	95.65	100	95.65	100
High Volatility Days																						
MSE		Avg	1.224	1.228	1.233	1.226	1.219	1.226	1.231	1.231	1.222	1.211	1.221	1.215	1.140	1.194	1.195	1.221	1.206	1.146	1.122	1.114
		Med	1.212	1.211	1.206	1.207	1.188	1.210	1.206	1.219	1.200	1.186	1.197	1.183	1.112	1.157	1.160	1.194	1.182	1.125	1.100	1.103
RC	5%		17.39	13.04	4.35	8.70	8.70	8.70	8.70	4.35	4.35	8.70	4.35	13.04	60.87	21.74	17.39	0.00	8.70	56.52	78.26	82.61
	10%		52.17	43.48	34.78	34.78	39.13	56.52	43.48	34.78	34.78	34.78	39.13	39.13	78.26	43.48	47.83	39.13	47.83	78.26	86.96	86.96
QLIKE		Avg	7.376	8.302	9.870	7.225	6.146	8.404	8.752	8.374	6.213	5.228	6.051	5.040	1.906	3.517	3.612	6.988	4.368	2.002	1.703	1.726
		Med	6.267	7.172	10.009	6.506	5.645	7.196	8.361	8.571	5.880	5.021	5.568	4.307	1.882	3.109	3.257	6.406	4.099	1.940	1.723	1.750
RC	5%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.35	0.00	8.70	0.00	0.00	0.00	0.00	13.04	13.04	30.44
	10%		8.70	0.00	4.35	4.35	0.00	8.70	0.00	0.00	0.00	0.00	8.70	0.00	13.04	4.35	4.35	4.35	0.00	13.04	30.44	47.83
MDA		Avg	1.067	1.068	1.067	1.067	1.066	1.067	1.068	1.068	1.066	1.065	1.066	1.066	1.044	1.067	1.065	1.064	1.064	1.047	1.025	1.016
		Med	1.058	1.058	1.058	1.058	1.058	1.058	1.058	1.058	1.058	1.058	1.058	1.058	1.049	1.058	1.058	1.058	1.058	1.049	1.022	1.000
RC	5%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.35	0.00	0.00	0.00	0.00	13.04	17.39	34.78
	10%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.35	0.00	0.00	0.00	0.00	13.04	17.39	39.13

Notes: The results present the out-of-sample volatility forecasting performance of FCNN models. The best value in each row is marked in bold.

^a The mean (or median) MSE ratio of ML to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement.

^b The mean (or median) QLIKE ratio of ML to CHAR for 23 tickers.

A ratio above one indicates degradation, while below one indicates improvement.

^c The mean (or median) MDA ratio of CHAR to ML for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement.

^d Percentage of tickers with the outstanding performance of FCNN against the HAR-family of models at the 5% and 10% significance levels.

cult to forecast even for HAR-family of models. This phenomenon may help explain part of the mixed evidence in the literature regarding the relative outperformance of ML models over HAR-family benchmarks, as our results show that changes in ML architecture can lead to improved forecasting performance during high volatility days relative to the LSTM.

7.6 Overfitting

To further strengthen our conclusions regarding overfitting, we carry out additional tests beyond those already presented. Overfitting and two approaches for addressing it are discussed in Subsection 4.1. Additionally, as outlined in Section 5, we incorporated MDA as a complementary performance metric alongside MSE and QLIKE. This addition provides a more nuanced evaluation of model performance, particularly in terms of capturing the directional dynamics of RV. Our findings consistently indicate that although ML models may not outperform benchmark models in terms of MSE and QLIKE, especially on days with high volatility, they exhibit notable improvements in MDA. This suggests that the ML models are effectively learning the underlying structure of RV. Furthermore, in Subsection 7.3, we expanded the model’s hyperparameter space by varying both the number of training epochs and units. Across these extended settings, we observed consistent results, even for the simplest ML models and the smallest number of epochs. Comparable outcomes were also obtained when the model architecture was entirely changed from LSTM to FCNN, as shown in Subsection 7.5, further reinforcing the conclusion that overfitting is unlikely to be a significant issue.

We now introduce an additional safeguard against overfitting, complementing the existing use of L^2 regularisation and dropout described in Subsection 4.1. Specifically, we reserve the last 20% of the training data from previous experiments as validation data. Following common practice, we apply early stopping with a patience of 7 epochs and restore the model weights from the epoch with the best validation performance. The early stopping range is consistent with that used in Subsection 7.3, spanning from 25 to 125 epochs. Accordingly, we re-train all models presented in Section 5. Figure 16 presents the results across all ML groups, including HAR, News, OB, and the combined News/OB. This figure is directly comparable to the ML section of Figure 4. Overall, the findings are consistent with those discussed in Section 5 for models without early stopping. Specifically, all models exhibit a consistent pattern: improved performance during periods of normal volatility, but deteriorated performance on high volatility days. As before, the

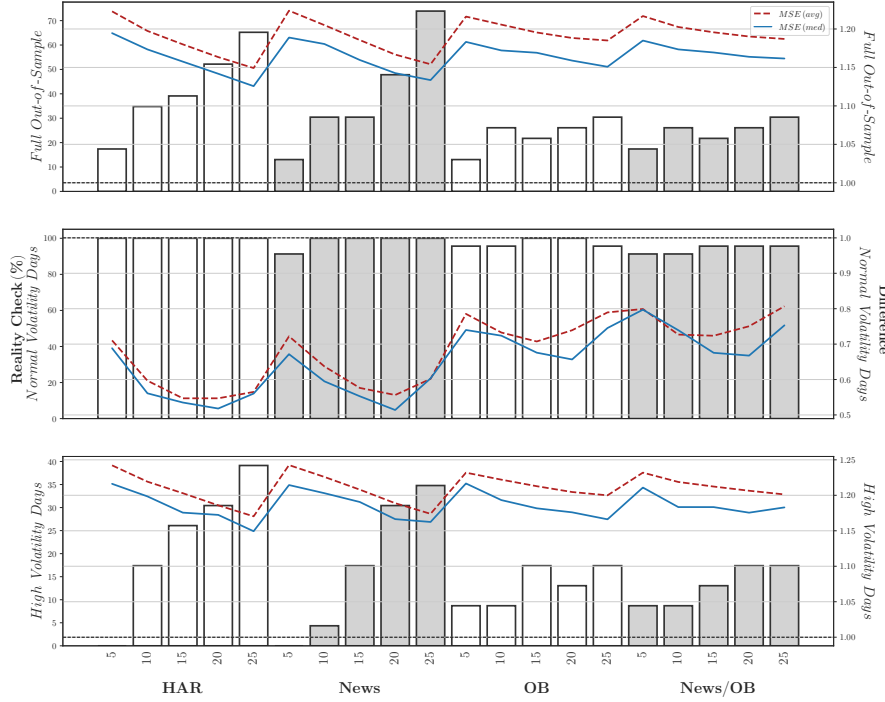
behaviour observed during the full out-of-sample period closely mirrors the pattern seen during high volatility days. The MDA results shown in Figure C11 are also consistent with our earlier findings, supporting the same conclusions as before.

To facilitate a more detailed comparison, Table 10 reports the differences between the results presented in Table 6 and Table 7 and the models with early stopping. Positive values indicate superior performance of the models with early stopping and are highlighted in bold. The primary finding is that the application of early stopping generally results in a decline in forecasting performance for both the OB-ML and News/OB-ML model groups across the majority of tests. Although slight performance improvements are occasionally observed when early stopping is applied to the HAR-ML and News-ML groups, these gains are limited in both frequency and magnitude. Overall, the evidence further substantiates our earlier conclusion that overfitting is not a primary factor underlying the observed results.

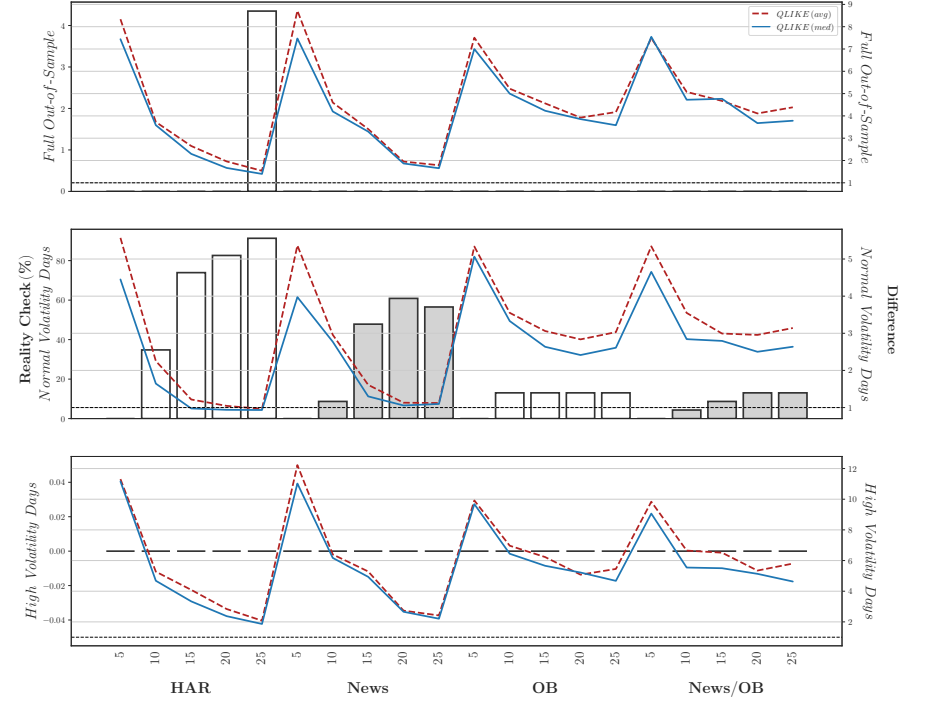
8 A Reflection on the Practical Usability of ML Models

All main tests in Section 5, as well as the robustness tests in Section 7, which consider different ML model types, hyperparameters, and information groups, lead to the same conclusion. ML models significantly improve RV forecasting performance on normal volatility days. However, they tend to degrade forecasting performance during high volatility days, which can adversely affect performance over the full out-of-sample period. As highlighted in Subsection 5.2, the stark contrast in model performance between normal and high volatility days suggests that ML models trained on full in-sample periods, which are dominated by normal volatility observations, are poorly suited for forecasting rare high volatility days. This distinction is further supported by our analyses of individual tickers in Subsection 5.4, which indicate that different modelling strategies are necessary for normal versus high volatility days. This raises the key question of whether to use HAR-family of models or ML models, and more importantly, how to effectively leverage the strengths of ML models for forecasting RV.

To address this, we propose a straightforward ensemble model that combines forecasts from both the HAR-family of models and ML models by averaging their predicted values. This simple ensemble method can help moderate model behaviour under varying market conditions, leveraging ML models during periods of normal volatility and relying on linear models during episodes of high



(a) Forecast Evaluation under MSE



(b) Forecast Evaluation under QLIKE

Notes: The bar chart is the percentage of tickers with the outstanding performance considering the MSE loss function in Figure 16a and the QLIKE loss function in Figure 16b at the 5% significance level of the RC compared to the HAR-family of models as the benchmark. The results are presented for each model group: HAR, News, OB, and News/OB, across varying numbers of units, specifically 5, 10, 15, 20, and 25 units. The values for the bar chart can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MSEs and QLIKEs of the ML model relative to the CHAR model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

Figure 16: ML Models Utilising Early Stopping

Table 10: Out-of-Sample Volatility Forecasting Performance of ML Models Utilising Early Stopping
14 September 2015 to 27 January 2022 (1,604 days)

			HAR-ML					News-ML					OB-ML					News/OB-ML				
Units			5	10	15	20	25	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25
Full Out-of-Sample																						
MSE	Avg	Med	0.002	0.001	0.002	-0.002	-0.003	0.006	-0.001	-0.003	-0.001	-0.004	-0.007	-0.027	-0.039	-0.044	-0.011	-0.001	-0.024	-0.039	-0.040	-0.049
			0.004	0.007	0.002	-0.003	-0.006	0.009	-0.001	0.000	-0.004	-0.011	-0.005	0.002	-0.033	-0.033	0.000	0.001	-0.019	-0.024	-0.017	-0.038
			4.35	0.00	0.00	-4.35	-8.70	-4.35	4.35	-8.70	-4.35	0.00	-13.04	-13.04	-34.78	-47.83	-47.83	0.00	-13.04	-26.09	-43.48	-47.83
RC	5%	10%	0.00	0.00	-8.70	8.70	-4.35	0.00	0.00	-4.35	4.35	0.00	-4.35	-4.35	-21.74	-30.44	-21.74	-13.04	0.00	-21.74	-21.74	-26.09
			-0.256	0.154	-0.133	-0.217	-0.147	-0.931	-0.339	-0.811	0.053	-0.262	-2.693	-2.884	-2.906	-2.399	-2.090	-1.310	-2.866	-2.993	-2.474	-2.930
			-0.562	-0.263	-0.112	-0.098	-0.093	0.125	-0.045	-0.822	-0.124	-0.301	-3.820	-3.001	-2.727	-2.415	-1.659	-2.406	-2.845	-3.238	-2.202	-2.397
QLIKE	Avg	Med	0.00	0.00	0.00	0.00	-8.70	0.00	0.00	0.00	-4.35	-8.70	0.00	-4.35	0.00	-8.70	-13.04	0.00	0.00	0.00	0.00	-8.70
			0.00	0.00	4.35	-8.70	-26.09	0.00	0.00	-4.35	0.00	-26.09	-4.35	-8.70	-13.04	-17.39	-21.74	0.00	-8.70	-13.04	-21.74	-17.39
			-0.016	0.011	0.005	0.005	0.005	-0.005	0.008	0.006	0.002	-0.001	-0.063	-0.058	-0.056	-0.047	-0.025	-0.036	-0.058	-0.058	-0.050	-0.043
MDA	Avg	Med	-0.009	0.017	0.007	0.002	0.004	-0.002	0.006	0.006	0.005	-0.005	-0.073	-0.064	-0.056	-0.041	-0.022	-0.061	-0.058	-0.069	-0.045	-0.040
			-8.70	0.00	0.00	0.00	0.00	0.00	4.35	-4.35	4.35	0.00	-30.43	-21.74	-26.09	-8.70	-13.04	-17.39	-21.74	-21.74	-26.09	-26.09
			-4.35	-4.35	0.00	0.00	0.00	4.35	0.00	4.35	0.00	0.00	-30.43	-21.74	-21.74	-8.70	-13.04	-17.39	-21.74	-13.04	-13.04	-21.74
Normal Volatility Days																						
MSE	Avg	Med	-0.011	0.021	0.026	0.024	0.019	-0.018	0.021	0.029	0.043	0.026	-0.245	-0.265	-0.233	-0.213	-0.235	-0.188	-0.260	-0.230	-0.234	-0.229
			-0.019	0.030	0.043	0.069	0.002	-0.032	0.040	0.045	0.077	0.052	-0.261	-0.338	-0.243	-0.185	-0.170	-0.305	-0.303	-0.247	-0.180	-0.210
			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-4.35	-4.35	0.00	0.00	-4.35	-8.70	-8.70	-4.35	-4.35
RC	5%	10%	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-4.35	-4.35	0.00	0.00	-4.35	-4.35	-4.35	0.00	0.00	-4.35
			-0.799	-0.121	0.046	-0.005	-0.038	-0.547	-0.276	-0.113	0.061	-0.086	-2.636	-2.397	-2.136	-1.885	-1.932	-1.394	-2.448	-2.040	-1.927	-2.166
			-0.729	-0.161	0.026	0.046	-0.006	0.124	-0.754	0.061	0.034	-0.087	-3.540	-2.458	-1.744	-1.465	-1.601	-2.324	-1.929	-1.889	-1.581	-1.661
QLIKE	Avg	Med	-4.35	4.35	8.70	4.35	-8.70	0.00	4.35	8.70	-4.35	-17.39	-39.13	-52.17	-73.91	-69.57	-69.57	-21.74	-60.87	-69.57	-73.91	-69.57
			-4.35	0.00	4.35	0.00	0.00	0.00	13.04	8.70	-13.04	-13.04	-39.13	-47.83	-73.91	-69.57	-69.57	-21.74	-65.22	-73.91	-78.26	-73.91
			-0.015	0.012	0.007	0.008	0.010	-0.005	0.010	0.008	0.005	0.003	-0.068	-0.056	-0.049	-0.035	-0.022	-0.044	-0.057	-0.050	-0.044	-0.030
MDA	Avg	Med	-0.006	0.019	0.016	0.013	0.011	-0.005	0.001	0.010	0.006	-0.004	-0.075	-0.060	-0.041	-0.041	-0.023	-0.072	-0.057	-0.057	-0.034	-0.027
			-8.70	4.35	0.00	0.00	0.00	0.00	4.35	0.00	0.00	0.00	-17.39	-4.35	-4.35	0.00	0.00	-8.70	-8.70	0.00	0.00	0.00
			-8.70	4.35	0.00	0.00	0.00	-4.35	4.35	0.00	0.00	0.00	-13.04	0.00	0.00	0.00	0.00	-8.70	0.00	0.00	0.00	0.00
High Volatility Days																						
MSE	Avg	Med	0.003	0.001	0.001	-0.004	-0.004	0.007	-0.002	-0.004	-0.003	-0.005	0.001	-0.017	-0.031	-0.037	-0.004	0.007	-0.014	-0.031	-0.032	-0.042
			0.002	0.000	0.007	-0.003	-0.008	0.007	0.001	-0.005	-0.004	-0.012	-0.020	0.006	-0.018	-0.024	0.003	-0.001	-0.001	-0.011	-0.008	-0.038
			17.39	13.04	4.35	8.70	8.70	8.70	8.70	4.35	4.35	8.70	4.35	13.04	60.87	21.74	17.39	0.00	8.70	56.52	78.26	82.61
RC	5%	10%	52.17	43.48	34.78	34.78	39.13	56.52	43.48	34.78	34.78	34.78	39.13	39.13	78.26	43.48	47.83	39.13	47.83	78.26	86.96	86.96
			0.114	0.338	-0.328	-0.411	-0.235	-1.198	-0.490	-1.560	0.086	-0.413	-2.956	-3.456	-3.844	-2.971	-2.413	-1.325	-3.344	-4.105	-3.118	-3.855
			-2.164	-0.003	0.011	-0.329	-0.169	0.247	0.061	-1.678	-0.232	-0.397	-4.719	-3.348	-3.417	-3.227	-1.818	-2.987	-2.240	-3.173	-3.088	-2.867
QLIKE	Avg	Med	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-4.35	0.00	0.00	0.00	-4.35	
			0.00	-4.35	0.00	4.35	0.00	0.00	0.00	0.00	-4.35	-13.04	-4.35	-4.35	0.00	-4.35	-4.35	-4.35	0.00	0.00	-4.35	-4.35
			0.001	0.000	-0.004	-0.005	-0.003	0.001	-0.002	-0.002	-0.005	-0.009	0.000	-0.002	-0.002	-0.007	-0.006	0.001	-0.001	-0.005	-0.009	-0.016
MDA	Avg	Med	0.000	0.000	0.000	-0.005	-0.008	0.000	0.000	0.000	-0.002	-0.004	0.000	0.000	-0.004	-0.004	-0.008	0.000	0.000	-0.004	-0.002	-0.008
			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-4.35
			0.00	0.00	0.00	0.00	-4.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Notes: The results report the difference in out-of-sample RV forecasting performance between the primary trained models in Section 5 and those trained with early stopping. Positive values indicate superior performance of the models with early stopping and are highlighted in bold. It should be noted that the presentation of values in this table differs from that in previous tables to facilitate interpretation and enhance the clarity of the comparison.

volatility. To further refine the ensemble, a weighted averaging scheme may be employed, allowing for differential weighting across model groups. These weights can be optimised using training data to enhance forecasting performance. However, for illustrative purposes, we adopt a conservative weighting strategy. Specifically, we assign a weight of 0.75 to the best-performing HAR-family model (CHAR) and 0.25 to the ML models, thereby favouring the more conventional model in the ensemble. This approach also holds practical value: it allows practitioners to leverage the predictive power of ML models without fully replacing conventional models, thereby benefiting from both ML and HAR models.

Table 11 reports the results of the ensemble models. These findings are directly comparable to those presented for the full out-of-sample period in Table 6, as well as for normal and high volatility days in Table 7. A comparison of ensemble models with the ML models used in our primary experiment clearly indicates that the ensemble models outperform individual ML approaches. Focusing on the full out-of-sample period, we observe that for nearly all models, the RC reaches or approaches 100%. Additionally, several ML models demonstrate modest improvements in MSE and QLIKE, with MDA improving by approximately 3%. On normal volatility days, as expected, we observe substantial improvements not only in RC but also in MSE, QLIKE, and MDA metrics. On average, MSE improves by around 30% and QLIKE by approximately 10% compared to the CHAR benchmark model, while MDA shows a 4% improvement. During high volatility days, improvements are more modest but still evident in several cases. Even in scenarios where no improvement is observed, the degradation in average and median values is significantly less pronounced compared to the results reported in Table 7. As in previous results, the OB-ML and News/OB-ML groups emerge as the best-performing models over the full out-of-sample period.

As outlined earlier, the approach employed here involves a simple ensemble model with pre-defined weights. While more optimal results can be achieved by calibrating these weights using training data on a rolling basis, even this basic implementation demonstrates that ML models can enhance forecasting performance alongside HAR-family of models. However, the findings also underscore an important caveat: exclusive reliance on ML models may be detrimental in certain scenarios, particularly during periods of high volatility. Therefore, any proposed switch between model classes should be supported by rigorous empirical evidence demonstrating consistent performance across both stable and turbulent market conditions. In the absence of such

Table 11: Out-of-Sample Volatility Forecasting Performance of Ensemble Models
14 September 2015 to 27 January 2022 (1,604 days)

			HAR-ML					News-ML					OB-ML					News/OB-ML				
Units			5	10	15	20	25	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25
Full Out-of-Sample																						
MSE ^a	Avg	Med	0.995	0.993	0.991	0.990	0.988	0.995	0.993	0.991	0.989	0.988	0.993	0.991	0.989	0.987	0.987	0.993	0.990	0.989	0.988	0.987
			1.000	1.000	0.998	0.997	0.995	1.000	1.000	0.996	0.996	0.994	0.997	0.998	0.995	0.994	0.993	0.997	0.998	0.995	0.994	0.993
RC ^d	5%	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	
		10%	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	
QLIKE ^b	Avg	Med	1.054	1.032	1.023	1.013	1.008	1.054	1.030	1.012	1.004	0.996	1.026	1.011	1.001	0.994	0.992	1.028	1.009	1.002	0.994	0.991
			1.048	1.025	1.019	1.010	1.004	1.050	1.022	1.011	1.003	0.998	1.025	1.009	1.001	0.994	0.991	1.024	1.006	1.003	0.995	0.993
RC	5%	43.48	60.87	65.22	78.26	95.65	47.83	69.57	78.26	86.96	86.96	65.22	78.26	91.30	100	100	60.87	73.91	82.61	86.96	86.96	
		10%	60.87	69.57	73.91	82.61	95.65	56.52	73.91	82.61	86.96	86.96	78.26	86.96	100	100	100	69.57	78.26	82.61	86.96	86.96
MDA ^c	Avg	Med	0.964	0.972	0.979	0.984	0.988	0.964	0.971	0.976	0.983	0.985	0.964	0.973	0.978	0.982	0.986	0.963	0.973	0.979	0.980	0.984
			0.961	0.974	0.979	0.983	0.988	0.966	0.971	0.976	0.984	0.987	0.967	0.975	0.978	0.981	0.988	0.964	0.974	0.979	0.980	0.983
RC	5%	100	100	100	95	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
		10%	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
Normal Volatility Days																						
MSE	Avg	Med	0.635	0.693	0.731	0.768	0.801	0.627	0.678	0.718	0.751	0.786	0.629	0.674	0.720	0.748	0.775	0.614	0.671	0.722	0.732	0.768
			0.644	0.697	0.726	0.763	0.816	0.635	0.677	0.715	0.741	0.783	0.620	0.669	0.733	0.752	0.775	0.615	0.662	0.714	0.741	0.765
RC	5%	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	
		10%	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
QLIKE	Avg	Med	0.865	0.906	0.933	0.948	0.966	0.859	0.894	0.918	0.935	0.954	0.845	0.874	0.903	0.913	0.930	0.832	0.874	0.904	0.907	0.925
			0.874	0.911	0.933	0.950	0.971	0.864	0.892	0.916	0.931	0.952	0.840	0.875	0.905	0.913	0.923	0.820	0.865	0.903	0.907	0.918
RC	5%	100	100	100	100	95.65	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
		10%	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
MDA	Avg	Med	0.943	0.957	0.968	0.977	0.985	0.942	0.953	0.964	0.974	0.980	0.947	0.961	0.969	0.976	0.982	0.944	0.960	0.971	0.973	0.980
			0.943	0.960	0.970	0.978	0.985	0.945	0.955	0.965	0.975	0.982	0.950	0.961	0.971	0.976	0.983	0.945	0.958	0.971	0.974	0.981
RC	5%	100	100	95.65	95.65	73.91	100	100	100	100	91.30	100	100	95.65	95.65	86.96	100	100	100	100	95.65	
		10%	100	100	95.65	95.65	91.30	100	100	100	100	95.65	100	100	95.65	95.65	91.30	100	100	100	100	95.65
High Volatility Days																						
MSE	Avg	Med	1.009	1.004	1.001	0.998	0.995	1.009	1.005	1.001	0.998	0.995	1.007	1.002	0.999	0.996	0.995	1.008	1.002	0.999	0.997	0.995
			1.017	1.012	1.007	1.004	1.002	1.016	1.011	1.006	1.004	1.002	1.014	1.010	1.006	1.004	1.002	1.014	1.010	1.006	1.004	1.002
RC	5%	56.52	60.87	65.22	82.61	91.30	56.52	60.87	73.91	86.96	86.96	56.52	65.22	65.22	82.61	86.96	56.52	60.87	69.57	78.26	82.61	
		10%	65.22	78.26	86.96	91.30	95.65	65.22	78.26	86.96	86.96	86.96	69.57	78.26	86.96	95.65	95.65	65.22	65.22	86.96	86.96	86.96
QLIKE	Avg	Med	1.237	1.154	1.106	1.074	1.046	1.242	1.162	1.102	1.069	1.035	1.206	1.147	1.097	1.075	1.054	1.221	1.142	1.098	1.079	1.058
			1.237	1.155	1.100	1.064	1.043	1.235	1.152	1.095	1.067	1.030	1.208	1.154	1.099	1.069	1.055	1.224	1.153	1.105	1.068	1.054
RC	5%	0.00	0.00	0.00	0.00	17.39	0.00	0.00	0.00	13.04	39.13	0.00	0.00	4.35	21.74	43.48	0.00	0.00	4.35	21.74	30.44	
		10%	0.00	0.00	0.00	0.00	21.74	0.00	0.00	13.04	21.74	52.17	0.00	0.00	8.70	34.78	43.48	0.00	4.35	8.70	26.09	30.44
MDA	Avg	Med	1.039	1.032	1.025	1.022	1.019	1.037	1.036	1.027	1.023	1.023	1.034	1.027	1.020	1.018	1.015	1.036	1.025	1.021	1.020	1.016
			1.037	1.027	1.022	1.013	1.022	1.034	1.034	1.034	1.024	1.019	1.034	1.022	1.013	1.013	1.013	1.037	1.022	1.022	1.013	1.014
RC	5%	8.70	13.04	26.09	30.44	60.87	13.04	13.04	21.74	21.74	39.13	17.39	13.04	34.78	52.17	65.22	13.04	13.04	30.44	39.13	52.17	
		10%	13.04	21.74	26.09	47.83	69.57	13.04	13.04	21.74	34.78	60.87	21.74	21.74	47.83	56.52	73.91	17.39	17.39	39.13	52.17	56.52

Notes: The results present the out-of-sample volatility forecasting performance of ensemble models. These ensemble models are constructed as weighted averages, combining the CHAR model, identified as the best-performing model within the HAR-family of models, with each ML model evaluated in Section 5. The ensemble assigns weights of 0.75 to the CHAR model and 0.25 to the respective ML model.

^a The mean (or median) MSE ratio of ensemble model to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^b The mean (or median) QLIKE ratio of ensemble model to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^c The mean (or median) MDA ratio of CHAR to ensemble model for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^d Percentage of tickers with the outstanding performance of ensemble model against the HAR-family of models at the 5% and 10% significance levels.

robustness, ensemble models offer a promising alternative by combining the respective strengths of each modelling approach.

9 Discussions and Conclusions

This study evaluates the performance of ML models, combined with a comprehensive set of features, for RV forecasting from 27 July 2007 to 27 January 2022. Three types of daily data are utilised: six HAR variables, 132 LOB variables, and nine news sentiment variables. The full sample period is divided into an in-sample training period, from 27 July 2007 to 11 September 2015, and an out-of-sample forecasting period, from 14 September 2015 to 27 January 2022. A diverse set of neural network-based ML models, incorporating different information sets and architectural designs, is used to train and evaluate over seven million model variants.

Empirical results show that ML models outperformed all HAR-family of models, and that LOB variables, relative to news sentiments and HAR variables, emerged as the most informative predictors for forecasting RV. However, this statistically significant improvement applies only to 90% of out-of-sample daily forecasts when the actual RV is not extremely high. For a small number of tickers, and for the remaining 10% of the out-of-sample forecasts when the actual RV is extremely high, HAR-family of models outperformed ML models. The underperformance of ML models during periods of high volatility affected their overall out-of-sample performance, suggesting that ML models should not be viewed as direct substitutes for conventional HAR-family of models. Our findings remain qualitatively the same when the forecasts are evaluated using MSE, QLIKE, MDA, or RC. Also, our findings emphasise not only the importance of capturing nonlinear relationships in the RV dynamics but also the importance of including a rich collection of predictors in a forecasting ML model to achieve better performance when using ML models.

The ML model is a black box, giving very little clue about which predictors contributed to good forecasting performance. Here, we implemented SHAP, an XAI technique, to identify the predictors that provided the greatest forecasting power. Results from SHAP suggest that, among the 147 input variables, mid prices at all LOB levels, mean bid and mean ask from the LOB, BPV from HAR, news count, and the ‘uncertainty’ sentiment are identified as the most important predictive variables for forecasting RV. In general, LOB variables have stronger predictive power

than HAR and news sentiment variables. There are also some performance variations across time. Notably, HAR variables, while generally of low importance, became highly informative during periods of extreme volatility, such as the COVID-19 crisis. Conversely, in more stable market conditions (pre-2018), news sentiment played a dominant role, while LOB variables gained importance post-2018 amid increased volatility. Once again, these changes over time highlight the importance of incorporating a broad and diverse range of predictors in ML models for RV forecasting. It also reinforces the important role of the HAR-family of models, and particularly their variables, in achieving accurate RV forecasting, especially on high volatility days.

To further validate the reliability of our findings, we conducted extensive robustness checks, including variations in model type, input variable selection, alternative loss functions, and measures to mitigate overfitting. These tests consistently confirmed the superiority of ML models over the HAR-family of models during normal volatility days, while highlighting their relative vulnerability during high volatility days. In response to this, we proposed a simple yet effective ensemble approach that combines forecasts from both ML and HAR-family of models. This ensemble method leverages the strengths of each model class, capturing nonlinear dynamics via ML while maintaining robustness during turbulent market periods through HAR, resulting in more stable and accurate forecasts across a wide range of market conditions. The success of the ensemble underscores the value of integrating diverse modelling approaches in volatility forecasting rather than relying exclusively on either traditional econometric methods or ML. Therefore, we stress that replacing conventional HAR-family of models with ML models can be dangerous in practice, and emphasise that this should be done with caution, where ensemble methods offer a simple alternative.

To summarise, our findings indicate that neither ML models nor HAR-family specifications exhibit uniform dominance in RV forecasting. Instead, relative performance depends critically on sample composition, prevailing market conditions, modelling design, and the choice of ML architecture. ML models exploit their flexibility and richer information sets to deliver superior performance primarily during normal volatility days; however, these gains are sensitive to model specification, hyperparameter tuning, and architectural choices. In contrast, HAR-family of models provide more stable and robust forecasts, particularly during high volatility days that are predominantly driven by firm-specific shocks and, to a lesser extent, by broader market-wide stress. This distinction helps explain the divergent conclusions in the literature regarding whether

ML models can serve as direct replacements for HAR-type benchmarks when these are implemented across different asset universes and sample timeframes. Overall, these results highlight the importance of regime awareness and careful model integration when developing and evaluating RV forecasting frameworks.

This study provides a new understanding of the complexities behind developing ML models for volatility forecasting. The tuning of the ML hyperparameters, the choice of the loss function, and the information content of a large volume of input variables can all affect the forecasting performance conditioned on the level of actual RV. This study is among the first comprehensive assessments to employ a rich feature set for RV forecasting and lays the foundation for future ML research in this area. Future research could extend the forecast evaluation to include multi-step-ahead RV predictions. Another promising direction is the implementation of switching methods to adjust models in response to evolving market conditions. Additionally, tree-based models such as XGBoost (Chen and Guestrin, 2016) and LGBM (Ke et al., 2017) offer valuable opportunities for further investigation. The application of Transformer-based architectures, which have recently gained significant attention (Vaswani et al., 2017), also merits further exploration. Finally, developing a universal model for RV forecasting by leveraging comprehensive market information represents an exciting avenue for future research.

A LOBSTER Dataset and Data Cleaning

The LOBSTER dataset is used to extract the HAR variables described in Subsection 3.1 and the LOB variables described in Subsection 3.3. Prior to computing these variables, the LOBSTER dataset is preprocessed using a set of modified cleaning steps based on Barndorff-Nielsen et al. (2009). These steps are applied to both the LOB and message data in the following way (the step names, such as P2, T4, ..., are consistent with the names in Barndorff-Nielsen et al. (2009)):

- **P2:** Delete entries with a bid, ask or transaction price equal to zero.
- **T4:** Delete entries with prices that are above the ‘ask’ plus the bid-ask spread, or below the ‘bid’ minus the bid-ask spread.
- **Q1:** When multiple quotes share the same timestamp, they are replaced by a single entry using the median bid price, median ask price, and the sum of all volumes from these multiple quotes. For messages with the same direction (buy or sell), the mentioned procedure is applied to the message data, and the last snapshot of the LOB is selected as the LOB associated with the merged message data. For messages with different directions, the message data and the LOB with the same direction are grouped, and the mentioned procedure is applied separately to the buy-side and sell-side.
- **Q2:** Delete entries for which the spread is negative.
- **Q3:** Delete entries for which the spread is more than 50 times the median spread on that day.
- **Q4:** Delete entries for which the mid-quote deviated by more than 10 mean absolute deviations from a rolling centred median (excluding the observation under consideration) of 50 observations (25 observations before and 25 after).

The summary statistics pertaining to the data cleaning steps are presented in Table A1.

Table A1: Data Cleaning Summary Statistics
Sample Period From 27 July 2007 to 27 January 2022

Name	Ticker	Sample Size	Removed (%)	P2 (%)	T4 (%)	Q1 (%)	Q2 (%)	Q3 (%)	Q4 (%)
Apple	AAPL	4,174,971,328	34.22	0.00	29.88	4.34	0.00	0.00	0.00
Microsoft	MSFT	3,827,824,574	35.88	0.01	30.18	5.69	0.00	0.00	0.00
Intel	INTC	2,807,965,330	38.59	0.01	31.79	6.78	0.01	0.00	0.01
Comcast	CMCSA	2,390,133,817	45.18	0.01	39.59	5.58	0.00	0.00	0.01
Qualcomm	QCOM	2,086,295,132	41.46	0.00	36.46	4.98	0.00	0.00	0.01
Cisco Systems	CSCO	2,296,179,428	40.46	0.01	33.50	6.94	0.00	0.00	0.01
eBay	EBAY	1,683,001,942	40.73	0.01	35.68	5.03	0.00	0.00	0.01
Gilead Sciences	GILD	1,404,574,567	41.68	0.00	38.41	3.25	0.00	0.00	0.01
Texas Instruments	TXN	1,485,049,597	39.45	0.00	35.14	4.29	0.00	0.00	0.01
Amazon.com	AMZN	1,201,210,867	23.06	0.00	19.89	3.15	0.00	0.00	0.02
Starbucks	SBUX	1,564,221,129	44.04	0.01	39.95	4.07	0.00	0.00	0.01
Nvidia	NVDA	1,548,447,223	35.47	0.01	30.40	5.05	0.00	0.00	0.01
Micron Technology	MU	2,110,482,619	35.99	0.00	31.11	4.86	0.00	0.00	0.01
Applied Materials	AMAT	1,616,466,522	39.70	0.01	34.41	5.27	0.00	0.00	0.01
NetApp	NTAP	1,015,914,054	44.99	0.01	41.14	3.82	0.00	0.00	0.02
Adobe	ADBE	1,083,392,595	37.76	0.01	34.35	3.39	0.00	0.00	0.02
Xilinx	XLNX	1,172,584,895	40.24	0.01	36.97	3.26	0.00	0.00	0.02
Amgen	AMGN	863,464,001	38.62	0.01	34.73	3.86	0.00	0.00	0.02
Vodafone Group	VOD	1,012,861,232	47.20	0.01	44.23	2.95	0.00	0.00	0.02
Cognizant	CTSH	928,987,253	46.22	0.01	43.28	2.91	0.00	0.00	0.02
KLA Corporation	KLAC	783,931,409	42.63	0.01	39.83	2.77	0.00	0.00	0.02
Paccar	PCAR	775,954,122	45.74	0.01	42.98	2.73	0.00	0.00	0.03
Autodesk	ADSK	803,552,017	41.73	0.01	38.96	2.74	0.00	0.00	0.02
Average			40.05	0.01	35.78	4.25	0.00	0.00	0.01

Notes: **P2:** Delete entries with a bid, ask or transaction price equal to zero, **T4:** Delete entries with prices that are above the ‘ask’ plus the bid-ask spread, or below the ‘bid’ minus the bid-ask spread, **Q1:** When multiple quotes share the same timestamp, they are replaced by a single entry using the median bid price, median ask price, the sum of all volumes, and the last snapshot of the LOB is selected as the LOB associated with the merged message data. For messages with different directions (buy or sell), the message data and the LOB with the same direction are grouped according to the buy side or sell side. The procedure mentioned above is then applied to the message data, and the last snapshot of the LOB of the group, **Q2:** Delete entries for which the spread is negative, **Q3:** Delete entries for which the spread is more than 50 times the median spread on that day, **Q4:** Delete entries for which the mid-quote deviated by more than 10 mean absolute deviations from a rolling centred median (excluding the observation under consideration) of 50 observations (25 observations before and 25 after).

Table A2: RV Descriptive Statistics (Without Cleaning)
Sample Period From 27 July 2007 to 27 January 2022

Ticker	Min	Max	1 st quantile	Median	3 rd quantile	Mean	STD	Kurtosis	Skewness	Correlation ^a
AAPL	0.101	229.529	0.898	1.737	3.702	4.623	12.579	110.333	9.093	0.9999
MSFT	0.096	216.486	0.828	1.458	2.810	3.240	8.119	194.967	11.294	0.9997
INTC	0.030	318.118	1.099	1.876	3.592	4.300	11.615	295.828	14.000	0.9999
CMCSA	0.006	237.387	0.913	1.631	3.344	3.833	9.773	190.221	11.459	0.9994
QCOM	0.123	368.449	1.025	1.980	4.140	5.076	15.363	197.106	12.012	0.9999
CSCO	0.038	343.946	0.884	1.564	3.031	4.117	13.170	213.266	12.287	0.9999
EBAY	0.215	259.723	1.328	2.263	4.361	5.111	12.745	142.560	10.028	0.9974
GILD	0.063	261.664	1.170	1.895	3.375	4.312	12.933	184.066	12.094	0.9999
TXN	0.183	289.765	1.046	1.895	3.713	4.006	9.928	310.664	14.275	0.9986
AMZN	0.066	551.566	1.307	2.342	4.833	6.203	19.355	246.159	12.802	0.9998
SBUX	0.048	265.554	0.864	1.594	3.441	4.209	11.227	161.691	10.615	0.9997
NVDA	0.159	1104.483	2.280	4.342	9.098	9.760	30.112	586.751	20.055	1.0000
MU	0.288	484.388	3.575	6.281	11.964	12.821	25.726	89.478	7.966	0.9990
AMAT	0.312	529.508	1.773	3.031	5.730	6.014	14.615	526.901	18.237	0.9999
NTAP	0.114	463.545	1.508	2.606	5.180	6.301	18.020	201.869	11.942	0.9998
ADBE	0.120	575.498	1.096	2.001	3.874	4.810	14.784	661.002	20.309	0.9998
XLNX	0.231	265.372	1.300	2.374	4.791	5.022	11.950	193.951	11.739	0.9990
AMGN	0.039	212.485	0.962	1.580	2.860	3.311	9.225	202.622	12.391	0.9995
VOD	0.043	217.091	0.684	1.334	3.081	3.693	9.500	149.567	10.122	0.9998
CTSH	0.186	493.255	0.979	1.743	4.085	5.218	15.843	339.283	14.667	0.9997
KLAC	0.149	499.806	1.451	2.701	5.412	5.820	16.308	384.071	16.516	0.9933
PCAR	0.029	389.021	1.157	2.172	4.657	5.137	12.123	309.547	12.928	0.9998
ADSK	0.268	696.615	1.642	2.770	5.176	6.660	22.503	389.706	16.604	0.9999

Notes: The descriptive statistics of RV without applying the cleaning steps are presented in this table, while the statistics after applying the cleaning steps are reported in Table 1. ^a Correlation between two sets of calculated RVs, with and without the implementation of the cleaning steps described in Appendix A.

B LSTM as a Unifying Framework

The LSTM architecture can be interpreted as a unifying framework that embeds a wide range of familiar models within a single, general formulation. By imposing simple restrictions on its gating mechanisms and state evolution, the LSTM reduces to well-known specifications, demonstrating that many standard models arise as special cases of its structure. In particular, fixing the gates at constant values yields dynamics equivalent to familiar linear and nonlinear models. Assume the forget gate is $G_f = \phi$, the input gate is set to $G_i = 1 - \phi$, and the output gate is fully open, $G_o = 1$. Additionally, linearise the cell activation function by replacing $\tanh(\cdot)$ in Equation (16) with the identity function. With a constant candidate value $\tilde{c}^t = \bar{c}$, Equation (15) simplifies to:

$$c^t = \phi c^{t-1} + (1 - \phi)\bar{c}, \quad (\text{B1})$$

and, since $a^t = c^t$ by Equation (16), the hidden output satisfies:

$$a^t = \phi a^{t-1} + (1 - \phi)\bar{c}. \quad (\text{B2})$$

Equation (B2) represents an AR(1) model with autoregressive coefficient ϕ and intercept $(1 - \phi)\bar{c}$. In this formulation, the gates serve the role of the autoregressive lag coefficient, while the candidate cell bias determines the long-run mean. As the next case, assume that $G_f = 1$, $G_i = 0$, $G_o = 1$, and, similar to the previous case, replace the $\tanh(\cdot)$ with the identity function. Under these assumptions, Equation (15) and Equation (16) reduce to:

$$c^t = c^{t-1}, \quad a^t = c^t, \quad (\text{B3})$$

which becomes the familiar random walk, $y_t = y_{t-1} + \varepsilon_t$, once an additive noise term ε_t is included.

We now consider a different case. A final limiting configuration shows that the LSTM cell collapses to an ordinary FCNN when the memory pathway is disabled and all gates are fixed. Assume that $G_f = 0$, $G_i = 1$, and $G_o = 1$. Also, for every t , replace the internal nonlinearity $\tanh(\cdot)$ by the identity, and eliminate the recurrent connection by zeroing the part of the candidate-state weight that multiplies the previous hidden state, $w_c = [0, w_x]$. Then, from Equation (11), this

yields:

$$\tilde{c}^t = w_x X^t + b_c. \quad (\text{B4})$$

Because $G_f = 0$ and $G_i = 1$, Equation (15) also reduces to:

$$c^t = \tilde{c}^t = w_x X^t + b_c. \quad (\text{B5})$$

With $G_o = 1$ and a linear output activation, Equation (16) simplifies to:

$$a^t = c^t = w_x X^t + b_c. \quad (\text{B6})$$

Equation (B6) is precisely the affine map produced by a single fully connected layer, and the hidden state no longer depends on past time steps. Stacking several such modified LSTM cells therefore reproduces a standard FCNN. These special cases highlight how LSTM nests a range of models, including AR(1), random walk, and FCNN, through simple gate configurations. This illustrates its flexibility in capturing both dynamic and static relationships within a unified framework.

C Complementary Results

Table C1: Correlation Between RV from LOBSTER and TAQ

Ticker	Correlation	Ticker	Correlation
AAPL	0.8441	MU	0.8402
MSFT	0.9856	AMAT	0.8855
INTC	0.8898	NTAP	0.9674
QCOM	0.9036	ADBE	0.9452
CSCO	0.9786	XLNX	0.9176
EBAY	0.8715	AMGN	0.8353
GILD	0.8310	VOD	0.7188
TXN	0.9785	CTSH	0.9702
AMZN	0.8248	KLAC	0.7689
SBUX	0.7870	PCAR	0.7441
NVDA	0.7870	ADSK	0.8736

Notes: This table reports the correlation between daily RV computed from LOBSTER and RV obtained from TAQ data. The procedures used to construct RV from TAQ data are consistent with the data-cleaning approach in Appendix A and the RV construction methodology in Section 2. For TAQ data, only cleaning steps that do not rely on LOB information are applied. The TAQ data are obtained from WRDS using two products. For 2007–2013 we rely on the trade and quote monthly product, in which trade timestamps are recorded at second level resolution. For 2014–2015 we use the millisecond trade and quote daily product, which provides millisecond timestamps. By contrast, LOBSTER data are recorded at up to nanosecond resolution. These differences in timestamp granularity may affect the statistics reported in this table. Due to limited access to TAQ data, the analysis is restricted to the training period from 27 July 2007 to 11 September 2015. The only missing ticker is CMCSA, which is excluded because it is not available in TAQ for the training period.

Table C2: Parameter Estimates of HAR-family of Models

	AR(1)	HAR	HAR-J	CHAR	SHAR	ARQ	HARQ	HARQ-F
β_0	3.6436 ^a (1.6305) ^b (100, 0) ^c	1.8528 (0.9216) (100, 0)	1.4401 (0.6446) (99.9, 0.1)	1.2273 (0.6357) (98.7, 1.3)	1.8444 (0.9213) (99.9, 0.1)	2.8734 (1.3585) (100, 0)	1.6127 (0.8758) (99.9, 0.1)	0.6807 (0.8382) (77.2, 22.8)
β_1	0.1428 (0.1090) (100, 0)	0.0394 (0.0382) (93.8, 6.2)	0.6544 (0.2743) (100, 0)			0.3653 (0.2028) (100, 0)	0.2136 (0.1093) (100, 0)	0.1695 (0.1193) (99.3, 0.7)
β_1^+					0.0625 (0.0942) (92.8, 7.2)			
β_1^-					0.0346 (0.0390) (88.4, 11.6)			
β^w		0.1471 (0.1805) (77.2, 22.8)	0.0523 (0.1305) (60.2, 39.8)		0.1449 (0.1774) (77.0, 23.0)		0.1123 (0.1568) (72.6, 27.4)	0.2076 (0.1731) (89.6, 10.4)
β^m		0.3566 (0.1754) (100, 0)	0.2510 (0.1562) (97.1, 2.9)		0.3527 (0.1767) (98.7, 1.3)		0.3090 (0.1675) (98.1, 1.9)	0.5367 (0.1956) (100, 0)
β_{jump}			-0.7353 (0.2840) (0, 100)					
β_{BPV}^d				0.4136 (0.2526) (98.3, 1.7)				
β_{BPV}^w				0.3434 (0.2926) (91.2, 8.8)				
β_{BPV}^m				0.4336 (0.3329) (92.5, 7.5)				
β_{1Q}						-0.0004 (0.0004) (0.4, 99.6)	-0.0003 (0.0003) (0.6, 99.4)	-0.0002 (0.0003) (2.0, 98.0)
β_Q^w								-0.0008 (0.0013) (22.0, 78.0)
β_Q^m								-0.0064 (0.0080) (7.4, 92.6)
<i>Adj. R²(avg)</i>	0.0318	0.0721	0.0984	0.1000	0.0731	0.0583	0.0818	0.0871
<i>Adj. R²(med)</i>	0.0104	0.0404	0.0673	0.0692	0.0406	0.0406	0.0505	0.0552
<i>MSE (avg)</i>	173.3761	165.6939	161.0308	160.7043	165.4823	168.4723	163.9610	162.7886
<i>MSE (med)</i>	127.6872	124.0916	120.5695	119.9260	124.0427	125.2517	121.4500	120.8497
<i>QLIKE (avg)</i>	0.6139	0.5619	0.5177	0.5167	0.5615	0.5773	0.5481	0.5599
<i>QLIKE (med)</i>	0.6223	0.5670	0.5187	0.5236	0.5656	0.5813	0.5452	0.5737
<i>MDA (avg)</i>	59.7570	61.5238	60.4922	60.8509	61.4839	59.0413	60.9705	61.5254
<i>MDA (med)</i>	59.6088	61.4181	60.3423	61.0758	61.3203	59.0709	60.9291	61.2225

Notes: We summarise the results by reporting the mean^[a], the standard deviation^[b], and the fractions of positive and negative coefficients^[c]. These statistics are computed from 36,892 daily coefficient estimates, corresponding to 1,604 trading days across 23 stocks. These values are aggregated across all 23 stocks, using either the overall average (or median). The HAR model follows Corsi (2009). HAR-J and CHAR are based on Andersen et al. (2007); Corsi and Reno (2009). SHAR follows Patton and Sheppard (2015). ARQ, HARQ, and HARQ-F follow Bollerslev et al. (2016). RV is computed using 5-minute returns. For each stock, MSE, QLIKE, and MDA are calculated as the average (or median) in-sample errors.

Table C3: HAR-family Out-of-Sample Forecasting Performance

			AR(1)	HAR	HAR-J	CHAR	SHAR	ARQ	HARQ	HARQ-F
Full Out-of-Sample										
MSE ^a		Avg	218.957	208.345	199.070	198.924	209.478	210.459	204.707	207.962
		Med	144.744	137.001	130.463	129.495	138.829	142.448	136.673	147.108
RC ^d	5%		52.17	73.91	95.65	100	73.91	47.83	69.57	82.61
	10%		86.96	95.65	100	100	95.65	69.57	86.96	95.65
QLIKE ^b		Avg	0.693	0.608	0.552	0.543	0.608	0.641	0.590	0.607
		Med	0.687	0.612	0.560	0.537	0.606	0.650	0.591	0.601
RC	5%		0.00	0.00	86.96	100	0.00	0.00	8.70	13.04
	10%		0.00	0.00	86.96	100	0.00	8.70	8.70	17.39
MDA ^c		Avg	59.961	61.230	60.276	60.639	61.146	59.269	60.701	60.954
		Med	59.888	60.699	59.638	60.137	60.761	59.139	60.200	60.699
RC	5%		52.17	100	47.83	73.91	95.65	34.78	52.17	91.30
	10%		69.57	100	69.57	82.61	100	39.13	73.91	95.65
Normal Volatility Days										
MSE		Avg	11.745	11.228	9.030	7.730	12.230	10.115	10.364	10.808
		Med	7.798	7.000	5.947	5.091	7.489	7.478	7.101	7.392
RC	5%		47.83	52.17	52.17	95.65	56.52	26.09	26.09	26.09
	10%		56.52	60.87	69.57	95.65	78.26	30.44	47.83	30.44
QLIKE		Avg	0.399	0.340	0.313	0.298	0.339	0.364	0.327	0.318
		Med	0.362	0.327	0.296	0.269	0.330	0.339	0.314	0.303
RC	5%		0.00	4.35	21.74	95.65	0.00	0.00	13.04	43.48
	10%		0.00	4.35	34.78	95.65	0.00	0.00	17.39	56.52
MDA		Avg	57.370	58.363	56.429	56.768	58.276	55.891	57.334	57.735
		Med	57.322	58.491	55.734	56.407	58.298	55.463	57.300	57.848
RC	5%		73.91	95.65	13.04	39.13	95.65	13.04	13.04	73.91
	10%		78.26	100	13.04	43.48	100	17.39	34.78	82.61
High Volatility Days										
MSE		Avg	2131.361	2028.059	1954.202	1964.630	2030.282	2060.872	1999.278	2027.084
		Med	1293.253	1223.783	1171.940	1163.489	1236.025	1266.151	1218.390	1334.427
RC	5%		47.83	69.57	100	91.30	69.57	69.57	91.30	100
	10%		91.30	91.30	100	100	95.65	91.30	91.30	100
QLIKE		Avg	3.329	3.008	2.690	2.739	3.015	3.126	2.946	3.184
		Med	3.275	2.905	2.618	2.691	2.914	3.109	2.902	3.108
RC	5%		43.48	52.17	95.65	100	47.83	69.57	43.48	26.09
	10%		73.91	56.52	95.65	100	56.52	82.61	52.17	52.17
MDA		Avg	51.243	52.609	54.261	54.508	52.561	51.423	52.598	52.835
		Med	51.282	53.012	54.386	54.839	53.012	51.534	52.601	52.941
RC	5%		0.00	4.35	95.65	86.96	8.70	21.74	21.74	39.13
	10%		0.00	21.74	100	95.65	13.04	26.09	30.43	39.13

Notes: The best value in each row is marked in bold. ^a Averaging (or taking the median of) the MSE across 23 stocks; lower values indicate better performance. ^b Averaging (or taking the median of) the QLIKE across 23 stocks; lower values indicate better performance. ^c Averaging (or taking the median of) the MDA across 23 stocks; higher values indicate better performance.

^d Percentage of tickers with outstanding performance of the specified model against the HAR-family of models, excluding the model under assessment, at the 5% and 10% significance levels.

Table C4: Sanity Check Filter Counts Across HAR-family of Models

Statistic	AR(1)	HAR	HAR-J	CHAR	SHAR	ARQ	HARQ	HARQ-F
Mean	0	0	2	0	0	1	1	3
Median	0	0	1	0	0	1	1	1
Min	0	0	0	0	0	0	0	0
Max	0	0	8	0	0	4	3	10
STD	0	0	2	0	0	1	1	3

Notes: This table reports summary statistics for the number of times the sanity check filter is applied for the HAR-family of models during the out-of-sample period. The statistics are computed across 23 individual stocks. For each stock, the count is obtained by recording the number of RV forecast occasions over the out-of-sample evaluation window in which the RV forecast falls outside the range of RV observed in the corresponding rolling in-sample window. In such cases, the forecasted RV is replaced by the in-sample mean RV. The maximum achievable value is 1,604, corresponding to the total number of out-of-sample trading days.

Table C5: Sanity Check Filter Counts Across ML Models

Statistic	HAR-ML					News-ML					OB-ML					News/OB-ML				
	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25
Mean	42	4	11	2	0	83	20	17	14	12	78	7	13	16	19	72	8	18	19	19
Median	25	0	0	0	0	70	10	4	8	6	11	3	6	5	7	23	4	4	3	7
Min	0	0	0	0	0	11	2	0	0	0	0	0	0	0	0	0	0	0	0	0
Max	269	48	207	31	0	199	104	55	66	88	811	45	65	95	89	396	53	137	144	135
STD	62	10	44	7	0	48	23	20	18	20	181	11	17	25	27	124	12	35	36	35

Notes: This table presents summary statistics on the frequency with which the sanity check filter is applied in the HAR-ML, News-ML, OB-ML, and News/OB-ML models, for alternative model specifications using 5, 10, 15, 20, and 25 units, over the out-of-sample period. The statistics are computed across 23 individual stocks. For each stock, the count is obtained by recording the number of RV forecast occasions over the out-of-sample evaluation window in which the RV forecast falls outside the range of RV observed in the corresponding rolling in-sample window. In such cases, the forecasted RV is replaced by the in-sample mean RV. The maximum achievable value is 1,604, corresponding to the total number of out-of-sample trading days.

Table C6: Benchmark Robustness: Out-of-Sample Performance Relative to HAR
14 September 2015 to 27 January 2022 (1,604 days)

		HAR-ML					News-ML					OB-ML					News/OB-ML				
Units		5	10	15	20	25	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25
Full Out-of-Sample																					
MSE ^a	Avg	1.168	1.143	1.127	1.107	1.093	1.173	1.148	1.128	1.112	1.097	1.153	1.124	1.103	1.091	1.084	1.160	1.124	1.103	1.097	1.086
	Med	1.143	1.123	1.103	1.082	1.066	1.148	1.128	1.108	1.090	1.076	1.129	1.098	1.086	1.078	1.065	1.129	1.108	1.085	1.080	1.070
QLIKE ^b	Avg	7.239	3.462	2.250	1.555	1.245	6.968	3.809	2.329	1.786	1.365	4.334	2.081	1.483	1.365	1.256	5.533	1.984	1.504	1.467	1.302
	Med	6.115	2.950	1.950	1.367	1.163	6.980	3.779	2.276	1.568	1.203	2.893	1.805	1.352	1.293	1.239	4.539	1.731	1.391	1.315	1.231
MDA ^c	Avg	1.025	1.003	0.982	0.978	0.979	1.026	1.009	0.984	0.974	0.971	0.976	0.955	0.948	0.952	0.965	1.002	0.952	0.947	0.954	0.961
	Med	1.029	0.995	0.976	0.978	0.981	1.013	0.999	0.985	0.970	0.973	0.955	0.951	0.951	0.956	0.965	0.970	0.949	0.947	0.957	0.960
Normal Volatility Days																					
MSE	Avg	0.509	0.459	0.427	0.422	0.430	0.511	0.485	0.448	0.445	0.464	0.390	0.343	0.343	0.377	0.417	0.443	0.343	0.354	0.374	0.413
	Med	0.458	0.430	0.450	0.467	0.439	0.438	0.447	0.454	0.451	0.497	0.367	0.296	0.330	0.376	0.419	0.380	0.302	0.319	0.374	0.415
QLIKE	Avg	4.183	1.857	1.110	0.912	0.814	4.219	2.342	1.316	1.045	0.910	2.338	1.008	0.811	0.834	0.845	3.416	0.969	0.835	0.901	0.855
	Med	3.365	1.317	0.864	0.865	0.824	3.558	1.760	1.212	0.948	0.879	1.286	0.795	0.804	0.848	0.854	2.129	0.806	0.820	0.833	0.853
MDA	Avg	0.981	0.968	0.952	0.954	0.963	0.983	0.971	0.951	0.946	0.950	0.930	0.913	0.915	0.926	0.947	0.955	0.910	0.916	0.923	0.942
	Med	0.979	0.970	0.947	0.954	0.964	0.974	0.965	0.945	0.942	0.940	0.919	0.912	0.925	0.933	0.948	0.925	0.912	0.924	0.931	0.937
High Volatility Days																					
MSE	Avg	1.203	1.179	1.163	1.143	1.127	1.207	1.183	1.163	1.146	1.130	1.192	1.165	1.142	1.128	1.118	1.197	1.164	1.142	1.135	1.120
	Med	1.178	1.160	1.146	1.130	1.115	1.178	1.167	1.155	1.130	1.115	1.165	1.139	1.117	1.122	1.095	1.175	1.145	1.123	1.116	1.103
QLIKE	Avg	10.394	5.098	3.411	2.207	1.682	10.089	5.391	3.407	2.558	1.828	6.446	3.191	2.178	1.923	1.693	7.848	3.033	2.207	2.042	1.782
	Med	8.386	4.030	3.078	1.877	1.557	9.862	5.502	2.903	2.074	1.631	4.668	2.968	2.105	1.804	1.621	5.898	3.030	2.112	1.832	1.611
MDA	Avg	1.031	1.027	1.025	1.021	1.024	1.031	1.026	1.024	1.021	1.019	1.031	1.027	1.025	1.021	1.016	1.031	1.026	1.022	1.018	1.012
	Med	1.034	1.029	1.033	1.022	1.029	1.034	1.034	1.025	1.025	1.025	1.034	1.034	1.029	1.023	1.014	1.034	1.034	1.034	1.022	1.014

Notes: This table reports out-of-sample performance measures using the standard HAR model as the benchmark and complements the results in Table 6 and Table 7, which use CHAR as the benchmark.

^a Averaging (or taking the median of) the MSE across 23 stocks; lower values indicate better performance.

^b Averaging (or taking the median of) the QLIKE across 23 stocks; lower values indicate better performance.

^c Averaging (or taking the median of) the MDA across 23 stocks; higher values indicate better performance.

Table C7: Out-of-Sample Volatility Forecasting Performance of ML Models (Increasing Window)
14 September 2015 to 27 January 2022 (1,604 days)

			HAR-ML					News-ML					OB-ML					News/OB-ML					
Units			5	10	15	20	25	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25	
Full Out-of-Sample																							
MSE ^a	Avg		1.253	1.238	1.079	1.067	1.141	1.259	1.241	1.215	1.183	1.070	1.139	1.105	1.157	1.192	1.105	1.204	1.137	1.098	1.187	1.172	
		Med	1.225	1.212	1.078	1.060	1.117	1.225	1.216	1.201	1.167	1.062	1.130	1.101	1.144	1.164	1.084	1.154	1.111	1.088	1.165	1.154	
			17.39	30.44	100	100	86.96	17.39	34.78	47.83	69.57	100	91.30	100	73.91	65.22	82.61	52.17	86.96	95.65	60.87	73.91	
RC ^d	5%		56.52	69.57	100	100	100	56.52	65.22	73.91	91.30	100	100	100	82.61	78.26	100	82.61	91.30	100	73.91	82.61	
			56.52	69.57	100	100	100	56.52	65.22	73.91	91.30	100	100	100	82.61	78.26	100	82.61	91.30	100	73.91	82.61	
			56.52	69.57	100	100	100	56.52	65.22	73.91	91.30	100	100	100	82.61	78.26	100	82.61	91.30	100	73.91	82.61	
QLIKE ^b	Avg		8.654	4.350	1.038	1.031	1.276	8.853	5.055	3.100	1.730	1.190	1.368	1.296	1.854	2.509	1.702	4.487	2.234	1.598	1.912	1.766	
		Med	7.455	3.993	1.031	1.033	1.274	8.143	4.387	2.856	1.550	1.149	1.360	1.332	1.542	1.846	1.555	2.699	1.367	1.353	1.844	1.480	
			0.00	0.00	86.96	86.96	43.48	0.00	0.00	0.00	4.35	43.48	17.39	26.09	8.70	0.00	0.00	4.35	13.04	13.04	0.00	4.35	
RC	5%		0.00	0.00	86.96	86.96	43.48	0.00	0.00	0.00	4.35	43.48	17.39	26.09	8.70	0.00	0.00	4.35	13.04	13.04	0.00	4.35	
			0.00	0.00	86.96	86.96	43.48	0.00	0.00	0.00	4.35	43.48	17.39	26.09	8.70	0.00	0.00	4.35	13.04	13.04	0.00	4.35	
			0.00	0.00	100	95.65	60.87	0.00	0.00	4.35	26.09	56.52	21.74	34.78	17.39	17.39	0.00	13.04	21.74	13.04	4.35	13.04	
MDA ^c	Avg		1.028	1.004	0.969	0.971	0.975	1.030	1.008	0.978	0.967	0.963	0.933	0.943	0.953	0.953	0.963	0.977	0.957	0.955	0.950	0.944	
		Med	1.021	0.994	0.970	0.971	0.977	1.039	0.986	0.972	0.969	0.959	0.933	0.951	0.951	0.948	0.967	0.972	0.949	0.949	0.950	0.940	
			47.83	69.57	100	100	95.65	52.17	69.57	95.65	100	100	100	100	100	95.65	91.30	95.65	73.91	91.30	100	100	100
RC	5%		47.83	69.57	100	100	95.65	52.17	69.57	95.65	100	100	100	100	100	95.65	91.30	95.65	73.91	91.30	100	100	100
			47.83	69.57	100	100	95.65	52.17	69.57	95.65	100	100	100	100	100	95.65	91.30	95.65	73.91	91.30	100	100	100
			47.83	73.91	100	100	95.65	56.52	73.91	100	100	100	100	100	100	95.65	91.30	100	73.91	91.30	100	100	100
Normal Volatility Days																							
MSE	Avg		0.894	0.738	0.761	0.826	0.766	0.954	0.793	0.716	0.667	1.054	0.549	0.751	0.621	0.635	1.027	0.695	0.777	0.913	0.561	0.568	
		Med	0.757	0.684	0.741	0.801	0.761	0.941	0.799	0.696	0.645	1.022	0.536	0.729	0.622	0.600	0.928	0.538	0.748	0.846	0.529	0.580	
			82.61	100	95.65	91.30	91.30	78.26	95.65	95.65	100	69.57	100	91.30	100	91.30	69.57	91.30	91.30	78.26	100	100	
RC	5%		86.96	100	100	91.30	91.30	78.26	95.65	95.65	100	69.57	100	100	100	91.30	73.91	91.30	91.30	82.61	100	100	
			86.96	100	100	91.30	91.30	78.26	95.65	95.65	100	69.57	100	100	100	91.30	73.91	91.30	91.30	82.61	100	100	
			86.96	100	100	91.30	91.30	78.26	95.65	95.65	100	69.57	100	100	100	91.30	73.91	91.30	91.30	82.61	100	100	
QLIKE	Avg		5.844	2.754	0.876	0.888	0.996	6.116	3.192	1.843	1.146	1.053	0.910	1.000	1.121	1.592	1.401	2.596	1.554	1.291	1.080	1.028	
		Med	4.594	2.116	0.873	0.881	0.972	4.950	2.290	1.578	1.041	1.076	0.889	1.013	1.086	1.017	1.354	1.157	1.090	1.298	0.996	1.039	
			0.00	8.70	91.30	91.30	69.57	0.00	0.00	8.70	47.83	56.52	86.96	78.26	60.87	60.87	21.74	52.17	56.52	30.44	60.87	69.57	
RC	5%		0.00	8.70	91.30	91.30	69.57	0.00	0.00	8.70	47.83	56.52	86.96	78.26	60.87	60.87	21.74	52.17	56.52	30.44	60.87	69.57	
			0.00	8.70	91.30	91.30	69.57	0.00	0.00	8.70	47.83	56.52	86.96	78.26	60.87	60.87	21.74	52.17	56.52	30.44	60.87	69.57	
			8.70	13.04	91.30	91.30	69.57	0.00	0.00	13.04	47.83	60.87	86.96	82.61	69.57	60.87	30.44	52.17	65.22	39.13	65.22	82.61	
MDA	Avg		0.966	0.947	0.958	0.959	0.947	0.969	0.953	0.925	0.921	0.934	0.889	0.907	0.903	0.901	0.926	0.918	0.916	0.920	0.896	0.895	
		Med	0.953	0.938	0.959	0.959	0.953	0.962	0.937	0.925	0.927	0.926	0.890	0.908	0.912	0.907	0.931	0.915	0.902	0.907	0.894	0.900	
			78.26	95.65	95.65	91.30	86.96	82.61	86.96	100	100	100	100	100	100	95.65	95.65	86.96	91.30	100	95.65	100	
RC	5%		78.26	95.65	95.65	91.30	86.96	82.61	86.96	100	100	100	100	100	100	95.65	95.65	86.96	91.30	100	95.65	100	
			78.26	95.65	95.65	91.30	86.96	82.61	86.96	100	100	100	100	100	100	95.65	95.65	86.96	91.30	100	95.65	100	
			78.26	95.65	95.65	91.30	86.96	91.30	95.65	100	100	100	100	100	100	95.65	95.65	95.65	91.30	100	100	100	
High Volatility Days																							
MSE	Avg		1.268	1.258	1.090	1.075	1.154	1.270	1.257	1.234	1.202	1.070	1.162	1.120	1.177	1.212	1.108	1.224	1.152	1.106	1.211	1.196	
		Med	1.231	1.225	1.082	1.068	1.136	1.227	1.223	1.213	1.180	1.050	1.142	1.113	1.157	1.180	1.076	1.192	1.120	1.089	1.180	1.168	
			13.04	17.39	100	100	73.91	13.04	17.39	39.13	47.83	95.65	60.87	86.96	65.22	30.44	69.57	30.44	73.91	86.96	39.13	56.52	
RC	5%		43.48	43.48	100	100	78.26	39.13	47.83	56.52	69.57	100	78.26	95.65	69.57	65.22	82.61	52.17	86.96	95.65	60.87	65.22	
			43.48	43.48	100	100	78.26	39.13	47.83	56.52	69.57	100	78.26	95.65	69.57	65.22	82.61	52.17	86.96	95.65	60.87	65.22	
			43.48	43.48	100	100	78.26	39.13	47.83	56.52	69.57	100	78.26	95.65	69.57	65.22	82.61	52.17	86.96	95.65	60.87	65.22	
QLIKE	Avg		11.871	6.118	1.210	1.186	1.550	12.134	7.021	4.541	2.343	1.359	1.891	1.638	2.677	3.585	2.071	6.903	3.061	1.984	2.824	2.644	
		Med	10.680	5.756	1.207	1.173	1.428	12.421	6.624	3.911	2.235	1.328	1.792	1.677	1.923	3.011	1.831	4.353	1.841	1.752	2.750	2.259	
			0.00	0.00	56.52	56.52	39.13	0.00	0.00	0.00	4.35	52.17	0.00	13.04	4.35	0.00	13.04	0.00	13.04	26.09	4.35	0.00	
RC	5%		0.00	0.00	56.52	56.52	39.13	0.00	0.00	0.00	4.35	52.17	0.00	13.04	4.35	0.00	13.04	0.00	13.04	26.09	4.35	0.00	
			0.00	0.00	56.52	56.52	39.13	0.00	0.00	0.00	4.35	52.17	0.00	13.04	4.35	0.00	13.04	0.00	13.04	26.09	4.35	0.00	
			0.00	0.00	65.22	69.57	60.87	0.00	0.00	0.00	26.09	65.22	4.35	26.09	8.70	0.00	21.74	4.35	30.44	43.48	4.35	4.35	
MDA	Avg		1.076	1.074	0.995	0.985	1.040	1.077	1.073	1.068	1.069	0.987	1.054	1.014	1.042	1.069	1.021	1.071	1.033	1.002	1.070	1.068	
		Med	1.067	1.071	1.000	0.978	1.035	1.067	1.067	1.058	1.069	0.988	1.054	1.000	1.044	1.058	1.012	1.058	1.018	0.987	1.058	1.058	
			0.00	0.00	95.65	95.65	43.48	0.00	0.00	0.00	4.35	52.17	0.00	21.74	4.35	0.00	21.74	0.00	17.39	26.09	4.35	4.35	
RC	5%		0.00	0.00	95.65	95.65	43.48	0.00	0.00	0.00	4.35	52.17	0.00	21.74	4.35	0.00	21.74	0.00	17.39	26.09	4.35	4.35	
			0.00	0.00	95.65	95.65	43.48	0.00	0.00	0.00	4.35	52.17	0.00	21.74	4.35	0.00	21.74	0.00	17.39	26.09	4.35	4.35	
			0.00	0.00	95.65	95.65	52.17	0.00	0.00	4.35	8.70	69.57	13.04	30.44	13.04	0.00	30.44	4.35	34.78	30.44	4.35	8.70	

Notes: The best value in each row is marked in bold. ^aThe mean (or median) MSE ratio of ML to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^bThe mean (or median) QLIKE ratio of ML to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^cThe mean (or median) MDA ratio of CHAR to ML for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^dPercentage of tickers with the outstanding performance of ML against the HAR-family of models at the 5% and 10% significance levels.

Table C8: Out-of-Sample Volatility Forecasting Performance of ML Models (Reduced In-Sample Training Period)
14 September 2015 to 27 January 2022 (1,604 days)

			HAR-ML					News-ML					OB-ML					News/OB-ML				
Units			5	10	15	20	25	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25
Full Out-of-Sample																						
MSE ^a	Avg		1.206	1.203	1.185	1.166	1.155	1.198	1.195	1.181	1.159	1.147	1.137	1.102	1.146	1.135	1.116	1.189	1.166	1.085	1.131	1.119
		Med	1.194	1.187	1.164	1.143	1.143	1.181	1.175	1.163	1.146	1.137	1.131	1.100	1.135	1.128	1.111	1.174	1.149	1.083	1.131	1.107
RC ^d	5%		17.39	26.09	30.44	47.83	47.83	21.74	30.44	39.13	56.52	60.87	60.87	91.30	60.87	69.57	78.26	34.78	39.13	95.65	69.57	82.61
		10%	47.83	47.83	69.57	78.26	82.61	56.52	60.87	65.22	82.61	86.96	82.61	100	82.61	86.96	91.30	56.52	73.91	100	91.30	91.30
QLIKE ^b	Avg		9.053	6.634	4.563	3.050	2.053	8.596	6.206	4.051	2.546	2.105	1.862	1.453	2.324	1.861	1.523	4.250	3.157	1.422	1.826	1.680
		Med	8.704	6.357	4.480	2.310	2.090	8.464	5.956	3.843	2.230	1.749	1.679	1.422	1.837	1.735	1.467	3.963	2.445	1.436	1.597	1.486
RC	5%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.35	0.00	0.00	13.04	0.00	0.00	0.00	0.00	8.70
		10%	0.00	0.00	0.00	0.00	17.39	0.00	0.00	0.00	0.00	8.70	0.00	8.70	8.70	8.70	21.74	0.00	4.35	4.35	0.00	13.04
MDA ^c	Avg		1.036	1.019	1.005	0.984	0.971	1.024	1.009	0.988	0.972	0.966	0.944	0.939	0.947	0.950	0.950	0.979	0.960	0.948	0.944	0.950
		Med	1.033	1.009	0.996	0.972	0.967	1.015	1.015	0.981	0.967	0.967	0.947	0.937	0.937	0.943	0.952	0.988	0.942	0.952	0.942	0.945
RC	5%		47.83	56.52	69.57	95.65	95.65	60.87	60.87	86.96	91.30	100	95.65	100	100	100	100	82.61	91.30	100	100	100
		10%	47.83	60.87	73.91	95.65	100	65.22	69.57	86.96	100	100	100	100	100	100	100	91.30	91.30	100	100	100
Normal Volatility Days																						
MSE	Avg		0.765	0.721	0.632	0.544	0.527	0.715	0.678	0.579	0.559	0.545	0.421	0.496	0.468	0.480	0.520	0.543	0.486	0.587	0.470	0.511
		Med	0.743	0.661	0.604	0.515	0.538	0.660	0.675	0.531	0.525	0.512	0.378	0.476	0.424	0.441	0.484	0.523	0.474	0.556	0.420	0.500
RC	5%		91.30	95.65	91.30	100	100	91.30	91.30	100	100	100	100	100	100	100	100	100	100	100	100	
		10%	91.30	95.65	95.65	100	100	95.65	91.30	100	100	100	100	100	100	100	100	100	100	100	100	100
QLIKE	Avg		6.281	4.347	3.005	1.726	1.188	6.096	4.366	2.549	1.566	1.317	1.126	1.017	1.498	1.169	1.014	2.632	2.201	1.140	1.204	1.219
		Med	6.259	3.656	3.085	1.321	1.085	5.670	3.862	2.693	1.451	1.110	1.027	0.986	1.070	1.050	0.958	2.232	1.358	1.029	1.033	1.007
RC	5%		0.00	0.00	0.00	17.39	47.83	0.00	0.00	13.04	26.09	56.52	73.91	82.61	56.52	60.87	69.57	8.70	39.13	52.17	69.57	65.22
		10%	0.00	0.00	0.00	26.09	52.17	0.00	0.00	13.04	39.13	56.52	73.91	82.61	69.57	69.57	73.91	13.04	39.13	60.87	73.91	73.91
MDA	Avg		0.981	0.963	0.953	0.934	0.927	0.968	0.955	0.935	0.924	0.921	0.891	0.904	0.898	0.906	0.914	0.916	0.904	0.919	0.901	0.914
		Med	0.983	0.961	0.943	0.929	0.932	0.967	0.975	0.931	0.916	0.921	0.895	0.900	0.896	0.900	0.914	0.925	0.900	0.923	0.901	0.907
RC	5%		78.26	86.96	91.30	100	100	82.61	91.30	95.65	100	100	100	100	100	100	100	100	100	100	100	100
		10%	86.96	91.30	91.30	100	100	82.61	100	100	100	100	100	100	100	100	100	100	100	100	100	100
High Volatility Days																						
MSE	Avg		1.223	1.221	1.206	1.190	1.179	1.216	1.214	1.204	1.183	1.170	1.164	1.125	1.172	1.160	1.139	1.213	1.191	1.104	1.157	1.142
		Med	1.218	1.218	1.193	1.179	1.173	1.189	1.199	1.196	1.177	1.161	1.154	1.131	1.166	1.164	1.139	1.204	1.190	1.109	1.150	1.141
RC	5%		4.35	13.04	17.39	34.78	34.78	8.70	13.04	21.74	34.78	34.78	39.13	47.83	34.78	34.78	47.83	17.39	26.09	69.57	43.48	39.13
		10%	39.13	43.48	43.48	52.17	52.17	34.78	39.13	43.48	52.17	60.87	52.17	73.91	56.52	52.17	69.57	39.13	47.83	86.96	56.52	69.57
QLIKE	Avg		11.643	8.924	6.007	4.234	2.888	11.079	8.000	5.509	3.448	2.813	2.538	1.870	3.081	2.494	2.002	5.833	4.092	1.684	2.401	2.108
		Med	10.574	8.520	5.673	3.364	2.800	10.413	8.209	5.069	3.041	2.351	2.368	1.852	2.641	2.403	1.858	5.273	3.552	1.646	2.274	1.948
RC	5%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.35	0.00	0.00	4.35	0.00	0.00	4.35	0.00	4.35
		10%	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.04	0.00	4.35	8.70	0.00	8.70	0.00	8.70	8.70	0.00	17.39
MDA	Avg		1.070	1.070	1.070	1.069	1.069	1.070	1.070	1.069	1.067	1.067	1.069	1.051	1.066	1.063	1.058	1.070	1.070	1.032	1.064	1.060
		Med	1.080	1.080	1.080	1.080	1.080	1.080	1.080	1.080	1.073	1.073	1.080	1.056	1.073	1.073	1.070	1.080	1.080	1.026	1.073	1.073
RC	5%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.70	0.00	0.00
		10%	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.70	0.00	4.35	4.35	0.00	0.00	13.04	4.35	8.70

Notes: The best value in each row is marked in bold. ^a The mean (or median) MSE ratio of ML to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^b The mean (or median) QLIKE ratio of ML to CHAR for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^c The mean (or median) MDA ratio of CHAR to ML for 23 tickers. A ratio above one indicates degradation, while below one indicates improvement. ^d Percentage of tickers with the outstanding performance of ML against the HAR-family of models at the 5% and 10% significance levels.

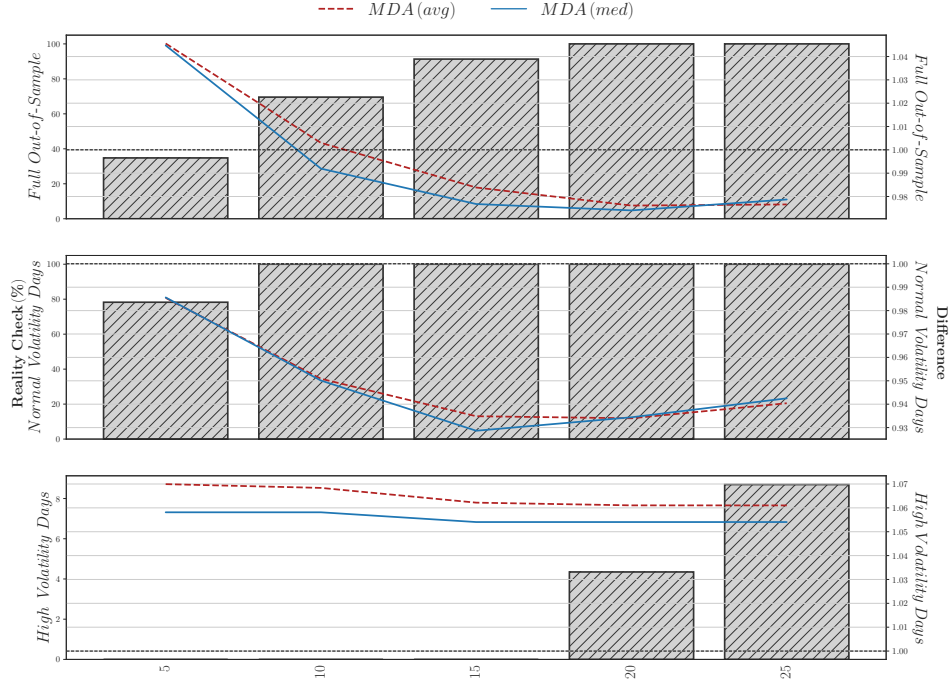


Figure C2: LSTM with $\Phi_t = \{RV_t, \dots, RV_{t-20}\}$ (MDA)

Notes: The bar chart is the percentage of tickers with outstanding performance considering the MDA loss function at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for every specified number of units of the ML model. The values for the bar chart can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MDAs of the CHAR model relative to the ML model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

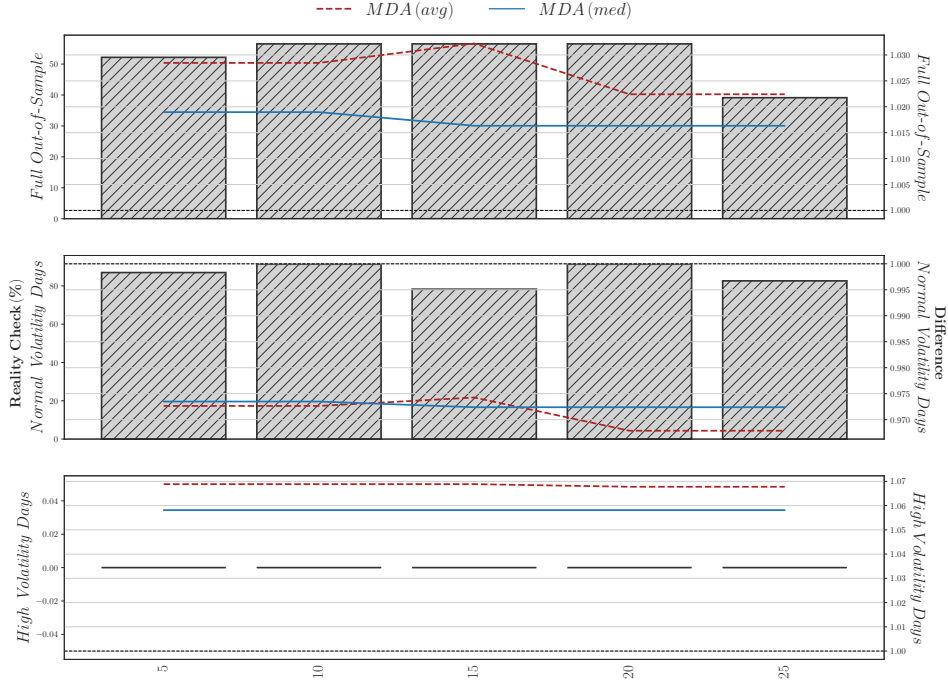


Figure C3: FCNN with $\Phi_t = \{RV_t, \overline{RV}_{t-1}^w, \overline{RV}_{t-1}^m\}$ (MDA)

Notes: The bar chart is the percentage of tickers with outstanding performance considering the MDA loss function at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for every specified number of units of the ML model. The values for the bar chart can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MDAs of the CHAR model relative to the ML model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

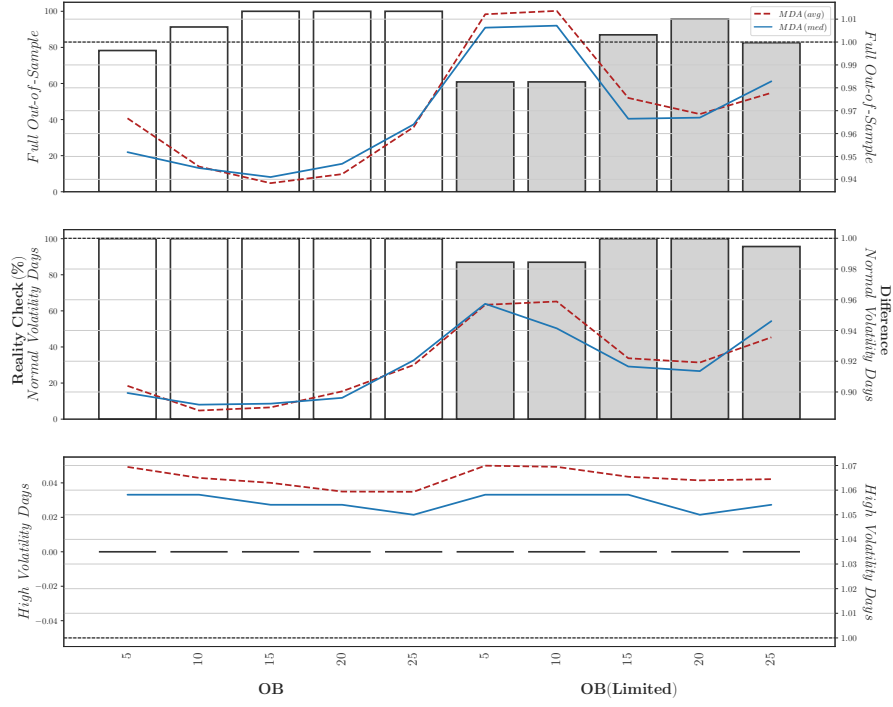


Figure C4: Comparison of OB-ML and OB-ML (Limited) Models (MDA)

Notes: The bar chart is the percentage of tickers with the outstanding performance considering the MDA loss function at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for every specified model. The first five bars represent the results obtained using the LOB information set in Subsection 3.3, while the second five bars are those derived from the limited LOB information set. The dashed (solid) line shows the average (median) out-of-sample MDAs of the CHAR model relative to the ML model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

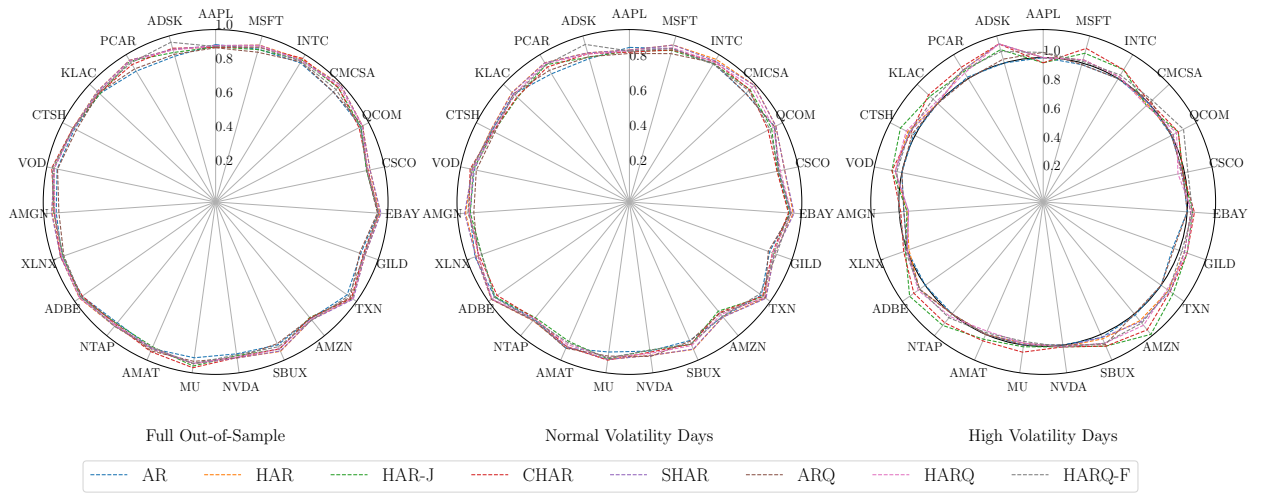
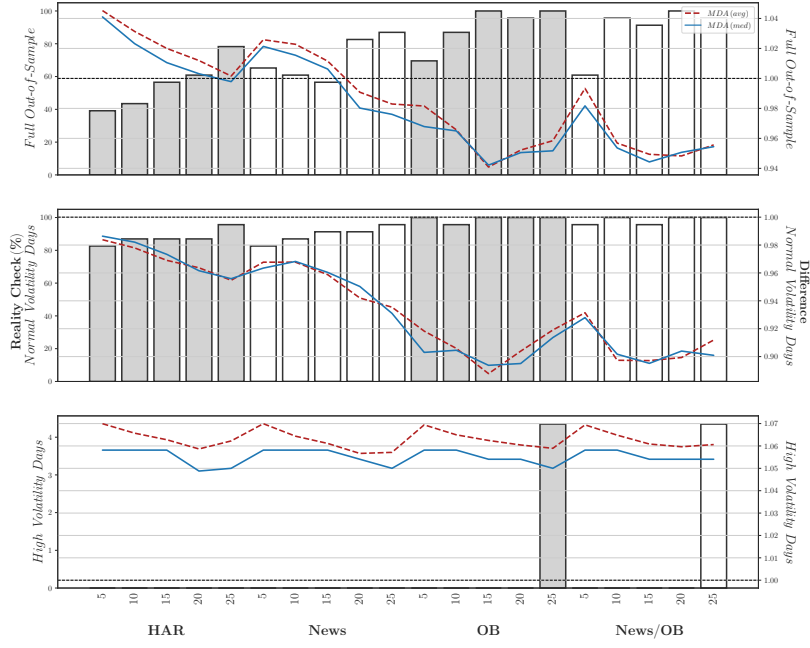
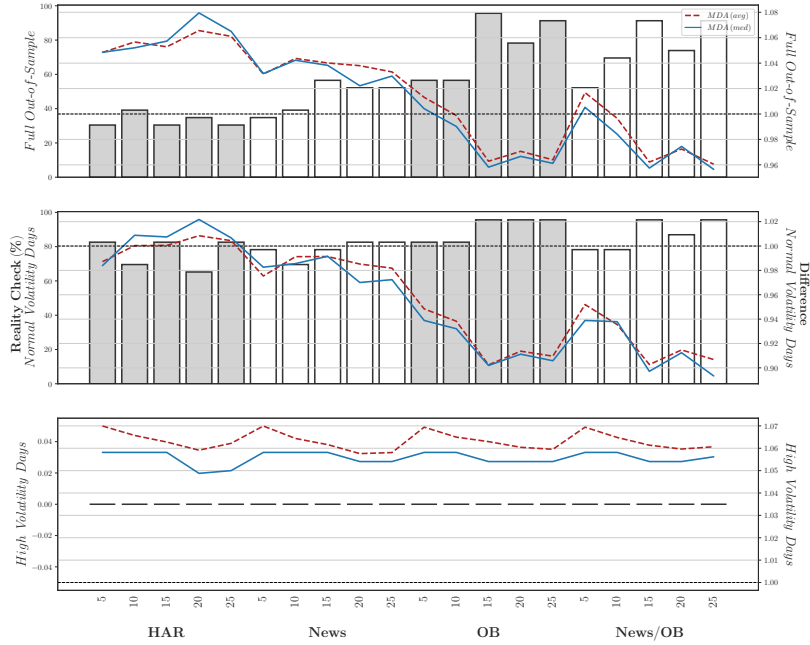


Figure C5: ρ_{MDA} between OB-ML (15 Units) and HAR-Family of Models

Notes: Each radar chart illustrates the performance of the AR(1), HAR, HAR-J, CHAR, SHAR, ARQ, HARQ, and HARQ-F models relative to the OB-ML model with 15 units across the specified tickers based on the MDA loss function. The evaluations are conducted over the full out-of-sample period (left figure), during normal volatility days (middle figure), and during high volatility days (right figure). A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The bold circle represents the baseline (no improvement, i.e., a value of one).



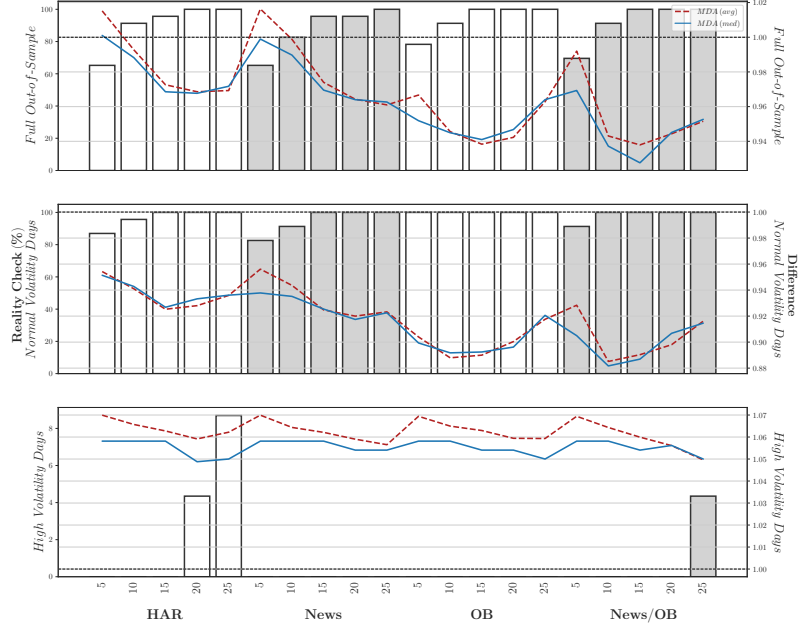
(a) Five Lags



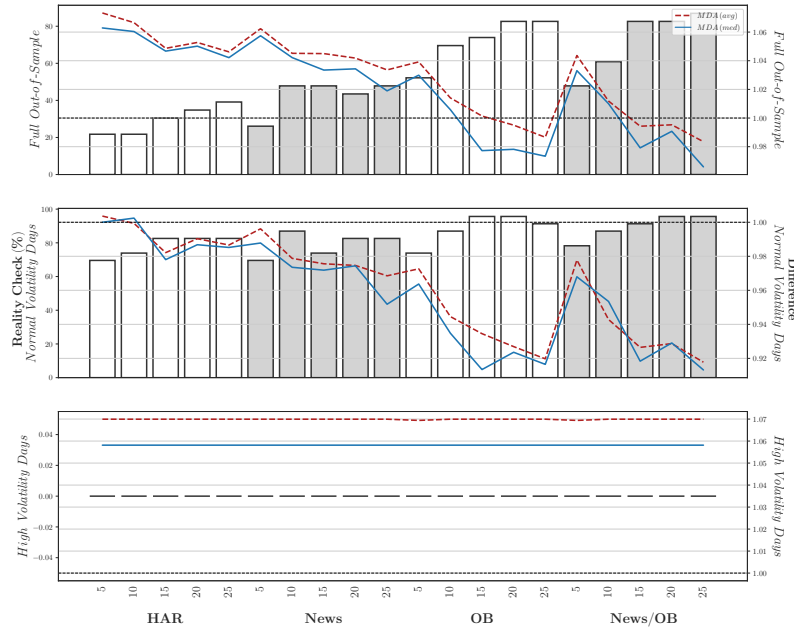
(b) One Lag

Figure C6: ML Models with Restricted Number of Lags (MDA)

Notes: Figure C6a and Figure C6b display the results considering five lags (last week) and one lag (last day). The bar chart is the percentage of tickers with the outstanding performance considering the MDA loss function at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for every HAR-ML, News-ML, OB-ML, and News/OB-ML group. The values for the bar chart can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MDAs of the CHAR model relative to the ML model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.



(a) MSE Loss Function



(b) QLIKE Loss Function

Figure C7: Minimising MSE vs QLIKE as the Loss Function in Training (MDA)

Notes: Figure C7a (Figure C7b) presents the results for the four groups of ML models (HAR-ML, News-ML, OB-ML, and News/OB-ML) trained using minimising MSE (QLIKE) as the loss function. The other model specifications are the same as in Subsection 4.2. The bar chart is the percentage of tickers with the outstanding performance at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for every specified ML model. The values for the bar chart can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MDAs of the CHAR model relative to the ML model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

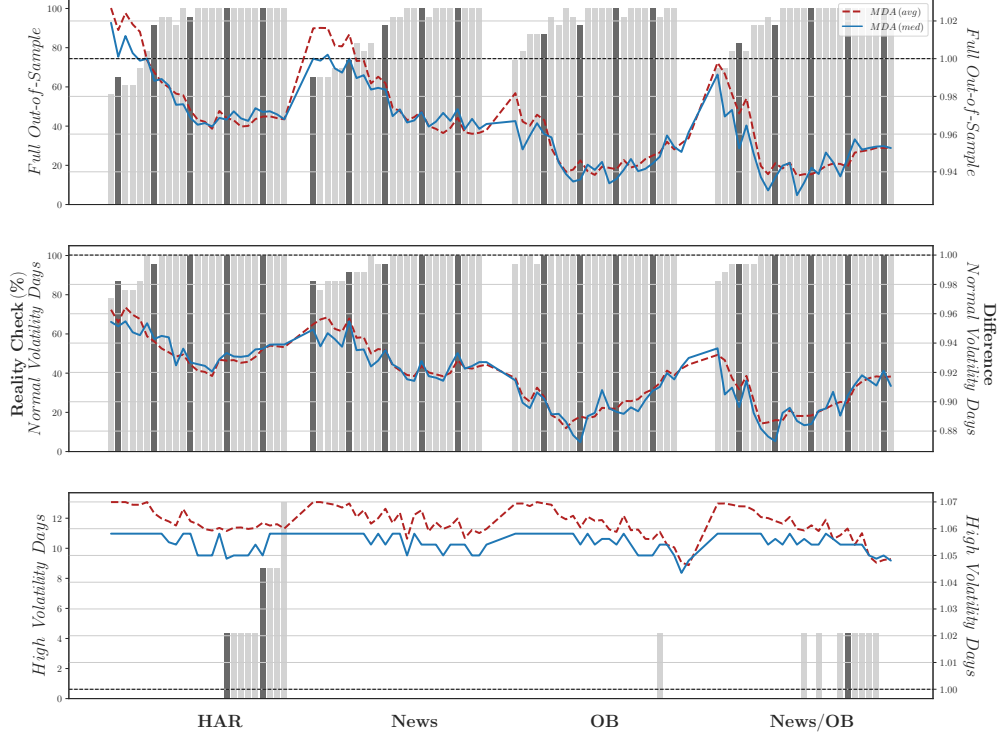


Figure C8: No. of Units vs. No. of Epochs (MDA)

Notes: From left to right, this figure consists of the HAR-ML, News-ML, OB-ML, and News/OB-ML groups. For every group, the results are shown in the following order from left to right (#units-#epochs): 5-25, 5-50, 5-75, 5-100, 5-125, 10-25, 10-50, 10-75, 10-100, 10-125, 15-25, 15-50, 15-75, 15-100, 15-125, 20-25, 20-50, 20-75, 20-100, 20-125, 25-25, 25-50, 25-75, 25-100, and 25-125. For the sake of clarity, these values are not shown in this figure. The bar chart is the percentage of tickers with the outstanding performance considering the MDA loss function at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for every specified ML model. The darker bar charts are the RC values of primary experiments in Section 5. The RC values can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MDAs of the CHAR model relative to the ML model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

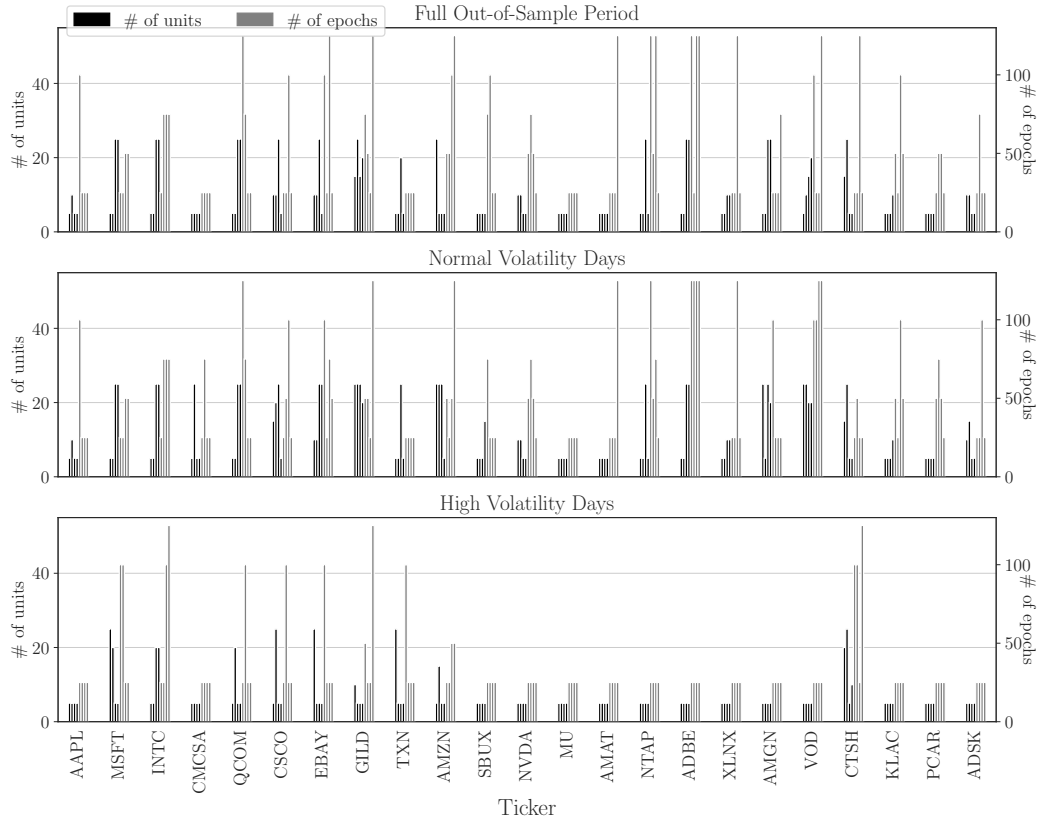


Figure C9: The Best-Performing Hyperparameters (#Units & #Epochs) (MDA)

Notes: There are two sets of four bars for each ticker that correspond to the four groups of ML models, viz. HAR-ML, News-ML, OB-ML, and News/OB-ML. The first four black bars correspond to their optimal number of units, while the next four grey bars correspond to their optimal number of epochs. The top, middle, and bottom figures correspond to the full out-of-sample period, normal volatility days, and high volatility days, respectively, based on the MDA loss function. The number of units (# of units) can be read from the left axis, and the number of epochs (# of epochs) can be read from the right axis.

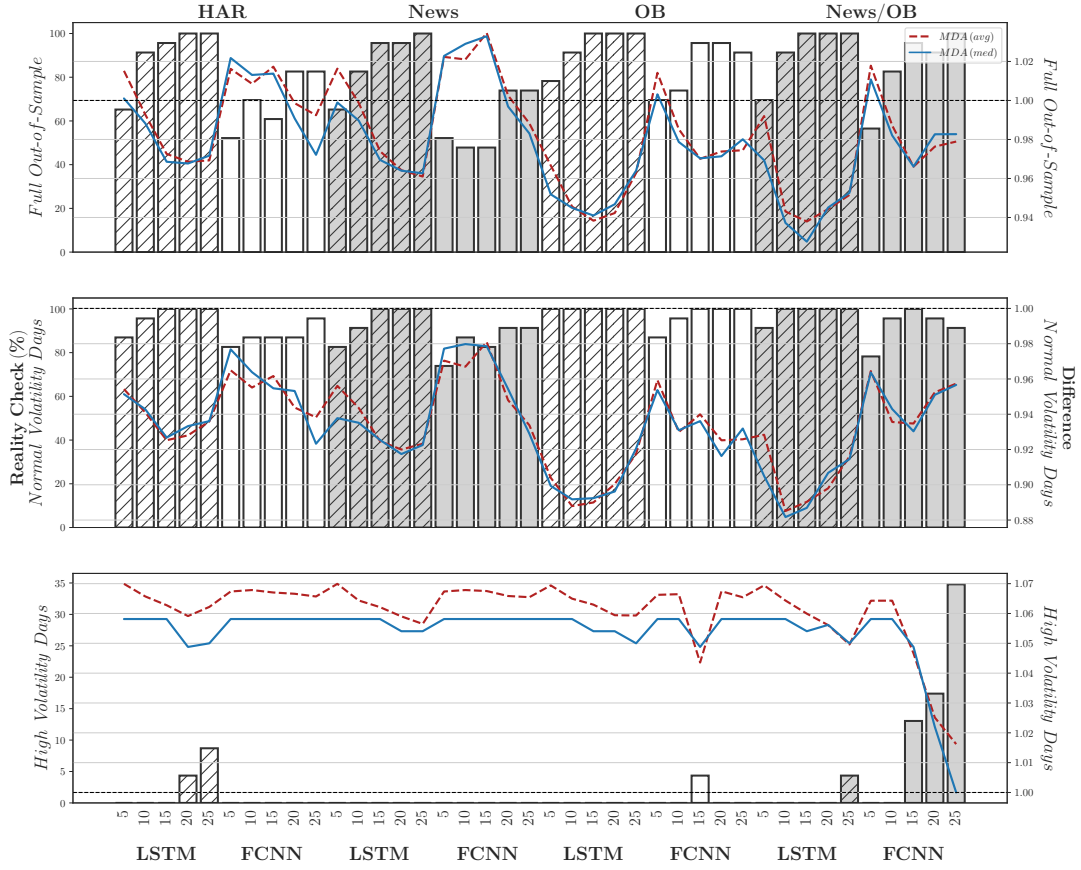


Figure C10: Comparison of LSTM and FCNN Models (MDA)

Notes: The bar chart is the percentage of tickers with the outstanding performance considering the MDA loss function at the 5% significance level of the RC compared to the HAR-family of models as the benchmark for every specified model. The first, second, third, and fourth sets of ten bars correspond to the HAR, News, OB, and News/OB groups, respectively. Within each set, the first five hatched bars represent results from the LSTM model, while the subsequent five bars represent results from the FCNN model. Each bar within these subsets reflects performance associated with a specific number of units. The values for the bar chart correspond to the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MDAs of the CHAR model relative to the ML model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

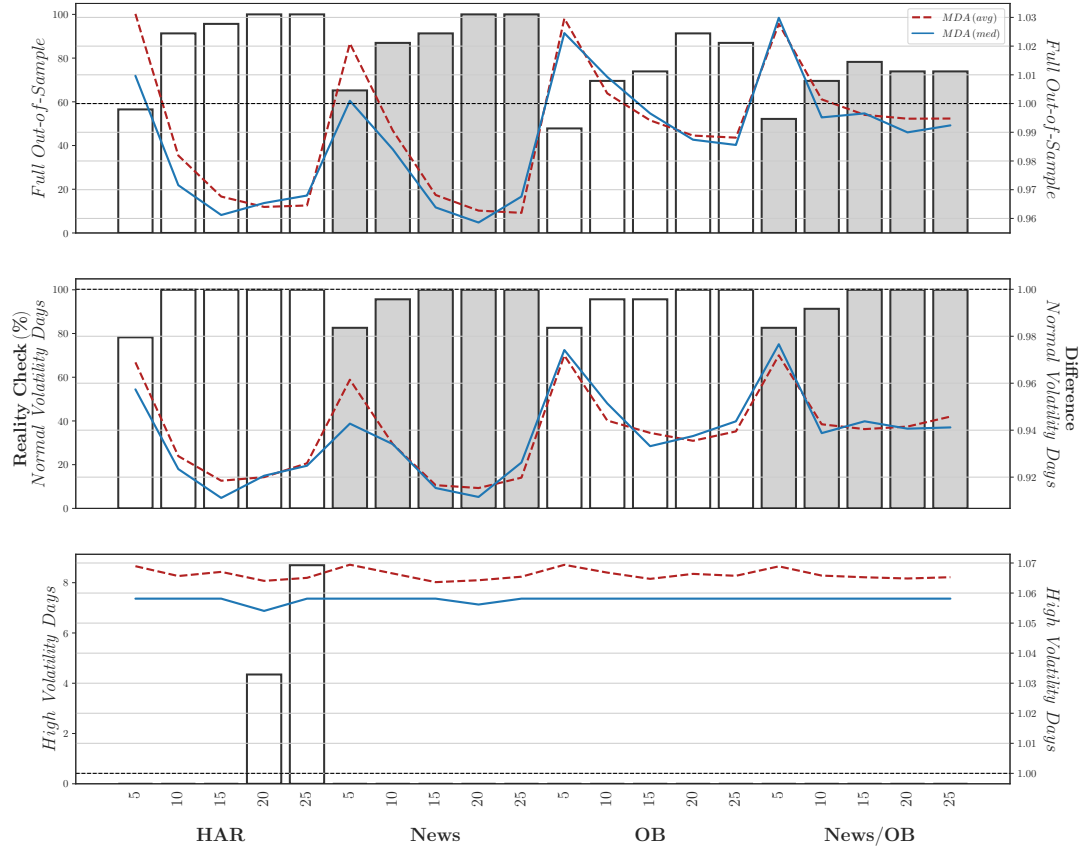


Figure C11: ML Models Utilising Early Stopping (MDA)

Notes: The bar chart is the percentage of tickers with the outstanding performance considering the MDA loss function at the 5% significance level of the RC compared to the HAR-family of models as the benchmark. The results are presented for each model group: HAR, News, OB, and News/OB, across varying numbers of units, specifically 5, 10, 15, 20, and 25 units. The values for the bar chart can be read from the left-hand axis. The dashed (solid) line shows the average (median) out-of-sample MDAs of the CHAR model relative to the ML model across 23 tickers. A value below one indicates improved performance of the ML model, while a value above one indicates degradation. The values for the dashed and solid lines can be read from the right-hand axis. The horizontal dashed line represents no improvement.

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